Ch.6 후반 · Ch.7 질적 정보(더미변수) — PPT 기준 필기본

교재: Wooldridge 5판 PDF p.202–267 · 강의 PPT: 8판 — 10.27 '계량경제학 1'(Ch.6 후반, A1–A12), 10.29 'econometrics_10.29'(Ch.7, S1–S38). 오른쪽 열 = PPT 필기 + '교수 설명' 칸(10.27·10.29 전사본, 10.29 판서 사진 6장).

1. 표시 읽는 법

노랑 하이라이트 PPT 슬라이드에 직접 나온 내용 — 1순위로 읽기
초록 박스 PPT에 나온 수식 · 추정 결과
파랑 밑줄 · REF PPT 주장의 교재 근거 — 이해가 막힐 때만
보라 · OBJ 학습목표(S2–4)에는 있으나 슬라이드 본문엔 없음
회색 · SKIP PPT 범위 밖 — 건너뛰기
표시 없음 PPT에서 다루지 않은 설명 문단 — 가볍게 훑기

여백 태그: **A3** = Ch.6 PPT 3번 슬라이드, **S14** = Ch.7 PPT 14번 슬라이드. 곡선을 따라가면 오른쪽 열의 필기로 연결됨. 주황 칸의 **[판서]** = 10.29 판서 사진에서 옮긴 내용. 하이라이트·박스는 PDF 주석이라 필기 앱에서 지우거나 바꿀 수 있음.

2. 슬라이드 ↔ 교재 매핑

Slide	내용	교재 쪽	비고
A1	Average Partial Effects	—	교재 5판에 없음 → 표지 A1 칸
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3. 학습목표 ↔ 슬라이드 갭 (전사본 확인 결과)

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- **목표 7.5** 절편 차이를 허용한 Chow 검정 → p.247–248 (보라) — **기본 Chow만 다룸 · 절편 허용 버전은 언급 없음**
- **목표 7.8** 이산 종속변수의 계수 해석 (7.7절) → p.256–257 (보라) — **슬라이드 없음 · 수업 언급 없음**

세 주제 모두 수업에서 따로 설명하지 않음 → 수업 기준 우선순위는 낮음. 학습목표에는 있으므로 시험 범위 포함 여부는 교수에게 확인 권장.

판(edition) 차이: 교재 PDF는 5판, PPT는 8판. ① **APE(A1)**와 **y(0)/y(1)·ATE·URA/RRA(S31–S37)**는 5판에 없음 → PPT와 교수 설명 칸이 유일한 자료. ② 수치 차이: 식 6.25 roe 계수(교재 19.63 / 슬라이드 19.03), 식 7.5 SE(교재 .30 / 슬라이드 .26). ③ 슬라이드 오류: S14의 n=2,725·R^2=.0422는 예제 7.12 수치(올바른 값 n=526, R^2=.461), S13의 %Δcolonial 식(교수도 수업에서 지적).

자료 상태: 10.27은 전사본 전체 + 판서 사진 없음. 10.29는 녹음이 S19 중간부터 시작(약 71분) + 판서 사진 6장 → S5–S18 복습분의 10.29 설명은 없음.

전사본 교정표: S test → **F test** · child test → **Chow test** · R-Spirit, R-Sphere → **R-squared** · multi-coordinator(ity) → **multicollinearity** · Hetschel's Schema 60, heterosexuality, Architecture Cassettes → **heteroskedasticity** · still a system → **still consistent** · Hester → **Head Start** · Dr. Vizs → **doctor visits** · omitted value of the virus → **omitted variable bias** · achievement effect → **treatment effect**

4. 공부 순서

1. Ch.6 후반(p.202–207, 212–213): 노랑 + 초록만. A7–A10(통제변수 선택)은 서술형 재료 — 주황 칸의 예시까지 읽기.
2. Ch.7 전반(p.227–243): S5–S19. 그림 7.1·7.2를 손으로 다시 그려보기(절편 이동 vs 기울기 차이).
3. Chow 검정(p.245–247): 판서의 3개 회귀(SSR_R, SSR1, SSR2)와 식 7.24를 직접 써보기.

4. LPM(p.248–252): 단점 3가지와 V(Y)=P(1−P) 유도는 [시험] 표시 — 주황 칸 S28 먼저.
5. 정책 평가(p.253–256): S30의 OVB 논리 → S31–S37은 교재에 없으므로 PPT + 주황 칸으로.
6. 보라 3곳은 마지막에 개념만. 회색은 스킵: 예측 신뢰구간 · 잔차 분석 (207–212) · 예제 6.7–6.8 (214–215) · 예제 7.2 (232) · 예제 7.7–7.8 (238–240) · 예제 7.9 (241) · 예제 7.11 (244–245) · 예제 7.12 (252–253) · 두 장의 연습문제 (218–226, 258–267).

5. 교수 설명 — 전체 · 교재에 없는 슬라이드

▶ 교수 설명 · GEN 공지 · 중간고사 재평가 · 다음 수업

10.27 수업 앞부분 (00:02–08:14) — 중간고사 재평가 투표

- 교수가 제시한 선택지: ① 중간고사 문제를 **과제(homework)**로 다시 내고 그 과제로 채점 ② 같은/비슷한 문제로 중간고사 재시험 ③ 아무것도 안 하고 응시자 전원 가산점
- 거수 투표 결과 → [중요] **① 과제로 결정**. 강의 웹사이트에 새 과제를 올림. 중간고사 문제 상당수를 그대로 냄. 해당 문제는 **중간고사가 아니라 과제 점수로 채점**
- [중요] 규칙: **AI 사용 금지**, 평소와 달리 **다른 학생과 논의도 금지**
- 학생 요청(과제가 시험보다 너무 쉽다 → 기말 대비용으로 더 어려운 과제를) → 교수: "그게 바로 이번 과제다. 어려운 문제들이 나간다"
- 기말고사 난이도도 거수: '같은 난이도(못 본 학생은 가산점)' 약 7명 vs '난이도 하향' 약 10명. 최종 결정은 전사본에 명확히 남지 않음 (전사 불명확)
- "중간고사가 어려웠다는 걸 안다. 못 봤어도 낙담하지 마라. **두 번째 기회**다."

10.29 수업: 녹음이 수업 중간(S19)부터 시작됨. 앞부분은 PPT 기준으로 필기.

- 수업 끝 공지 (01:11:44): [중요] **다음 주에 Ch.8을 마치고 RStudio로 돌아가 이분산-일치 표준오차 계산과 이분산 처리 실습**

출석: 두 수업 모두 '질문을 던지고 생각할 시간을 주는 동안' 출석을 부름 (10.27: S14 기준집단 질문, 10.29: S30 OVB 질문).

▶ 교수 설명 · A1 Average Partial Effects (10.27 PPT 1번 슬라이드)

(10.27 08:14–11:10) 교호항·이차항이 있으면 "x1이 늘 때 y가 어떻게 되나?"의 답이 **"it depends"** — x2의 값에 따라 달라짐. 이를 한 숫자로 요약하는 방법이 **평균 부분효과(APE)**.

- 절차: ① 부분효과를 먼저 구한다 ② 표본 전체에서 **평균**낸다
- 교호항 모형 $y = \beta0+\beta1x1+\beta2x2+\beta3x1x2$: x1의 부분효과 = $\beta1+\beta3x2$ → 추정치를 넣고 x2 자리에 **표본평균 $\bar{x}2$** → $APE = \beta1 + \beta3 \cdot \bar{x}2$
- 이차항 모형 $y = \beta0+\beta1x1+\beta2x1^2$: 부분효과 = $\beta1+2\beta2x1$ → $APE = \beta1 + 2\beta2 \cdot \bar{x}1$ ("정확히는 x1의 표본평균")
- 교재 5판에는 APE 용어가 없음(8판에서 추가). 6.2절 교호항 설명(p.198–200, 이 PDF 범위 밖)의 '관심 있는 값에서 부분효과 평가'와 같은 발상
- (식의 계수 번호는 전사본에서 불명확 → PPT 정의에 맞춰 정리)

▶ 교수 설명 · CH8 Ch.8 도입: 이분산 (10.29 후반, 교재 PDF 범위 밖)

(10.29 01:02:22–01:11:44) Ch.7을 마치고 **Ch.8 이분산** 도입 (PPT: econometrics_10.29_2.pptx S4–S6). 교재 PDF 범위 밖이라 여기에 정리.

지금까지 이분산은 "있으면 망한다"는 **유령 이야기**처럼만 다뤘음. 이 장은 **"그럼 어떻게 싸울 것인가"** — 물론 없는 게 최선이지만, 있을 때 쓸 방법을 배움.

좋은 소식

- OLS는 이분산 하에서도 여전히 **불편·일치**
- R^2 해석도 그대로

나쁜 소식

- OLS 추정량의 **분산 공식이 틀려짐** → 통상의 **t검정·F검정은 수정 필요**
- OLS가 더 이상 **BLUE(최량선형불편추정량)가 아닐 수 있음** → 더 효율적인(분산이 작은) 선형추정량이 있을 수 있음

[판서] 추정량 분포 그림: 둘 다 참값 βj 중심(불편)이지만, 이분산일 때 OLS βj보다 **분포가 더 좁은 '다른 βj'**가 존재할 수 있음. "OLS는 여전히 불편·일치지만 더 나은 불편·일치 추정량이 있을 수 있다."

Robust의 뜻: "우리가 쓴 가정이 틀려도 여전히 괜찮은가? 괜찮으면 robust." → 이분산이 있어도 타당한 OLS 표준오차 = **이분산-robust 표준오차**.

- [판서] s.e.(βj,OLS) → **HC1, HC2, HC3, ⋯** 여러 버전. HC = **Heteroskedasticity-Consistent**
- 'consistent'이므로 **대표본**에서 타당 (일치성 = 표본이 무한히 커질 때 맞음)
- 버전별 차이는 R 실습 때 다룸

Adjusted *R*-Squared

Most regression packages will report, along with the *R*-squared, a statistic called the **adjusted *R*-squared**. Because the adjusted *R*-squared is reported in much applied work, and because it has some useful features, we cover it in this subsection.

To see how the usual *R*-squared might be adjusted, it is usefully written as

$$R^2 = 1 - (\text{SSR}/n)/(\text{SST}/n), \qquad [6.20]$$

where SSR is the sum of squared residuals and SST is the total sum of squares; compared with equation (3.28), all we have done is divide both SSR and SST by n. This expression reveals what R^2 is actually estimating. Define σ_y^2 as the population variance of y and let σ_u^2 denote the population variance of the error term, u. (Until now, we have used σ^2 to denote σ_u^2, but it is helpful to be more specific here.) The **population *R*-squared** is defined as $\rho^2 = 1 - \sigma_u^2/\sigma_y^2$; this is the proportion of the variation in y in the population explained by the independent variables. This is what R^2 is supposed to be estimating.

R^2 estimates σ_u^2 by SSR/n, which we know to be biased. So why not replace SSR/n with SSR/$(n - k - 1)$? Also, we can use SST/$(n - 1)$ in place of SST/n, as the former is the unbiased estimator of σ_y^2. Using these estimators, we arrive at the adjusted *R*-squared:

$$\begin{aligned}\bar{R}^2 &= 1 - [\text{SSR}/(n - k - 1)]/[\text{SST}/(n - 1)] \\ &= 1 - \hat{\sigma}^2/[\text{SST}/(n - 1)],\end{aligned} \qquad [6.21]$$

because $\hat{\sigma}^2 = \text{SSR}/(n - k - 1)$. Because of the notation used to denote the adjusted *R*-squared, it is sometimes called *R-bar squared*.

The adjusted *R*-squared is sometimes called the *corrected R-squared*, but this is not a good name because it implies that $\bar{R}^2$ is somehow better than R^2 as an estimator of the population *R*-squared. Unfortunately, $\bar{R}^2$ is *not* generally known to be a better estimator. It is tempting to think that $\bar{R}^2$ corrects the bias in R^2 for estimating the population *R*-squared, ρ^2, but it does not: the ratio of two unbiased estimators is not an unbiased estimator.

The primary attractiveness of $\bar{R}^2$ is that it imposes a penalty for adding additional independent variables to a model. We know that R^2 can never fall when a new independent variable is added to a regression equation: this is because SSR never goes up (and usually falls) as more independent variables are added. But the formula for $\bar{R}^2$ shows that it depends explicitly on k, the number of independent variables. If an independent variable is added to a regression, SSR falls, but so does the df in the regression, $n - k - 1$. SSR/$(n - k - 1)$ can go up or down when a new independent variable is added to a regression.

An interesting algebraic fact is the following: If we add a new independent variable to a regression equation, $\bar{R}^2$ increases if, and only if, the t statistic on the new variable is greater than one in absolute value. (An extension of this is that $\bar{R}^2$ increases when a group of variables is added to a regression if, and only if, the F statistic for joint significance of the new variables is greater than unity.) Thus, we see immediately that using $\bar{R}^2$ to decide whether a certain independent variable (or set of variables) belongs in a model gives us a different answer than standard t or F testing (because a t or F statistic of unity is not statistically significant at traditional significance levels).

It is sometimes useful to have a formula for $\bar{R}^2$ in terms of R^2. Simple algebra gives

$$\bar{R}^2 = 1 - (1 - R^2)(n - 1)/(n - k - 1). \qquad [6.22]$$

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A2 **A2 R² 일반론**(교재 p.200–201, 이 PDF 범위 밖): ① 높은 R^2 ≠ 인과 해석 ② 낮은 R^2여도 부분효과는 정밀하게 추정 가능. R^2는 '설명된 변동 비율'일 뿐, 추정량의 정확도를 말하지 않음.

A3 $R^2 = 1 - (SSR/n)/(SST/n)$. n으로 나눠 보면 R^2가 무엇을 추정하는지 보임.

A3 **모집단 R^2** $= 1 - \sigma^2u/\sigma^2y$. 표본 R^2는 σ^2u를 SSR/n(편의 있음)으로 추정.

A3 **Adjusted R^2**: SSR/n → SSR/(n−k−1), SST/n → SST/(n−1) (자유도 반영). 분자는 $\hat{\sigma}^2$.

A4 **핵심 성질**: 변수 추가에 별점. R^2는 변수를 넣으면 절대 안 내려가지만 (SSR↓), R^2는 df(n−k−1)도 줄어 오를 수도 내릴 수도 있음.

A4 새 변수 추가 시 R^2↑ ⇔ **$|t| > 1$** (변수 묶음이면 $F > 1$).

REF 주의: $|t|>1$은 통상 유의수준($|t|\approx 2$)보다 느슨함 → R^2 기준 선택과 t·F 검정 결과가 다를 수 있음.

A4 $R^2 = 1 - (1-R^2)(n-1)/(n-k-1)$. R^2와 n, k만 알면 계산 가능.

▶ 교수 설명 (수업 전사본 · 판서)

A2 · General remarks on R-squared

(10.27 11:20–12:45) R^2 = 데이터의 변동 중 **설명된 비율**.

- R^2가 낮아도 인과관계가 있을 수 있고, 높아도 인과관계가 없을 수 있음. "R^2를 보고 인과를 말할 수 없다. R^2가 하는 일이 아니다."
- R^2가 낮아도 추정량은 정밀할 수 있음. **[중요] R^2는 추정량이 얼마나 정확한지 말해주지 않는다.**

A3 · Adjusted R-squared (1/2)

(10.27 12:46–19:00) R^2를 올리고 싶으면? 예: 국가별 **출생체중**을 설명 → 교육·문화는 좋은 설명변수.

- 더 올리려고 주식시장 지수를, 나아가 **주식 1만 개의 가격**을 독립변수로 넣으면? → **우연히** 종속변수와 상관된 것들이 생겨 R^2가 올라감
- 하지만 **좋은 설명을 한 게 아님**. 변수가 수백만 개면 일부는 반드시 y와 상관됨

왜 adjusted R^2인가: 이걸 감안하기 위해. 독립변수를 너무 많이 넣으면 R^2를 깎는 **벌점**을 둠.

For example, if $R^2 = .30$, $n = 51$, and $k = 10$, then $\bar{R}^2 = 1 - .70(50)/40 = .125$. Thus, for small n and large k, $\bar{R}^2$ can be substantially below R^2. In fact, if the usual R-squared is small, and $n - k - 1$ is small, $\bar{R}^2$ can actually be negative! For example, you can plug in $R^2 = .10$, $n = 51$, and $k = 10$ to verify that $\bar{R}^2 = -.125$. A negative $\bar{R}^2$ indicates a very poor model fit relative to the number of degrees of freedom.

The adjusted R-squared is sometimes reported along with the usual R-squared in regressions, and sometimes $\bar{R}^2$ is reported in place of R^2. It is important to remember that it is R^2, not $\bar{R}^2$, that appears in the F statistic in (4.41). The same formula with $\bar{R}^2_r$ and $\bar{R}^2_{ur}$ is *not* valid.

Using Adjusted *R*-Squared to Choose between Nonnested Models

In Section 4.5, we learned how to compute an F statistic for testing the joint significance of a group of variables; this allows us to decide, at a particular significance level, whether at least one variable in the group affects the dependent variable. This test does not allow us to decide *which* of the variables has an effect. In some cases, we want to choose a model without redundant independent variables, and the adjusted R-squared can help with this.

In the major league baseball salary example in Section 4.5, we saw that neither *hrunsyr* nor *rbisyr* was individually significant. These two variables are highly correlated, so we might want to choose between the models

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 hrunsyr + u$$

and

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 rbisyr + u.$$

These two equations are **nonnested models** because neither equation is a special case of the other. The F statistics we studied in Chapter 4 only allow us to test *nested* models: one model (the restricted model) is a special case of the other model (the unrestricted model). See equations (4.32) and (4.28) for examples of restricted and unrestricted models. One possibility is to create a composite model that contains *all* explanatory variables from the original models and then to test each model against the general model using the F test. The problem with this process is that either both models might be rejected or neither model might be rejected (as happens with the major league baseball salary example in Section 4.5). Thus, it does not always provide a way to distinguish between models with nonnested regressors.

In the baseball player salary regression, $\bar{R}^2$ for the regression containing *hrunsyr* is .6211, and $\bar{R}^2$ for the regression containing *rbisyr* is .6226. Thus, based on the adjusted R-squared, there is a very slight preference for the model with *rbisyr*. But the difference is practically very small, and we might obtain a different answer by controlling for some of the variables in Computer Exercise C5 in Chapter 4. (Because both nonnested models contain five parameters, the usual R-squared can be used to draw the same conclusion.)

Comparing $\bar{R}^2$ to choose among different nonnested sets of independent variables can be valuable when these variables represent different functional forms. Consider two models relating R&D intensity to firm sales:

$$rdintens = \beta_0 + \beta_1 \log(sales) + u. \quad [6.23]$$

$$rdintens = \beta_0 + \beta_1 sales + \beta_2 sales^2 + u. \quad [6.24]$$

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p.203 · PPT 필기 A4 A5

A4 — A4 **R^2는 음수 가능**. 예: R^2=.10, n=51, k=10 → R^2 = 1 − .90(50)/40 = −.125.

REF — REF F통계량(4.41)에는 R^2가 아니라 **R^2**를 써야 함.

A5 — A5 **A5 = 이 소절.**

A5 — A5 **비내포(nonnested)**: 어느 쪽도 다른 쪽의 특수경우가 아님 → F검정으로 직접 비교 불가.

REF — REF A5 슬라이드 노트 질문의 답: **nested** = 제약모형이 비제약모형의 특수경우(계수=0 제약). F검정은 nested에서만.

A5 — A5 R^2 비교가 쓸모 있는 경우: 설명변수의 함수형이 다른 모형(종속변수는 같아야 함).

A5 — A5 모형 1: rdintens on log(sales) — 모수 2개.

A5 — A5 모형 2: rdintens on sales, $sales^2$ — 모수 3개.

▶ 교수 설명 (수업 전사본 · 판서)

A4 · Adjusted R-squared (2/2)

(10.27 17:49–20:44) k = 독립변수 개수. 식에서 **k가 커지면 adjusted R^2는 내려감**.

- R^2가 높은 이유가 '좋은 설명'이 아니라 '변수가 너무 많아서'일 수 있다는 점을 반영
- adjusted R^2가 올라가려면 새 변수의 **t통계량이 1보다 커야** 함 → y와 상관이 아주 낮거나 무관한 변수는 올리지 못함
- adjusted R^2는 **음수**가 될 수도 있음

The first model captures a diminishing return by including *sales* in logarithmic form; the second model does this by using a quadratic. Thus, the second model contains one more parameter than the first.

When equation (6.23) is estimated using the 32 observations on chemical firms in RDCHEM.RAW, R^2 is .061, and R^2 for equation (6.24) is .148. Therefore, it appears that the quadratic fits much better. But a comparison of the usual R-squareds is unfair to the first model because it contains one fewer parameter than (6.24). That is, (6.23) is a more parsimonious model than (6.24).

Everything else being equal, simpler models are better. Since the usual R-squared does not penalize more complicated models, it is better to use $\bar{R}^2$. $\bar{R}^2$ for (6.23) is .030, while $\bar{R}^2$ for (6.24) is .090. Thus, even after adjusting for the difference in degrees of freedom, the quadratic model wins out. The quadratic model is also preferred when profit margin is added to each regression.

There is an important limitation in using $\bar{R}^2$ to choose between nonnested models: we cannot use it to choose between different functional forms for the dependent variable. This is unfortunate, because we often want to decide on whether y or $\log(y)$ (or maybe some other transformation) should be used as the dependent variable based on goodness-of-fit. But neither R^2 nor $\bar{R}^2$ can be used for this purpose. The reason is simple: these R-squareds measure the explained proportion of the total variation in whatever dependent variable we are using in the regression, and different functions of the dependent variable will have different amounts of variation to explain. For example, the total variations in y and $\log(y)$ are not the same, and are often very different. Comparing the adjusted R-squareds from regressions with these different forms of the dependent variables does not tell us anything about which model fits better; they are fitting two separate dependent variables.

EXPLORING FURTHER 6.4

Explain why choosing a model by maximizing $\bar{R}^2$ or minimizing $\hat{\sigma}$ (the standard error of the regression) is the same thing.

EXAMPLE 6.4 CEO COMPENSATION AND FIRM PERFORMANCE

Consider two estimated models relating CEO compensation to firm performance:

$$\widehat{salary} = 830.63 + .0163\, sales + 19.63\, roe \quad [6.25]$$
$$(223.90)\quad (.0089)\qquad (11.08)$$
$$n = 209,\ R^2 = .029,\ \bar{R}^2 = .020$$

and

$$\widehat{lsalary} = 4.36 + .275\, lsales + .0179\, roe \quad [6.26]$$
$$(0.29)\quad (.033)\qquad (.0040)$$
$$n = 209,\ R^2 = .282,\ \bar{R}^2 = .275,$$

where *roe* is the return on equity discussed in Chapter 2. For simplicity, *lsalary* and *lsales* denote the natural logs of *salary* and *sales*. We already know how to interpret these different estimated equations. But can we say that one model fits better than the other?

The R-squared for equation (6.25) shows that *sales* and *roe* explain only about 2.9% of the variation in CEO salary in the sample. Both *sales* and *roe* have marginal statistical significance.

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p.204 · PPT 필기 A5 A6

A5 R²: .061 vs .148 → 그냥 비교하면 모수가 적은 모형 1에 불공평.

A5 R²: **.030 vs .090** → 자유도를 조정해도 이차모형이 우세.

A6 **한계**: 종속변수의 형태가 다르면(y vs log y) R^2, R^2 모두 비교 불가.

A6 이유: R^2는 '그 회귀의 종속변수' 총변동 중 설명 비율. y와 log(y)는 설명할 변동량 자체가 다름.

A6 **A6 = 예제 6.4** (CEO 보수).

A6 수준모형: R^2=.029, R^2=.020. (교재 roe 계수 19.63, 슬라이드 19.03 — 판 차이 · 오타로 보임.)

A6 로그모형: R^2=.282, R^2=.275.

▶ 교수 설명 (수업 전사본 · 판서)

A5 · Nonnested models

(10.27 20:44–23:08) 지금까지 모형 비교는 **내포(nested) 모형**이었음: 어떤 계수가 0이면 한 모형이 다른 모형이 됨 → 한쪽이 다른 쪽의 특수경우 → **F검정**(전사본 'S test')을 썼음.

- 이 슬라이드의 두 모형은 그런 관계가 **없음** → 그 검정을 하기 어려움
- 그래도 말할 수 있는 것: 두 번째 모형은 독립변수가 하나 더 많은데도 **R^2와 adjusted R^2 모두 더 큼** → 두 번째 모형이 더 나은 설명

Equation (6.26) shows that log(*sales*) and *roe* explain about 28.2% of the variation in log(*salary*). In terms of goodness-of-fit, this much higher *R*-squared would seem to imply that model (6.26) is much better, but this is not necessarily the case. The total sum of squares for *salary* in the sample is 391,732,982, while the total sum of squares for log(*salary*) is only 66.72. Thus, there is much less variation in log(*salary*) that needs to be explained.

At this point, we can use features other than R^2 or $\bar{R}^2$ to decide between these models. For example, log(*sales*) and *roe* are much more statistically significant in (6.26) than are *sales* and *roe* in (6.25), and the coefficients in (6.26) are probably of more interest. To be sure, however, we will need to make a valid goodness-of-fit comparison.

In Section 6.4, we will offer a goodness-of-fit measure that does allow us to compare models where *y* appears in both level and log form.

Controlling for Too Many Factors in Regression Analysis

In many of the examples we have covered, and certainly in our discussion of omitted variables bias in Chapter 3, we have worried about omitting important factors from a model that might be correlated with the independent variables. It is also possible to control for too *many* variables in a regression analysis.

If we overemphasize goodness-of-fit, we open ourselves to controlling for factors in a regression model that should not be controlled for. To avoid this mistake, we need to remember the ceteris paribus interpretation of multiple regression models.

To illustrate this issue, suppose we are doing a study to assess the impact of state beer taxes on traffic fatalities. The idea is that a higher tax on beer will reduce alcohol consumption, and likewise drunk driving, resulting in fewer traffic fatalities. To measure the ceteris paribus effect of taxes on fatalities, we can model *fatalities* as a function of several factors, including the beer *tax*:

$$fatalities = \beta_0 + \beta_1 tax + \beta_2 miles + \beta_3 percmale + \beta_4 perc16_21 + \ldots,$$

where

miles = total miles driven.
percmale = percentage of the state population that is male.
perc16_21 = percentage of the population between ages 16 and 21, and so on.

Notice how we have not included a variable measuring per capita beer consumption. Are we committing an omitted variables error? The answer is no. If we control for beer consumption in this equation, then how would beer taxes affect traffic fatalities? In the equation

$$fatalities = \beta_0 + \beta_1 tax + \beta_2 beercons + \ldots,$$

β_1 measures the difference in fatalities due to a one percentage point increase in *tax*, holding *beercons* fixed. It is difficult to understand why this would be interesting. We should not be controlling for differences in *beercons* across states, unless we want to test for some sort of indirect effect of beer taxes. Other factors, such as gender and age distribution, should be controlled for.

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p.205 · PPT 필기 A6 A7

A6
A6 SST: salary **391,732,982** vs log(salary) **66.72** → 로그는 설명할 변동이 훨씬 작음. R^2가 높다고 더 좋은 모형이라 못 함.

REF
REF 해결책은 6.4절: corr(y, ŷ)²로 비교 (p.214–215, PPT 범위 밖).

A7
A7 **A7–A8 = 이 소절** (과잉 통제, over controlling).

A7
A7 **예 1**: 맥주세 → (음주량↓) → 교통사고 사망. 경로: tax → beercons → fatalities.

A7
A7 beercons를 통제하면 '음주량이 같을 때 세금의 효과'가 됨 → 보고 싶은 경로를 스스로 막음. **매개변수(효과의 통로)는 통제하지 않는다.**

▶ 교수 설명 (수업 전사본 · 판서)

A6 · Different dependent variables

(10.27 23:08–25:40) 종속변수에 log를 씌우면 **변동이 줄어듦**. R^2는 계속 '변동'에 관한 이야기.

- 종속변수가 다르고 변동이 줄었으므로, R^2가 더 높아도 **"더 나은 설명"이라고 말할 수 없음**

As a second example, suppose that, for a developing country, we want to estimate the effects of pesticide usage among farmers on family health expenditures. In addition to pesticide usage amounts, should we include the number of doctor visits as an explanatory variable? No. Health expenditures include doctor visits, and we would like to pick up all effects of pesticide use on health expenditures. If we include the number of doctor visits as an explanatory variable, then we are only measuring the effects of pesticide use on health expenditures other than doctor visits. It makes more sense to use number of doctor visits as a dependent variable in a separate regression on pesticide amounts.

The previous examples are what can be called **over controlling** for factors in multiple regression. Often this results from nervousness about potential biases that might arise by leaving out an important explanatory variable. But it is important to remember the ceteris paribus nature of multiple regression. In some cases, it makes no sense to hold some factors fixed precisely because they should be allowed to change when a policy variable changes.

Unfortunately, the issue of whether or not to control for certain factors is not always clear-cut. For example, Betts (1995) studies the effect of high school quality on subsequent earnings. He points out that, if better school quality results in more education, then controlling for education in the regression along with measures of quality will underestimate the return to quality. Betts does the analysis with and without years of education in the equation to get a range of estimated effects for quality of schooling.

To see explicitly how pursuing high R-squareds can lead to trouble, consider the housing price example from Section 4.5 that illustrates the testing of multiple hypotheses. In that case, we wanted to test the rationality of housing price assessments. We regressed log(*price*) on log(*assess*), log(*lotsize*), log(*sqrft*), and *bdrms* and tested whether the latter three variables had zero population coefficients while log(*assess*) had a coefficient of unity. But what if we change the purpose of the analysis and estimate a *hedonic price model*, which allows us to obtain the marginal values of various housing attributes? Should we include log(*assess*) in the equation? The adjusted R-squared from the regression with log(*assess*) is .762, while the adjusted R-squared without it is .630. Based on goodness-of-fit only, we should include log(*assess*). But this is incorrect if our goal is to determine the effects of lot size, square footage, and number of bedrooms on housing values. Including log(*assess*) in the equation amounts to holding one measure of value fixed and then asking how much an additional bedroom would change another measure of value. This makes no sense for valuing housing attributes.

If we remember that different models serve different purposes, and we focus on the ceteris paribus interpretation of regression, then we will not include the wrong factors in a regression model.

Adding Regressors to Reduce the Error Variance

We have just seen some examples of where certain independent variables should not be included in a regression model, even though they are correlated with the dependent variable. From Chapter 3, we know that adding a new independent variable to a regression can exacerbate the multicollinearity problem. On the other hand, since we are taking something out of the error term, adding a variable generally reduces the error variance. Generally, we cannot know which effect will dominate.

However, there is one case that is clear: we should always include independent variables that affect y and are *uncorrelated* with all of the independent variables of interest. Why?

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A7 A7 A8 A9

A7 A7

REF

A8 A8

A9 A9 A9

A7 **예 2**: 농약 사용 → 의료비. 병원 방문 횟수는 의료비의 일부(효과의 통로) → 통제하면 '방문 외 의료비'만 측정.

A7 용어: **over controlling**. 원인은 누락변수 편의에 대한 과도한 걱정. 기준은 ceteris paribus 해석 — '정책이 바뀌면 같이 변해야 하는 변수'는 고정하면 안 됨.

REF 애매한 경우의 실무 대응: 포함/제외 두 버전을 모두 돌려 효과의 범위를 제시.

A8 **A8**: 회귀의 목적이 통제변수를 결정. 목적이 '주택 속성의 가치'라면 log(assess)를 넣으면 안 됨.

A8 R^2는 .762(포함) vs .630(제외) — 적합도만 보면 넣어야 하지만 틀린 선택. 감정가를 넣는 건 '감정가의 타당성'을 볼 때만.

A9 **A9–A10 = 이 소절**.

A9 변수 추가의 두 효과: ① 다중공선성 악화(R^2j↑ → 분산↑) ② 오차분산 σ^2↓ → 분산↓. $Var(\beta j)=\sigma^2/[SSTj(1-R^2j)]$.

A9 **확실한 경우**: y에 영향을 주면서 관심 변수와 무상관인 변수 → 항상 포함.

▶ 교수 설명 (수업 전사본 · 판서)

A7 · Controlling for too many factors (1/2)

(10.27 26:00–27:35) **통제변수** 이야기. 맥주세가 교통사고 사망에 미치는 효과를 볼 때 맥주 소비를 넣어도 되나? → **안 됨**.

- 맥주세 → 소비 → 사망. 소비를 통제하면 **분석하려는 효과를 숨김**. [중요] "보고 싶은 효과 자체를 통제할 수는 없다."
- 농약 → 의료비에서 병원 방문도 마찬가지: 병원 방문은 **분석하려는 효과의 일부**

A8 · Controlling for too many factors (2/2)

(10.27 27:35–30:00) "**왜 회귀를 돌리는가**에 달렸다."

- 목적이 '방이 많은 집이 더 비싼가'라면 감정가(price assessment)를 통제하면 안 됨: 방이 많으면 감정가도 높고, 그게 보고 싶은 효과의 일부
- 감정가를 넣어야 할 때: **감정가가 주택가격을 잘 예측하는지**(감정가가 좋은 척도인지) 보려는 경우 — 이때는 분석하는 대상 자체가 다름

Because adding such a variable does not induce multicollinearity in the population (and therefore multicollinearity in the sample should be negligible), but it will reduce the error variance. In large sample sizes, the standard errors of all OLS estimators will be reduced.

As an example, consider estimating the individual demand for beer as a function of the average county beer price. It may be reasonable to assume that individual characteristics are uncorrelated with county-level prices, and so a simple regression of beer consumption on county price would suffice for estimating the effect of price on individual demand. But it is possible to get a more precise estimate of the price elasticity of beer demand by including individual characteristics, such as age and amount of education. If these factors affect demand and are uncorrelated with price, then the standard error of the price coefficient will be smaller, at least in large samples.

As a second example, consider the grants for computer equipment given at the beginning of Section 6.3. If, in addition to the grant variable, we control for other factors that can explain college GPA, we can probably get a more precise estimate of the effect of the grant. Measures of high school grade point average and rank, SAT and ACT scores, and family background variables are good candidates. Because the grant amounts are randomly assigned, all additional control variables are uncorrelated with the grant amount; in the sample, multicollinearity between the grant amount and other independent variables should be minimal. But adding the extra controls might significantly reduce the error variance, leading to a more precise estimate of the grant effect. Remember, the issue is not unbiasedness here: we obtain an unbiased and consistent estimator whether or not we add the high school performance and family background variables. The issue is getting an estimator with a smaller sampling variance.

Unfortunately, cases where we have information on additional explanatory variables that are uncorrelated with the explanatory variables of interest are somewhat rare in the social sciences. But it is worth remembering that when these variables are available, they can be included in a model to reduce the error variance without inducing multicollinearity.

6.4 Prediction and Residual Analysis

In Chapter 3, we defined the OLS predicted or fitted values and the OLS residuals. **Predictions** are certainly useful, but they are subject to sampling variation, because they are obtained using the OLS estimators. Thus, in this section, we show how to obtain confidence intervals for a prediction from the OLS regression line.

From Chapters 3 and 4, we know that the residuals are used to obtain the sum of squared residuals and the *R*-squared, so they are important for goodness-of-fit and testing. Sometimes, economists study the residuals for particular observations to learn about individuals (or firms, houses, etc.) in the sample.

Confidence Intervals for Predictions

Suppose we have estimated the equation

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_k x_k. \qquad [6.27]$$

When we plug in particular values of the independent variables, we obtain a prediction for y, which is an estimate of the *expected value* of y given the particular values for the explanatory variables. For emphasis, let $c_1, c_2, \ldots, c_k$ denote particular values for each of

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p.207 · PPT 필기 A9 A10

A9 — A9 이유: 공선성은 안 늘고(R^2j 그대로) σ^2만 줄어듦 → 표준오차↓.

A10 — A10 **A10 예**: 개인 맥주 수요 on 지역 맥주 가격. 개인 특성(나이·교육)은 가격과 무상관이라 가정 가능.

A10 — A10 개인 특성 추가 → 가격탄력성 추정의 SE↓ (편의 때문이 아니라 정밀도 때문에 넣는 것).

REF — REF 무작위 배정이면 통제변수 없이도 불편·일치. 통제변수의 역할은 분산 축소.

A9 — A9 단, 이런 '무상관이면서 설명력 있는' 변수는 사회과학에서 드묾.

SKIP — SKIP 6.4절 앞부분(예측 신뢰구간 · 잔차 분석, p.207–212) — PPT 미포함.

▶ 교수 설명 (수업 전사본 · 판서)

A9 · Adding regressors to reduce error variance

(10.27 32:50–37:08) 표준편차·표준오차를 줄이거나 늘리는 요인 (3장 분산 공식 복습).

- **다중공선성**이 있으면 분산이 커짐 (이미 배움)
- **오차분산 / 잔차제곱합(SSR)**이 줄면 분산이 작아짐 → 더 좋은 설명을 주면 오차가 작아짐
- 오차를 잘 설명하는 독립변수를 추가하면 이 부분이 줄어 **표준오차 감소**. 슬라이드에서 표준오차가 줄어든 이유 = SSR이 줄었기 때문
- 우리는 표준오차가 **작기를** 원함 → [중요] 찾아야 할 변수: **다른 설명변수와 무상관**이면서 **오차를 설명**하는 변수. 다만 찾기 어려움

A10 · Example: beer consumption · session FE

(10.27 37:08–41:08) 맥주 가격과 **개인 특성**: 반드시 회귀에 넣어야 할 후보.

- 개인 특성과 맥주 가격은 **무상관**으로 보임
- 개인 특성은 맥주 소비를 설명함 — "맥주를 정말 좋아하는 사람과 싫어하는 사람이 있다"
- → 표준편차·표준오차 감소

세션 고정효과(session fixed effects)도 같은 논리 (10.08 수업 실험 논문):

- 사람들이 세션에 **무작위 배정** → 세션은 다른 독립변수와 무상관 (관심도가 높든 낮든 배정과 무관)
- 세션마다 약간씩 다른 일이 있었으므로 **결과(종속변수)는 설명**함
- [중요] "이런 예를 찾아라": **종속변수에는 관련 있고, 다른 독립변수와는 관련 없는** 변수

SKIP

the k independent variables; these may or may not correspond to an actual data point in our sample. The parameter we would like to estimate is

$$\begin{aligned}\theta_0 &= \beta_0 + \beta_1 c_1 + \beta_2 c_2 + \ldots + \beta_k c_k \\ &= \mathrm{E}(y|x_1 = c_1, x_2 = c_2, \ldots, x_k = c_k).\end{aligned} \qquad [6.28]$$

The estimator of θ_0 is

$$\hat{\theta}_0 = \hat{\beta}_0 + \hat{\beta}_1 c_1 + \hat{\beta}_2 c_2 + \ldots + \hat{\beta}_k c_k. \qquad [6.29]$$

In practice, this is easy to compute. But what if we want some measure of the uncertainty in this predicted value? It is natural to construct a confidence interval for θ_0, which is centered at $\hat{\theta}_0$.

To obtain a confidence interval for θ_0, we need a standard error for $\hat{\theta}_0$. Then, with a large df, we can construct a 95% confidence interval using the rule of thumb $\hat{\theta}_0 \pm 2 \cdot \mathrm{se}(\hat{\theta}_0)$. (As always, we can use the exact percentiles in a t distribution.)

How do we obtain the standard error of $\hat{\theta}_0$? This is the same problem we encountered in Section 4.4: we need to obtain a standard error for a linear combination of the OLS estimators. Here, the problem is even more complicated, because all of the OLS estimators generally appear in $\hat{\theta}_0$ (unless some c_j are zero). Nevertheless, the same trick that we used in Section 4.4 will work here. Write $\beta_0 = \theta_0 - \beta_1 c_1 - \ldots - \beta_k c_k$ and plug this into the equation

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u$$

to obtain

$$y = \theta_0 + \beta_1(x_1 - c_1) + \beta_2(x_2 - c_2) + \ldots + \beta_k(x_k - c_k) + u. \qquad [6.30]$$

In other words, we subtract the value c_j from each observation on x_j, and then we run the regression of

$$y_i \text{ on } (x_{i1} - c_1), \ldots, (x_{ik} - c_k),\ i = 1, 2, \ldots, n. \qquad [6.31]$$

The predicted value in (6.29) and, more importantly, its standard error, are obtained from the *intercept* (or constant) in regression (6.31).

As an example, we obtain a confidence interval for a prediction from a college GPA regression, where we use high school information.

EXAMPLE 6.5 CONFIDENCE INTERVAL FOR PREDICTED COLLEGE GPA

Using the data in GPA2.RAW, we obtain the following equation for predicting college GPA:

$$\begin{aligned}\widehat{colgpa} &= 1.493 + .00149\, sat - .01386\, hsperc \\ &\qquad (0.075)\ \ (.00007)\qquad (.00056) \\ &\quad - .06088\, hsize + .00546\, hsize^2 \qquad [6.32] \\ &\qquad (.01650)\qquad\ (.00227) \\ n &= 4{,}137,\ R^2 = .278,\ \bar{R}^2 = .277,\ \hat{\sigma} = .560,\end{aligned}$$

p.208 · PPT 필기

SKIP 예측의 신뢰구간(예제 6.5–6.6) · 잔차 분석 — PPT 미포함. p.212 'Predicting y When log(y)…'에서 다시 시작.

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where we have reported estimates to several digits to reduce round-off error. What is predicted college GPA, when $sat = 1{,}200$, $hsperc = 30$, and $hsize = 5$ (which means 500)? This is easy to get by plugging these values into equation (6.32): $\widehat{colgpa} = 2.70$ (rounded to two digits). Unfortunately, we cannot use equation (6.32) directly to get a confidence interval for the expected *colgpa* at the given values of the independent variables. One simple way to obtain a confidence interval is to define a new set of independent variables: $sat0 = sat - 1{,}200$, $hsperc0 = hsperc - 30$, $hsize0 = hsize - 5$, and $hsizesq0 = hsize^2 - 25$. When we regress *colgpa* on these new independent variables, we get

$$\begin{gathered}\widehat{colgpa} = 2.700 + .00149\, sat0 - .01386\, hsperc0 \\ (0.020)\ \ (.00007)\qquad (.00056) \\ - .06088\, hsize0 + .00546\, hsizesq0 \\ (.01650)\qquad\quad (.00227) \\ n = 4{,}137,\ R^2 = .278,\ \bar{R}^2 = .277,\ \hat{\sigma} = .560.\end{gathered}$$

The only difference between this regression and that in (6.32) is the intercept, which is the prediction we want, along with its standard error, .020. It is not an accident that the slope coefficents, their standard errors, *R*-squared, and so on are the same as before; this provides a way to check that the proper transformations were done. We can easily construct a 95% confidence interval for the expected college GPA: $2.70 \pm 1.96(.020)$ or about 2.66 to 2.74. This confidence interval is rather narrow due to the very large sample size.

Because the variance of the intercept estimator is smallest when each explanatory variable has zero sample mean (see Question 2.5 for the simple regression case), it follows from the regression in (6.31) that the variance of the prediction is smallest at the mean values of the x_j. (That is, $c_j = \bar{x}_j$ for all j.) This result is not too surprising, since we have the most faith in our regression line near the middle of the data. As the values of the c_j get farther away from the $\bar{x}_j$, $\text{Var}(\hat{y})$ gets larger and larger.

The previous method allows us to put a confidence interval around the OLS estimate of $\text{E}(y|x_1, \ldots, x_k)$ for any values of the explanatory variables. In other words, we obtain a confidence interval for the *average* value of y for the subpopulation with a given set of covariates. But a confidence interval for the average person in the subpopulation is not the same as a confidence interval for a particular unit (individual, family, firm, and so on) from the population. In forming a confidence interval for an unknown outcome on y, we must account for another very important source of variation: the variance in the unobserved error, which measures our ignorance of the unobserved factors that affect y.

Let y^0 denote the value for which we would like to construct a confidence interval, which we sometimes call a **prediction interval**. For example, y^0 could represent a person or firm not in our original sample. Let $x_1^0, \ldots, x_k^0$ be the new values of the independent variables, which we assume we observe, and let u^0 be the unobserved error. Therefore, we have

$$y^0 = \beta_0 + \beta_1 x_1^0 + \beta_2 x_2^0 + \ldots + \beta_k x_k^0 + u^0. \qquad [6.33]$$

As before, our best prediction of y^0 is the expected value of y^0 given the explanatory variables, which we estimate from the OLS regression line: $\hat{y}^0 = \hat{\beta}_0 + \hat{\beta}_1 x_1^0 + \hat{\beta}_2 x_2^0 + \ldots + \hat{\beta}_k x_k^0$. The **prediction error** in using $\hat{y}^0$ to predict y^0 is

$$\hat{e}^0 = y^0 - \hat{y}^0 = (\beta_0 + \beta_1 x_1^0 + \ldots + \beta_k x_k^0) + u^0 - \hat{y}^0. \qquad [6.34]$$

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Now, $E(\hat{y}^0) = E(\hat{\beta}_0) + E(\hat{\beta}_1)x_1^0 + E(\hat{\beta}_2)x_2^0 + \ldots + E(\hat{\beta}_k)x_k^0 = \beta_0 + \beta_1 x_1^0 + \ldots + \beta_k x_k^0$, because the $\hat{\beta}_j$ are unbiased. (As before, these expectations are all conditional on the sample values of the independent variables.) Because u^0 has zero mean, $E(\hat{e}^0) = 0$. We have shown that the expected prediction error is zero.

In finding the variance of $\hat{e}^0$, note that u^0 is uncorrelated with each $\hat{\beta}_j$, because u^0 is uncorrelated with the errors in the sample used to obtain the $\hat{\beta}_j$. By basic properties of covariance (see Appendix B), u^0 and $\hat{y}^0$ are uncorrelated. Therefore, the **variance of the prediction error** (conditional on all in-sample values of the independent variables) is the sum of the variances:

$$\text{Var}(\hat{e}^0) = \text{Var}(\hat{y}^0) + \text{Var}(u^0) = \text{Var}(\hat{y}^0) + \sigma^2, \qquad [6.35]$$

where $\sigma^2 = \text{Var}(u^0)$ is the error variance. There are two sources of variation in $\hat{e}^0$. The first is the sampling error in $\hat{y}^0$, which arises because we have estimated the β_j. Because each $\hat{\beta}_j$ has a variance proportional to $1/n$, where n is the sample size, $\text{Var}(\hat{y}^0)$ is proportional to $1/n$. This means that, for large samples, $\text{Var}(\hat{y}^0)$ can be very small. By contrast, σ^2 is the variance of the error in the population; it does not change with the sample size. In many examples, σ^2 will be the dominant term in (6.35).

Under the classical linear model assumptions, the $\hat{\beta}_j$ and u^0 are normally distributed, and so $\hat{e}^0$ is also normally distributed (conditional on all sample values of the explanatory variables). Earlier, we described how to obtain an unbiased estimator of $\text{Var}(\hat{y}^0)$, and we obtained our unbiased estimator of σ^2 in Chapter 3. By using these estimators, we can define the standard error of $\hat{e}^0$ as

$$\text{se}(\hat{e}^0) = \{[\text{se}(\hat{y}^0)]^2 + \hat{\sigma}^2\}^{1/2}. \qquad [6.36]$$

Using the same reasoning for the t statistics of the $\hat{\beta}_j$, $\hat{e}^0/\text{se}(\hat{e}^0)$ has a t distribution with $n - (k + 1)$ degrees of freedom. Therefore,

$$P[-t_{.025} \leq \hat{e}^0/\text{se}(\hat{e}^0) \leq t_{.025}] = .95,$$

where $t_{.025}$ is the 97.5th percentile in the t_{n-k-1} distribution. For large $n - k - 1$, remember that $t_{.025} \approx 1.96$. Plugging in $\hat{e}^0 = y^0 - \hat{y}^0$ and rearranging gives a 95% prediction interval for y^0:

$$\hat{y}^0 \pm t_{.025} \cdot \text{se}(\hat{e}^0); \qquad [6.37]$$

as usual, except for small *df*, a good rule of thumb is $\hat{y}^0 \pm 2\text{se}(\hat{e}^0)$. This is wider than the confidence interval for $\hat{y}^0$ itself because of $\hat{\sigma}^2$ in (6.36); it often is much wider to reflect the factors in u^0 that we have not accounted for.

EXAMPLE 6.6 CONFIDENCE INTERVAL FOR FUTURE COLLEGE GPA

Suppose we want a 95% CI for the future college GPA of a high school student with *sat* = 1,200, *hsperc* = 30, and *hsize* = 5. In Example 6.5, we obtained a 95% confidence interval for the *average* college GPA among all students with the particular characteristics *sat* = 1,200, *hsperc* = 30, and *hsize* = 5. Now, we want a 95% confidence interval for any *particular* student with these characteristics. The 95% prediction interval must account for the variation in the individual, unobserved characteristics that affect college performance. We have everything we need to obtain a CI for *colgpa*. $\text{se}(\hat{y}^0) = .020$ and

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$\hat{\sigma} = .560$ and so, from (6.36), $\text{se}(\hat{e}^0) = [(.020)^2 + (.560)^2]^{1/2} \approx .560$. Notice how small $\text{se}(\hat{y}^0)$ is relative to $\hat{\sigma}$: virtually all of the variation in $\hat{e}^0$ comes from the variation in u^0. The 95% CI is $2.70 \pm 1.96(.560)$ or about 1.60 to 3.80. This is a wide confidence interval and shows that, based on the factors we included in the regression, we cannot accurately pin down an individual's future college grade point average. (In one sense, this is good news, as it means that high school rank and performance on the SAT do not preordain one's performance in college.) Evidently, the unobserved characteristics that affect college GPA vary widely among individuals with the same observed SAT score and high school rank.

Residual Analysis

Sometimes, it is useful to examine individual observations to see whether the actual value of the dependent variable is above or below the predicted value; that is, to examine the residuals for the individual observations. This process is called **residual analysis**. Economists have been known to examine the residuals from a regression in order to aid in the purchase of a home. The following housing price example illustrates residual analysis. Housing price is related to various observable characteristics of the house. We can list all of the characteristics that we find important, such as size, number of bedrooms, number of bathrooms, and so on. We can use a sample of houses to estimate a relationship between price and attributes, where we end up with a predicted value and an actual value for each house. Then, we can construct the residuals, $\hat{u}_i = y_i - \hat{y}_i$. The house with the most negative residual is, at least based on the factors we have controlled for, the most underpriced one relative to its *observed* characteristics. Of course, a selling price substantially below its predicted price could indicate some undesirable feature of the house that we have failed to account for, and which is therefore contained in the unobserved error. In addition to obtaining the prediction and residual, it also makes sense to compute a confidence interval for what the future selling price of the home could be, using the method described in equation (6.37).

Using the data in HPRICE1.RAW, we run a regression of *price* on *lotsize*, *sqrft*, and *bdrms*. In the sample of 88 homes, the most negative residual is −120.206, for the 81st house. Therefore, the asking price for this house is $120,206 below its predicted price.

There are many other uses of residual analysis. One way to rank law schools is to regress median starting salary on a variety of student characteristics (such as median LSAT scores of entering class, median college GPA of entering class, and so on) and to obtain a predicted value and residual for each law school. The law school with the largest residual has the highest predicted value added. (Of course, there is still much uncertainty about how an individual's starting salary would compare with the median for a law school overall.) These residuals can be used along with the costs of attending each law school to determine the best value; this would require an appropriate discounting of future earnings.

Residual analysis also plays a role in legal decisions. A *New York Times* article entitled "Judge Says Pupil's Poverty, Not Segregation, Hurts Scores" (6/28/95) describes an important legal case. The issue was whether the poor performance on standardized tests in the Hartford School District, relative to performance in surrounding suburbs, was due to poor school quality at the highly segregated schools. The judge concluded that "the disparity in test scores does not indicate that Hartford is doing an inadequate or poor job in educating its students or that its schools are failing, because the predicted scores based upon the relevant socioeconomic factors are about at the levels that one would expect." This conclusion is based on a regression analysis of average or median scores on

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EXPLORING FURTHER 6.5

How would you use residual analysis to determine which professional athletes are overpaid or underpaid relative to their performance?

socioeconomic characteristics of various school districts in Connecticut. The judge's conclusion suggests that, given the poverty levels of students at Hartford schools, the actual test scores were similar to those predicted from a regression analysis: the residual for Hartford was not sufficiently negative to conclude that the schools themselves were the cause of low test scores.

Predicting y When log(y) Is the Dependent Variable

Because the natural log transformation is used so often for the dependent variable in empirical economics, we devote this subsection to the issue of predicting y when log(y) is the dependent variable. As a byproduct, we will obtain a goodness-of-fit measure for the log model that can be compared with the R-squared from the level model.

To obtain a prediction, it is useful to define $logy = \log(y)$; this emphasizes that it is the log of y that is predicted in the model

$$logy = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u. \quad [6.38]$$

In this equation, the x_j might be transformations of other variables; for example, we could have $x_1 = \log(sales)$, $x_2 = \log(mktval)$, $x_3 = ceoten$ in the CEO salary example.

Given the OLS estimators, we know how to predict $logy$ for any value of the independent variables:

$$\widehat{logy} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_k x_k. \quad [6.39]$$

Now, since the exponential undoes the log, our first guess for predicting y is to simply exponentiate the predicted value for log(y): $\hat{y} = \exp(\widehat{logy})$. This does not work; in fact, it will systematically *underestimate* the expected value of y. In fact, if model (6.38) follows the CLM assumptions MLR.1 through MLR.6, it can be shown that

$$E(y|\boldsymbol{x}) = \exp(\sigma^2/2)\cdot\exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k),$$

where $\boldsymbol{x}$ denotes the independent variables and σ^2 is the variance of u. [If $u \sim$ Normal(0,σ^2), then the expected value of exp(u) is exp(σ^2/2).] This equation shows that a simple adjustment is needed to predict y:

$$\hat{y} = \exp(\hat{\sigma}^2/2)\exp(\widehat{logy}), \quad [6.40]$$

where $\hat{\sigma}^2$ is simply the unbiased estimator of σ^2. Because $\hat{\sigma}$, the standard error of the regression, is always reported, obtaining predicted values for y is easy. Because $\hat{\sigma}^2 > 0$, $\exp(\hat{\sigma}^2/2) > 1$. For large $\hat{\sigma}^2$, this adjustment factor can be substantially larger than unity.

The prediction in (6.40) is not unbiased, but it is consistent. There are no unbiased predictions of y, and in many cases, (6.40) works well. However, it does rely on the normality of the error term, u. In Chapter 5, we showed that OLS has desirable properties, even when u is not normally distributed. Therefore, it is useful to have a prediction that does not rely on normality. If we just assume that u is independent of the explanatory variables, then we have

$$E(y|\boldsymbol{x}) = \alpha_0\exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k), \quad [6.41]$$

where α_0 is the expected value of exp(u), which must be greater than unity.

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SKIP

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p.212 · PPT 필기 A11

A11 A11 = 이 소절.

A11 log(y) = β0 + β1x1 + ⋯ + βkxk + u ⇒ y = exp(β0+…+βkxk) · exp(u).

A11 단순 지수변환 ŷ=exp(logŷ)는 과소추정. 이유: E[exp(u)] > 1 (u의 평균이 0이어도 exp는 볼록).

REF 정규성 가정 시: ŷ = exp(σ̂²/2) · exp(logŷ). 슬라이드는 정규성 없이 쓰는 아래 식(6.41–6.43)을 사용.

A11 일치하지만 편의가 있는 추정량의 첫 사례 (A11 슬라이드 노트 'consistent but biased').

A11 u가 x와 독립이면 E(y|x) = α0 · exp(β0+…+βkxk), **α0 = E[exp(u)]**.

▶ 교수 설명 (수업 전사본 · 판서)

A11 · Predicting y when log(y) is dependent

(10.27 41:08–47:10) "log y일 때 y를 어떻게 예측하나?" 첫 추측은 당연히 exp를 씌우는 식. **교재의 주장은 보정한 식이 더 낫다**는 것. 다만 이 식에도 문제는 있음.

- [중요] 이 과목에서 처음 보는 **'일치하지만 편의가 있는 (consistent but biased)' 추정량**. 앞으로 더 나옴
- 왜 불편추정량 대신 편의추정량을 쓰나? → **편의의 크기를 모르는 경우가 많기 때문**
- 편의의 크기 b를 안다면: E(β) = β + b → **β − b** 가 불편추정량이 됨. "편의 크기를 알았다면 그냥 불편추정량을 만들었을 것" — 대부분은 b를 몰라서 불가능

Given an estimate $\hat{\alpha}_0$, we can predict y as

$$\hat{y} = \hat{\alpha}_0 \exp(\widehat{logy}), \qquad [6.42]$$

which again simply requires exponentiating the predicted value from the log model and multiplying the result by $\hat{\alpha}_0$.

Two approaches suggest themselves for estimating α_0 without the normality assumption. The first is based on $\alpha_0 = \mathrm{E}[\exp(u)]$. To estimate α_0 we replace the population expectation with a sample average and then we replace the unobserved errors, u_i, with the OLS residuals, $\hat{u}_i = \log(y_i) - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \ldots - \hat{\beta}_k x_{ik}$. This leads to the method of moments estimator (see Appendix C)

$$\hat{\alpha}_0 = n^{-1} \sum_{i=1}^{n} \exp(\hat{u}_i). \qquad [6.43]$$

Not surprisingly, $\hat{\alpha}_0$ is a consistent estimator of α_0, but it is not unbiased because we have replaced u_i with $\hat{u}_i$ inside a nonlinear function. This version of $\hat{\alpha}_0$ is a special case of what Duan (1983) called a **smearing estimate**. Because the OLS residuals have a zero sample average, it can be shown that, for any data set, $\hat{\alpha}_0 > 1$. (Technically, $\hat{\alpha}_0$ would equal one if all the OLS residuals were zero, but this never happens in any interesting application.) That $\hat{\alpha}_0$ is necessarily greater than one is convenient because it must be that $\alpha_0 > 1$.

A different estimate of α_0 is based on a simple regression through the origin. To see how it works, define $m_i = \exp(\beta_0 + \beta_1 x_{i1} + \ldots + \beta_k x_{ik})$, so that, from equation (6.41), $\mathrm{E}(y_i|m_i) = \alpha_0 m_i$. If we could observe the m_i, we could obtain an unbiased estimator of α_0 from the regression y_i on m_i without an intercept. Instead, we replace the β_j with their OLS estimates and obtain $\hat{m}_i = \exp(\widehat{logy}_i)$, where, of course, the $\widehat{logy}_i$ are the fitted values from the regression $logy_i$ on $x_{i1}, \ldots, x_{ik}$ (with an intercept). Then $\check{\alpha}_0$ [to distinguish it from $\hat{\alpha}_0$ in equation (6.43)] is the OLS slope estimate from the simple regression y_i on $\hat{m}_i$ (no intercept):

$$\check{\alpha}_0 = \left(\sum_{i=1}^{n} \hat{m}_i^2 \right)^{-1} \left(\sum_{i=1}^{n} \hat{m}_i y_i \right). \qquad [6.44]$$

We will call $\check{\alpha}_0$ the regression estimate of α_0. Like $\hat{\alpha}_0$, $\check{\alpha}_0$ is consistent but not unbiased. Interestingly, $\check{\alpha}_0$ is not guaranteed to be greater than one, although it will be in most applications. If $\check{\alpha}_0$ is less than one, and especially if it is much less than one, it is likely that the assumption of independence between u and the x_j is violated. If $\check{\alpha}_0 < 1$, one possibility is to just use the estimate in (6.43), although this may simply be masking a problem with the linear model for $\log(y)$. We summarize the steps:

Predicting y When the Dependent Variable Is $\log(y)$:

1. Obtain the fitted values, $\widehat{logy}_i$, and residuals, $\hat{u}_i$, from the regression $logy$ on $x_1, \ldots, x_k$.
2. Obtain $\hat{\alpha}_0$ as in equation (6.43) or $\check{\alpha}_0$ in equation (6.44).
3. For given values of $x_1, \ldots, x_k$, obtain $\widehat{logy}$ from (6.42).
4. Obtain the prediction $\hat{y}$ from (6.42) (with $\hat{\alpha}_0$ or $\check{\alpha}_0$).

We now show how to predict CEO salaries using this procedure.

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A11 — A11 예측식: ŷ = α̂0 · exp(logŷ).

A11 — A11 α0 추정: 모평균 → 표본평균, 오차 u → 잔차 û로 대체 (적률법).

A11 — A11 **α̂0 = (1/n)Σexp(ûi)** (smearing estimate). 슬라이드 마지막 식의 괄호 부분.

A11 — A11 α̂0는 일치추정량이지만 불편은 아님: 비선형 함수 안에 u 대신 û를 넣었기 때문. 항상 α̂0 > 1.

SKIP — SKIP 원점 통과 회귀로 구하는 대안 추정치(6.44) — PPT 미포함.

REF — REF 절차 요약: ① log y 회귀 → 적합값·잔차 ② α̂0 계산 ③ 주어진 x에서 logŷ ④ ŷ = α̂0 · exp(logŷ).

EXAMPLE 6.7 PREDICTING CEO SALARIES

The model of interest is

$$\log(salary) = \beta_0 + \beta_1 \log(sales) + \beta_2 \log(mktval) + \beta_3 ceoten + u,$$

so that β_1 and β_2 are elasticities and $100 \cdot \beta_3$ is a semi-elasticity. The estimated equation using CEOSAL2.RAW is

$$\widehat{lsalary} = 4.504 + .163\ lsales + .109\ lmktval + .0117\ ceoten \quad [6.45]$$
$$(.257)\ \ (.039)\qquad (.050)\qquad (.0053)$$
$$n = 177,\ R^2 = .318,$$

where, for clarity, we let *lsalary* denote the log of *salary*, and similarly for *lsales* and *lmktval*. Next, we obtain $\hat{m}_i = \exp(\widehat{lsalary}_i)$ for each observation in the sample.

The Duan smearing estimate from (6.43) is about $\hat{\alpha}_0 = 1.136$, and the regression estimate from (6.44) is $\check{\alpha}_0 = 1.117$. We can use either estimate to predict *salary* for any values of *sales*, *mktval*, and *ceoten*. Let us find the prediction for *sales* = 5,000 (which means \$5 billion because *sales* is in millions), *mktval* = 10,000 (or \$10 billion), and *ceoten* = 10. From (6.45), the prediction for *lsalary* is $4.504 + .163 \cdot \log(5{,}000) + .109 \cdot \log(10{,}000) + .0117(10) \approx 7.013$, and $\exp(7.013) \approx 1{,}110.983$. Using the estimate of α_0 from (6.43), the predicted salary is about 1,262.077, or \$1,262,077. Using the estimate from (6.44) gives an estimated salary of about \$1,240,968. These differ from each other by much less than each differs from the naive prediction of \$1,110,983.

SKIP

We can use the previous method of obtaining predictions to determine how well the model with $\log(y)$ as the dependent variable explains y. We already have measures for models when y is the dependent variable: the R-squared and the adjusted R-squared. The goal is to find a goodness-of-fit measure in the $\log(y)$ model that can be compared with an R-squared from a model where y is the dependent variable.

There are different ways to define a goodness-of-fit measure after retransforming a model for $\log(y)$ to predict y. Here we present an approach that is easy to implement and that gives the same value whether we estimate α_0 as in (6.40), (6.43), or (6.44). To motivate the measure, recall that in the linear regression equation estimated by OLS,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \ldots + \hat{\beta}_k x_k, \quad [6.46]$$

the usual R-squared is simply the square of the correlation between y_i and $\hat{y}_i$ (see Section 3.2). Now, if instead we compute fitted values from (6.42)—that is, $\hat{y}_i = \hat{\alpha}_0 m_i$ for all observations i—then it makes sense to use the square of the correlation between y_i and these fitted values as an R-squared. Because correlation is unaffected if we multiply by a constant, it does not matter which estimate of α_0 we use. In fact, this R-squared measure for y [not $\log(y)$] is just the squared correlation between y_i and $\hat{m}_i$. We can compare this directly with the R-squared from equation (6.46). [Because the R-squared calculation does not depend on the estimate of α_0, it does not allow us to choose among (6.40), (6.43), and (6.44). But we know that (6.44) minimizes the sum of squared residuals between y_i and $\hat{m}_i$, without a constant. In other words, given the $\hat{m}_i$, $\check{\alpha}_0$ is chosen to produce the best fit based on sum of squared residuals. We are interested here in choosing between the linear model for y and $\log(y)$, and so an R-squared measure that does not depend on how we estimate α_0 is suitable.]

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SKIP 예제 6.7 (CEO 연봉 예측) — PPT 미포함. 감만 잡기: α̂0=1.136 → 단순 지수변환보다 약 14% 높게 보정.

REF A6의 미해결 문제(y vs log y 모형 비교)의 답: 로그모형에서 만든 ŷ와 y의 **상관계수 제곱**을 수준모형 R^2와 비교.

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SKIP 예제 6.8 · log 모형의 예측구간 — PPT 미포함.

EXAMPLE 6.8 PREDICTING CEO SALARIES

After we obtain the $\hat{m}_i$, we just obtain the correlation between $salary_i$ and $\hat{m}_i$; it is .493. The square of it is about .243, and this is a measure of how well the log model explains the variation in *salary*, not log(*salary*). [The R^2 from (6.45), .318, tells us that the log model explains about 31.8% of the variation in log(*salary*).]

As a competing linear model, suppose we estimate a model with all variables in levels:

$$salary = \beta_0 + \beta_1 sales + \beta_2 mktval + \beta_3 ceoten + u. \qquad [6.47]$$

The key is that the dependent variable is *salary*. We could use logs of *sales* or *mktval* on the right-hand side, but it makes more sense to have all dollar values in levels if one (*salary*) appears as a level. The *R*-squared from estimating this equation using the same 177 observations is .201. Thus, the log model explains more of the variation in *salary*, and so we prefer it to (6.47) on goodness-of-fit grounds. The log model is also preferred because it seems more realistic and its parameters are easier to interpret.

If we maintain the full set of classical linear model assumptions in the model (6.38), we can easily obtain prediction intervals for $y^0 = \exp(\beta_0 + \beta_1 x_1^0 + \ldots + \beta_k x_k^0 + u^0)$ when we have estimated a linear model for log(*y*). Recall that $x_1^0, x_2^0, \ldots, x_k^0$ are known values and u^0 is the unobserved error that partly determines y^0. From equation (6.37), a 95% prediction interval for $logy^0 = \log(y^0)$ is simply $\widehat{logy}^0 \pm t_{.025} \cdot \text{se}(\hat{e}^0)$, where $\text{se}(\hat{e}^0)$ is obtained from the regression of log(*y*) on $x_1, \ldots, x_k$ using the original *n* observations. Let $c_l = -t_{.025} \cdot \text{se}(\hat{e}^0)$ and $c_u = t_{.025} \cdot \text{se}(\hat{e}^0)$ be the lower and upper bounds of the prediction interval for $logy^0$. That is, $P(c_l \le logy^0 \le c_u) = .95$. Because the exponential function is strictly increasing, it is also true that $P[\exp(c_l) \le \exp(logy^0) \le \exp(c_u)] = .95$,that is, $P[\exp(c_l) \le y^0 \le \exp(c_u)] = .95$. Therefore, we can take $\exp(c_l)$ and $\exp(c_u)$ as the lower and upper bounds, respectively, for a 95% prediction interval for y^0. For large *n*, $t_{.025} = 1.96$, and so a 95% prediction interval for y^0 is $\exp[-1.96 \cdot \text{se}(\hat{e}^0)] \exp(\hat{\beta}_0 + \mathbf{x}^0\hat{\boldsymbol{\beta}})$ to $\exp[-1.96 \cdot \text{se}(\hat{e}^0)] \exp(\hat{\beta}_0 + \mathbf{x}^0\hat{\boldsymbol{\beta}})$, where $\mathbf{x}^0\hat{\boldsymbol{\beta}}$ is shorthand for $\hat{\beta}_1 x_1^0 + \ldots + \hat{\beta}_k x_k^0$. Remember, the $\hat{\beta}_j$ and $\text{se}(\hat{e}^0)$ are obtained from the regression with log(*y*) as the dependent variable. Because we assume normality of *u* in (6.38), we probably would use (6.40) to obtain a point prediction for y^0. Unlike in equation (6.37), this point prediction will not lie halfway between the upper and lower bounds $\exp(c_l)$ and $\exp(c_u)$. One can obtain different 95% prediction intervalues by choosing different quantiles in the t_{n-k-1} distribution. If $q_{\alpha 1}$ and $q_{\alpha 2}$ are quantiles with $\alpha_2 - \alpha_1 = .95$, then we can choose $c_l = q_{\alpha 1}\text{se}(\hat{e}^0)$ and $c_u = q_{\alpha 2}\text{se}(\hat{e}^0)$.

As an example, consider the CEO salary regression, where we make the prediction at the same values of *sales*, *mktval*, and *ceoten* as in Example 6.7. The standard error of the regression for (6.43) is about .505, and the standard error of $\widehat{logy}^0$ is about .075. Therefore, using equation (6.36), $\text{se}(\hat{e}^0) \approx .511$; as in the GPA example, the error variance swamps the estimation error in the parameters, even though here the sample size is only 177. A 95% prediction interval for $salary^0$ is $\exp[-1.96 \cdot (.511)] \exp(7.013)$ to $\exp[1.96 \cdot (.511)] \exp(7.013)$, or about 408.071 to 3,024.678, that is, $408,071 to $3,024,678. This very wide 95% prediction interval for CEO salary at the given sales, market value, and tenure values shows that there is much else that we have not included in the regression that determines salary. Incidentally, the point prediction for salary, using (6.40), is about $1,262,075—higher than the predictions using the other estimates of α_0 and closer to the lower bound than the upper bound of the 95% prediction interval.

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Summary

In this chapter, we have covered some important multiple regression analysis topics.

Section 6.1 showed that a change in the units of measurement of an independent variable changes the OLS coefficient in the expected manner: if x_j is multiplied by c, its coefficient is divided by c. If the dependent variable is multiplied by c, *all* OLS coefficients are multiplied by c. Neither t nor F statistics are affected by changing the units of measurement of any variables.

We discussed beta coefficients, which measure the effects of the independent variables on the dependent variable in standard deviation units. The beta coefficients are obtained from a standard OLS regression after the dependent and independent variables have been transformed into z-scores.

We provided a detailed discussion of functional form, including the logarithmic transformation, quadratics, and interaction terms. It is helpful to summarize some of our conclusions.

CONSIDERATIONS WHEN USING LOGARITHIMS

1. The coefficients have percentage change interpretations. We can be ignorant of the units of measurement of any variable that appears in logarithmic form, and changing units from, say, dollars to thousands of dollars has no effect on a variable's coefficient when that variable appears in logarithmic form.
2. Logs are often used for dollar amounts that are always positive, as well as for variables such as population, especially when there is a lot of variation. They are used less often for variables measured in years, such as schooling, age, and experience. Logs are used infrequently for variables that are already percents or proportions, such as an unemployment rate or a pass rate on a test.
3. Models with $\log(y)$ as the dependent variable often more closely satisfy the classical linear model assumptions. For example, the model has a better chance of being linear, homoskedasticity is more likely to hold, and normality is often more plausible.
4. In many cases, taking the log greatly reduces the variation of a variable, making OLS estimates less prone to outlier influence. However, in cases where y is a fraction and close to zero for many observations, $\log(y_i)$ can have much more variability than y_i. For values y_i very close to zero, $\log(y_i)$ is a negative number very large in magnitude.
5. If $y \geq 0$ but $y = 0$ is possible, we cannot use $\log(y)$. Sometimes $\log(1 + y)$ is used, but interpretation of the coefficients is difficult.
6. For large changes in an explanatory variable, we can compute a more accurate estimate of the percentage change effect.
7. It is harder (but possible) to predict y when we have estimated a model for $\log(y)$.

CONSIDERATIONS WHEN USING QUADRATICS

1. A quadratic function in an explanatory variable allows for an increasing or decreasing effect.
2. The turning point of a quadratic is easily calculated, and it should be calculated to see if it makes sense.
3. Quadratic functions where the coefficients have the opposite sign have a strictly positive turning point; if the signs of the coefficients are the same, the turning point is at a negative value of x.
4. A seemingly small coefficient on the square of a variable can be practically important in what it implies about a changing slope. One can use a t test to see if the quadratic is statistically significant, and compute the slope at various values of x to see if it is practically important.

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REF Ch.6 Summary. 로그·이차·교호항 정리는 10.15 수업 범위(중간고사) 복습용.

5. For a model quadratic in a variable x, the coefficient on x measures the partial effect starting from $x = 0$, as can be seen in equation (6.11). If zero is not a possible or interesting value of x, one can center x about a more interesting value, such as the average in the sample, before computing the square. Computing Exercise 6.12 provides an example.

CONSIDERATIONS WHEN USING INTERACTIONS

1. Interaction terms allow the partial effect of an explanatory variable, say x_1, to depend on the level of another variable, say x_2—and vice versa.
2. Interpreting models with interactions can be tricky. The coefficient on x_1, say β_1, measures the partial effect of x_1 on y when $x_2 = 0$, which may be impossible or uninteresting. Centering x_1 and x_2 around interesting values before constructing the interaction term typically leads to an equation that is visually more appealing.
3. A standard t test can be used to determine if an interaction term is statistically significant. Computing the partial effects at different values of the explanatory variables can be used to determine the practical importance of interactions.

We introduced the adjusted R-squared, $\bar{R}^2$, as an alternative to the usual R-squared for measuring goodness-of-fit. Whereas R^2 can never fall when another variable is added to a regression, $\bar{R}^2$ penalizes the number of regressors and can drop when an independent variable is added. This makes $\bar{R}^2$ preferable for choosing between nonnested models with different numbers of explanatory variables. Neither R^2 nor $\bar{R}^2$ can be used to compare models with different dependent variables. Nevertheless, it is fairly easy to obtain goodness-of-fit measures for choosing between y and $\log(y)$ as the dependent variable, as shown in Section 6.4.

In Section 6.3, we discussed the somewhat subtle problem of relying too much on R^2 or $\bar{R}^2$ in arriving at a final model: it is possible to control for too many factors in a regression model. For this reason, it is important to think ahead about model specification, particularly the ceteris paribus nature of the multiple regression equation. Explanatory variables that affect y and are uncorrelated with all the other explanatory variables can be used to reduce the error variance without inducing multicollinearity.

In Section 6.4, we demonstrated how to obtain a confidence interval for a prediction made from an OLS regression line. We also showed how a confidence interval can be constructed for a future, unknown value of y.

Occasionally, we want to predict y when $\log(y)$ is used as the dependent variable in a regression model. Section 6.4 explains this simple method. Finally, we are sometimes interested in knowing about the sign and magnitude of the residuals for particular observations. Residual analysis can be used to determine whether particular members of the sample have predicted values that are well above or well below the actual outcomes.

Key Terms

Adjusted R-Squared	Over Controlling	Resampling Method
Beta Coefficients	Population R-Squared	Residual Analysis
Bootstrap	Prediction Error	Smearing Estimate
Bootstrap Standard Error	Prediction Interval	Standardized Coefficients
Interaction Effect	Predictions	Variance of the Prediction Error
Nonnested Models	Quadratic Functions	

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p.217 · PPT 필기 A4 A7 A9 A11

A4 **A4 Summary로 A3–A6 한 번에 정리**: R^2는 변수 수에 벌점 → 비내포 모형 선택에 유용. 종속변수가 다르면 R^2·R^2 모두 불가.

A7 **A7** A7–A8 요약: 과잉 통제.

A9 **A9** A9–A10 요약.

A11 **A11** 요약.

KEY **KEY** Key Terms 중 노랑 = PPT에서 다룬 용어.

SKIP

SKIP Ch.6 연습문제 · Computer Exercises · 부록 6A(부트스트랩), p.218–226 — 필기 범위 밖.

Problems

1 The following equation was estimated using the data in CEOSAL1.RAW:

$$\widehat{\log(salary)} = 4.322 + .276\log(sales) + .0215\,roe - .00008\,roe^2$$
$$(.324)\ \ (.033)\qquad\qquad (.0129)\qquad\ (.00026)$$
$$n = 209,\ R^2 = .282.$$

This equation allows *roe* to have a diminishing effect on log(*salary*). Is this generality necessary? Explain why or why not.

2 Let $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k$ be the OLS estimates from the regression of y_i on $x_{i1}, \ldots, x_{ik}$, $i=1, 2, \ldots, n$. For nonzero constants $c_1, \ldots, c_k$, argue that the OLS intercept and slopes from the regression of $c_0 y_i$ on $c_1 x_{i1}, \ldots, c_k x_{ik}$, $i = 1, 2, \ldots, n$, are given by $\tilde{\beta}_0 = c_0\hat{\beta}_0$, $\tilde{\beta}_1 = (c_0/c_1)\hat{\beta}_1, \ldots, \tilde{\beta}_k = (c_0/c_k)\hat{\beta}_k$. [*Hint*: Use the fact that the $\hat{\beta}_j$ solve the first order conditions in (3.13), and the $\tilde{\beta}_j$ must solve the first order conditions involving the rescaled dependent and independent variables.]

3 Using the data in RDCHEM.RAW, the following equation was obtained by OLS:

$$\widehat{rdintens} = 2.613 + .00030\,sales - .0000000070\,sales^2$$
$$(.429)\ \ (.00014)\qquad\ (.0000000037)$$
$$n = 32,\ R^2 = .1484.$$

(i) At what point does the marginal effect of *sales* on *rdintens* become negative?
(ii) Would you keep the quadratic term in the model? Explain.
(iii) Define *salesbil* as sales measured in billions of dollars: *salesbil* = *sales*/1,000. Rewrite the estimated equation with *salesbil* and $salesbil^2$ as the independent variables. Be sure to report standard errors and the *R*-squared. [*Hint*: Note that $salesbil^2 = sales^2/(1{,}000)^2$.]
(iv) For the purpose of reporting the results, which equation do you prefer?

4 The following model allows the return to education to depend upon the total amount of both parents' education, called *pareduc*:

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 educ\cdot pareduc + \beta_3 exper + \beta_4 tenure + u.$$

(i) Show that, in decimal form, the return to another year of education in this model is

$$\Delta\log(wage)/\Delta educ = \beta_1 + \beta_2 pareduc.$$

What sign do you expect for β_2? Why?
(ii) Using the data in WAGE2.RAW, the estimated equation is

$$\widehat{\log(wage)} = 5.65 + .047\,educ + .00078\,educ\cdot pareduc +$$
$$(.13)\ \ (.010)\qquad\ (.00021)$$
$$.019\,exper + .010\,tenure$$
$$(.004)\qquad\ (.003)$$
$$n = 722,\ R^2 = .169.$$

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(Only 722 observations contain full information on parents' education.) Interpret the coefficient on the interaction term. It might help to choose two specific values for *pareduc*—for example, *pareduc* = 32 if both parents have a college education, or *pareduc* = 24 if both parents have a high school education—and to compare the estimated return to *educ*.

(iii) When *pareduc* is added as a separate variable to the equation, we get:

$$\widehat{\log(wage)} = 4.94 + .097\,educ + .033\,pareduc - .0016\,educ{\cdot}pareduc$$
$$(.38)\quad (.027)\qquad (.017)\qquad\quad (.0012)$$
$$+ .020\,exper + .010\,tenure$$
$$(.004)\qquad\quad (.003)$$
$$n = 722,\ R^2 = .174.$$

Does the estimated return to education now depend positively on parent education? Test the null hypothesis that the return to education does not depend on parent education.

5 In Example 4.2, where the percentage of students receiving a passing score on a tenth-grade math exam (*math*10) is the dependent variable, does it make sense to include *sci11*—the percentage of eleventh graders passing a science exam—as an additional explanatory variable?

6 When $atndrte^2$ and *ACT*·*atndrte* are added to the equation estimated in (6.19), the *R*-squared becomes .232. Are these additional terms jointly significant at the 10% level? Would you include them in the model?

7 The following three equations were estimated using the 1,534 observations in 401K.RAW:

$$\widehat{prate} = 80.29 + 5.44\,mrate + .269\,age - .00013\,totemp$$
$$(.78)\quad (.52)\qquad (.045)\qquad (.00004)$$
$$R^2 = .100,\ \bar{R}^2 = .098.$$

$$\widehat{prate} = 97.32 + 5.02\,mrate + .314\,age - 2.66\log(totemp)$$
$$(1.95)\quad (0.51)\qquad (.044)\qquad (.28)$$
$$R^2 = .144,\ \bar{R}^2 = .142.$$

$$\widehat{prate} = 80.62 + 5.34\,mrate + .290\,age - .00043\,totemp$$
$$(.78)\quad (.52)\qquad (.045)\qquad (.00009)$$
$$+ .0000000039\,totemp^2$$
$$(.0000000010)$$
$$R^2 = .108,\ \bar{R}^2 = .106.$$

Which of these three models do you prefer? Why?

8 Suppose we want to estimate the effects of alcohol consumption (*alcohol*) on college grade point average (*colGPA*). In addition to collecting information on grade point averages and alcohol usage, we also obtain attendance information (say, percentage of lectures attended, called *attend*). A standardized test score (say, *SAT*) and high school GPA (*hsGPA*) are also available.

(i) Should we include *attend* along with *alcohol* as explanatory variables in a multiple regression model? (Think about how you would interpret $\beta_{alcohol}$.)

(ii) Should *SAT* and *hsGPA* be included as explanatory variables? Explain.

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9 If we start with (6.38) under the CLM assumptions, assume large n, and ignore the estimation error in the $\hat{\beta}_j$, a 95% prediction interval for y^0 is $[\exp(-1.96\hat{\sigma})\exp(\widehat{logy^0}), \exp(1.96\hat{\sigma})\exp(\widehat{logy^0})]$. The point prediction for y^0 is $\hat{y}^0 = \exp(\hat{\sigma}^2/2)\exp(\widehat{logy^0})$.

(i) For what values of $\hat{\sigma}$ will the point prediction be in the 95% prediction interval? Does this condition seem likely to hold in most applications?

(ii) Verify that the condition from part (i) is satisfied in the CEO salary example.

Computer Exercises

C1 Use the data in KIELMC.RAW, only for the year 1981, to answer the following questions. The data are for houses that sold during 1981 in North Andover, Massachusetts; 1981 was the year construction began on a local garbage incinerator.

(i) To study the effects of the incinerator location on housing price, consider the simple regression model

$$\log(price) = \beta_0 + \beta_1 \log(dist) + u,$$

where *price* is housing price in dollars and *dist* is distance from the house to the incinerator measured in feet. Interpreting this equation causally, what sign do you expect for β_1 if the presence of the incinerator depresses housing prices? Estimate this equation and interpret the results.

(ii) To the simple regression model in part (i), add the variables log(*intst*), log(*area*), log(*land*), *rooms*, *baths*, and *age*, where *intst* is distance from the home to the interstate, *area* is square footage of the house, *land* is the lot size in square feet, *rooms* is total number of rooms, *baths* is number of bathrooms, and *age* is age of the house in years. Now, what do you conclude about the effects of the incinerator? Explain why (i) and (ii) give conflicting results.

(iii) Add $[\log(intst)]^2$ to the model from part (ii). Now what happens? What do you conclude about the importance of functional form?

(iv) Is the square of log(*dist*) significant when you add it to the model from part (iii)?

C2 Use the data in WAGE1.RAW for this exercise.

(i) Use OLS to estimate the equation

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u$$

and report the results using the usual format.

(ii) Is $exper^2$ statistically significant at the 1% level?

(iii) Using the approximation

$$\%\Delta\widehat{wage} \approx 100(\hat{\beta}_2 + 2\hat{\beta}_3 exper)\Delta exper,$$

find the approximate return to the fifth year of experience. What is the approximate return to the twentieth year of experience?

(iv) At what value of *exper* does additional experience actually lower predicted log(*wage*)? How many people have more experience in this sample?

C3 Consider a model where the return to education depends upon the amount of work experience (and vice versa):

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 educ \cdot exper + u.$$

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(i) Show that the return to another year of education (in decimal form), holding *exper* fixed, is $\beta_1 + \beta_3 exper$.

(ii) State the null hypothesis that the return to education does not depend on the level of *exper*. What do you think is the appropriate alternative?

(iii) Use the data in WAGE2.RAW to test the null hypothesis in (ii) against your stated alternative.

(iv) Let θ_1 denote the return to education (in decimal form), when $exper = 10$: $\theta_1 = \beta_1 + 10\beta_3$. Obtain $\hat{\theta}_1$ and a 95% confidence interval for θ_1. (*Hint:* Write $\beta_1 = \theta_1 - 10\beta_3$ and plug this into the equation; then rearrange. This gives the regression for obtaining the confidence interval for θ_1.)

C4 Use the data in GPA2.RAW for this exercise.

(i) Estimate the model

$$sat = \beta_0 + \beta_1 hsize + \beta_2 hsize^2 + u,$$

where *hsize* is the size of the graduating class (in hundreds), and write the results in the usual form. Is the quadratic term statistically significant?

(ii) Using the estimated equation from part (i), what is the "optimal" high school size? Justify your answer.

(iii) Is this analysis representative of the academic performance of *all* high school seniors? Explain.

(iv) Find the estimated optimal high school size, using log(*sat*) as the dependent variable. Is it much different from what you obtained in part (ii)?

C5 Use the housing price data in HPRICE1.RAW for this exercise.

(i) Estimate the model

$$\log(price) = \beta_0 + \beta_1 \log(lotsize) + \beta_2 \log(sqrft) + \beta_3 bdrms + u$$

and report the results in the usual OLS format.

(ii) Find the predicted value of log(*price*), when $lotsize = 20{,}000$, $sqrft = 2{,}500$, and $bdrms = 4$. Using the methods in Section 6.4, find the predicted value of *price* at the same values of the explanatory variables.

(iii) For explaining variation in *price*, decide whether you prefer the model from part (i) or the model

$$price = \beta_0 + \beta_1 lotsize + \beta_2 sqrft + \beta_3 bdrms + u.$$

C6 Use the data in VOTE1.RAW for this exercise.

(i) Consider a model with an interaction between expenditures:

$$voteA = \beta_0 + \beta_1 prtystrA + \beta_2 expendA + \beta_3 expendB + \beta_4 expendA \cdot expendB + u.$$

What is the partial effect of *expendB* on *voteA*, holding *prtystrA* and *expendA* fixed? What is the partial effect of *expendA* on *voteA*? Is the expected sign for β_4 obvious?

(ii) Estimate the equation in part (i) and report the results in the usual form. Is the interaction term statistically significant?

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(iii) Find the average of *expendA* in the sample. Fix *expendA* at 300 (for \$300,000). What is the estimated effect of another \$100,000 spent by Candidate B on *voteA*? Is this a large effect?

(iv) Now fix *expendB* at 100. What is the estimated effect of $\Delta expendA = 100$ on *voteA*? Does this make sense?

(v) Now, estimate a model that replaces the interaction with *shareA*, Candidate A's percentage share of total campaign expenditures. Does it make sense to hold both *expendA* and *expendB* fixed, while changing *shareA*?

(vi) (Requires calculus) In the model from part (v), find the partial effect of *expendB* on *voteA*, holding *prtystrA* and *expendA* fixed. Evaluate this at $expendA = 300$ and $expendB = 0$ and comment on the results.

C7 Use the data in ATTEND.RAW for this exercise.

(i) In the model of Example 6.3, argue that

$$\Delta stndfnl/\Delta priGPA \approx \beta_2 + 2\beta_4 priGPA + \beta_6 atndrte.$$

Use equation (6.19) to estimate the partial effect when $priGPA = 2.59$ and $atndrte = 82$. Interpret your estimate.

(ii) Show that the equation can be written as

$$stndfnl = \theta_0 + \beta_1 atndrte + \theta_2 priGPA + \beta_3 ACT + \beta_4 (priGPA - 2.59)^2 + \beta_5 ACT^2 + \beta_6 priGPA(atndrte - 82) + u,$$

where $\theta_2 = \beta_2 + 2\beta_4(2.59) + \beta_6(82)$. (Note that the intercept has changed, but this is unimportant.) Use this to obtain the standard error of $\hat{\theta}_2$ from part (i).

(iii) Suppose that, in place of $priGPA(atndrte - 82)$, you put $(priGPA - 2.59) \cdot (atndrte - 82)$. Now how do you interpret the coefficients on *atndrte* and *priGPA*?

C8 Use the data in HPRICE1.RAW for this exercise.

(i) Estimate the model

$$price = \beta_0 + \beta_1 lotsize + \beta_2 sqrft + \beta_3 bdrms + u$$

and report the results in the usual form, including the standard error of the regression. Obtain predicted price, when we plug in $lotsize = 10{,}000$, $sqrft = 2{,}300$, and $bdrms = 4$; round this price to the nearest dollar.

(ii) Run a regression that allows you to put a 95% confidence interval around the predicted value in part (i). Note that your prediction will differ somewhat due to rounding error.

(iii) Let $price^0$ be the unknown future selling price of the house with the characteristics used in parts (i) and (ii). Find a 95% CI for $price^0$ and comment on the width of this confidence interval.

C9 The data set NBASAL.RAW contains salary information and career statistics for 269 players in the National Basketball Association (NBA).

(i) Estimate a model relating points-per-game (*points*) to years in the league (*exper*), *age*, and years played in college (*coll*). Include a quadratic in *exper*; the other variables should appear in level form. Report the results in the usual way.

(ii) Holding college years and age fixed, at what value of experience does the next year of experience actually reduce points-per-game? Does this make sense?

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(iii) Why do you think *coll* has a negative and statistically significant coefficient? (*Hint:* NBA players can be drafted before finishing their college careers and even directly out of high school.)

(iv) Add a quadratic in *age* to the equation. Is it needed? What does this appear to imply about the effects of age, once experience and education are controlled for?

(v) Now regress log(*wage*) on *points*, *exper*, $exper^2$, *age*, and *coll*. Report the results in the usual format.

(vi) Test whether *age* and *coll* are jointly significant in the regression from part (v). What does this imply about whether age and education have separate effects on wage, once productivity and seniority are accounted for?

C10 Use the data in BWGHT2.RAW for this exercise.

(i) Estimate the equation

$$\log(bwght) = \beta_0 + \beta_1 npvis + \beta_2 npvis^2 + u$$

by OLS, and report the results in the usual way. Is the quadratic term significant?

(ii) Show that, based on the equation from part (i), the number of prenatal visits that maximizes log(*bwght*) is estimated to be about 22. How many women had at least 22 prenatal visits in the sample?

(iii) Does it make sense that birth weight is actually predicted to decline after 22 prenatal visits? Explain.

(iv) Add mother's age to the equation, using a quadratic functional form. Holding *npvis* fixed, at what mother's age is the birth weight of the child maximized? What fraction of women in the sample are older than the "optimal" age?

(v) Would you say that mother's age and number of prenatal visits explain a lot of the variation in log(*bwght*)?

(vi) Using quadratics for both *npvis* and *age*, decide whether using the natural log or the level of *bwght* is better for predicting *bwght*.

C11 Use APPLE.RAW to verify some of the claims made in Section 6.3.

(i) Run the regression *ecolbs* on *ecoprc*, *regprc* and report the results in the usual form, including the *R*-squared and adjusted *R*-squared. Interpret the coefficients on the price variables and comment on their signs and magnitudes.

(ii) Are the price variables statistically significant? Report the *p*-values for the individual *t* tests.

(iii) What is the range of fitted values for *ecolbs*? What fraction of the sample reports *ecolbs* = 0? Comment.

(iv) Do you think the price variables together do a good job of explaining variation in *ecolbs*? Explain.

(v) Add the variables *faminc*, *hhsize* (household size), *educ*, and *age* to the regression from part (i). Find the *p*-value for their joint significance. What do you conclude?

(vi) Run separate simple regressions of *ecolbs* on *ecoprc* and then *ecolbs* on *regprc*. How do the simple regression coefficients compare with the multiple regression from part (i)? Find the correlation coefficient between *ecoprc* and *regprc* to help explain your findings.

C12 Use the subset of 401KSUBS.RAW with *fsize* = 1; this restricts the analysis to single-person households; see also Computer Exercise C8 in Chapter 4.

(i) What is the youngest age of people in this sample? How many people are at that age?

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(ii) In the model

$$nettfa = \beta_0 + \beta_1 inc + \beta_2 age + \beta_3 age^2 + u,$$

what is the literal interpretation of β_2? By itself, is it of much interest?

(iii) Estimate the model from part (ii) and report the results in standard form. Are you concerned that the coefficient on *age* is negative? Explain.

(iv) Because the youngest people in the sample are 25, it makes sense to think that, for a given level of income, the lowest average amount of net total financial assets is at age 25. Recall that the partial effect of *age* on *nettfa* is $\beta_2 + 2\beta_3 age$, so the partial effect at age 25 is $\beta_2 + 2\beta_3(25) = \beta_2 + 50\beta_3$; call this θ_2. Find $\hat{\theta}_2$ and obtain the two-sided *p*-value for testing H_0: $\theta_2 = 0$. You should conclude that $\hat{\theta}_2$ is small and very statistically insignificant. [*Hint:* One way to do this is to estimate the model $nettfa = \alpha_0 + \beta_1 inc + \theta_2 age + \beta_3 (age - 25)^2 + u$, where the intercept, α_0, is different from β_0. There are other ways, too.]

(v) Because the evidence against H_0: $\theta_2 = 0$ is very weak, set it to zero and estimate the model

$$nettfa = \alpha_0 + \beta_1 inc + \beta_3 (age - 25)^2 + u.$$

In terms of goodness-of-fit, does this model fit better than that in part (ii)?

(vi) For the estimated equation in part (v), set $inc = 30$ (roughly, the average value) and graph the relationship between *nettfa* and *age*, but only for $age \geq 25$. Describe what you see.

(vii) Check to see whether including a quadratic in *inc* is necessary.

C13 Use the data in MEAP00.RAW to answer this question.

(i) Estimate the model

$$math4 = \beta_0 + \beta_2 lexppp + \beta_2 lenroll + \beta_3 lunch + u$$

by OLS, and report the results in the usual form. Is each explanatory variable statistically significant at the 5% level?

(ii) Obtain the fitted values from the regression in part (i). What is the range of fitted values? How does it compare with the range of the actual data on *math4*?

(iii) Obtain the residuals from the regression in part (i). What is the building code of the school that has the largest (positive) residual? Provide an interpretation of this residual.

(iv) Add quadratics of all explanatory variables to the equation, and test them for joint significance. Would you leave them in the model?

(v) Returning to the model in part (i), divide the dependent variable and each explanatory variable by its sample standard deviation, and rerun the regression. (Include an intercept unless you also first subtract the mean from each variable.) In terms of standard deviation units, which explanatory variable has the largest effect on the math pass rate?

C14 Use the data in BENEFITS.RAW to answer this question. It is a school-level data set at the K–5 level on average teacher salary and benefits. See Example 4.10 for background.

(i) Regress *lavgsal* on *bs* and report the results in the usual form. Can you reject H_0: $\beta_{bs} = 0$ against a two-sided alternative? Can you reject H_0: $\beta_{bs} = -1$ against H_1: $\beta_{bs} > -1$? Report the *p*-values for both tests.

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(ii) Define $lbs = \log(bs)$. Find the range of values for *lbs* and find its standard deviation. How do these compare to the range and standard deviation for *bs*?

(iii) Regress *lavgsal* on *lbs*. Does this fit better than the regression from part (i)?

(iv) Estimate the equation
$lavgsal = \beta_0 + \beta_1 bs + \beta_2 lenroll + \beta_3 lstaff + \beta_4 lunch + u$
and report the results in the usual form. What happens to the coefficient on *bs*? Is it now statistically different from zero?

(v) Interpret the coefficient on *lstaff*. Why do you think it is negative?

(vi) Add $lunch^2$ to the equation from part (iv). Is it statistically significant? Compute the turning point (minimum value) in the quadratic, and show that it is within the range of the observed data on *lunch*. How many values of *lunch* are higher than the calculated turning point?

(vii) Based on the findings from part (vi), describe how teacher salaries relate to school poverty rates. In terms of teacher salary, and holding other factors fixed, is it better to teach at a school with $lunch = 0$ (no poverty), $lunch = 50$, or $lunch = 100$ (all kids eligible for the free lunch program)?

APPENDIX 6A

6A. A Brief Introduction to Bootstrapping

In many cases where formulas for standard errors are hard to obtain mathematically, or where they are thought not to be very good approximations to the true sampling variation of an estimator, we can rely on a **resampling method**. The general idea is to treat the observed data as a population that we can draw samples from. The most common resampling method is the **bootstrap**. (There are actually several versions of the bootstrap, but the most general, and most easily applied, is called the *nonparametric bootstrap*, and that is what we describe here.)

Suppose we have an estimate, $\hat{\theta}$, of a population parameter, θ. We obtained this estimate, which could be a function of OLS estimates (or estimates that we cover in later chapters), from a random sample of size n. We would like to obtain a standard error for $\hat{\theta}$ that can be used for constructing t statistics or confidence intervals. Remarkably, we can obtain a valid standard error by computing the estimate from different random samples drawn from the original data.

Implementation is easy. If we list our observations from 1 through n, we draw n numbers randomly, with replacement, from this list. This produces a new data set (of size n) that consists of the original data, but with many observations appearing multiple times (except in the rather unusual case that we resample the original data). Each time we randomly sample from the original data, we can estimate θ using the same procedure that we used on the original data. Let $\hat{\theta}^{(b)}$ denote the estimate from bootstrap sample b. Now, if we repeat the resampling and estimation m times, we have m new estimates, $\{\hat{\theta}^{(b)}: b = 1, 2, \ldots, m\}$. The **bootstrap standard error** of $\hat{\theta}$ is just the sample standard deviation of the $\hat{\theta}^{(b)}$, namely,

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$$\text{bse}(\hat{\theta}) = \left[(m-1)^{-1}\sum_{b=1}^{m}(\hat{\theta}^{(b)} - \bar{\hat{\theta}})^2\right]^{1/2}, \qquad [6.48]$$

where $\bar{\hat{\theta}}$ is the average of the bootstrap estimates.

If obtaining an estimate of θ on a sample of size n requires little computational time, as in the case of OLS and all the other estimators we encounter in this text, we can afford to choose m—the number of bootstrap replications—to be large. A typical value is $m = 1{,}000$, but even $m = 500$ or a somewhat smaller value can produce a reliable standard error. Note that the size of m—the number of times we resample the original data—has nothing to do with the sample size, n. (For certain estimation problems beyond the scope of this text, a large n can force one to do fewer bootstrap replications.) Many statistics and econometrics packages have built-in bootstrap commands, and this makes the calculation of bootstrap standard errors simple, especially compared with the work often required to obtain an analytical formula for an asymptotic standard error.

One can actually do better in most cases by using the bootstrap sample to compute p-values for t statistics (and F statistics), or for obtaining confidence intervals, rather than obtaining a bootstrap standard error to be used in the construction of t statistics or confidence intervals. See Horowitz (2001) for a comprehensive treatment.

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CHAPTER 7

Multiple Regression Analysis with Qualitative Information: Binary (or Dummy) Variables

In previous chapters, the dependent and independent variables in our multiple regression models have had *quantitative* meaning. Just a few examples include hourly wage rate, years of education, college grade point average, amount of air pollution, level of firm sales, and number of arrests. In each case, the magnitude of the variable conveys useful information. In empirical work, we must also incorporate *qualitative* factors into regression models. The gender or race of an individual, the industry of a firm (manufacturing, retail, and so on), and the region in the United States where a city is located (South, North, West, and so on) are all considered to be qualitative factors.

Most of this chapter is dedicated to qualitative *independent* variables. After we discuss the appropriate ways to describe qualitative information in Section 7.1, we show how qualitative explanatory variables can be easily incorporated into multiple regression models in Sections 7.2, 7.3, and 7.4. These sections cover almost all of the popular ways that qualitative independent variables are used in cross-sectional regression analysis.

In Section 7.5, we discuss a binary dependent variable, which is a particular kind of qualitative dependent variable. The multiple regression model has an interesting interpretation in this case and is called the linear probability model. While much maligned by some econometricians, the simplicity of the linear probability model makes it useful in many empirical contexts. We will describe its drawbacks in Section 7.5, but they are often secondary in empirical work.

7.1 Describing Qualitative Information

Qualitative factors often come in the form of binary information: a person is female or male; a person does or does not own a personal computer; a firm offers a certain kind of employee pension plan or it does not; a state administers capital punishment or it does not. In all of these examples, the relevant information can be captured by defining a **binary variable** or a **zero-one variable**. In econometrics, binary variables are most commonly called **dummy variables**, although this name is not especially descriptive.

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S5 **질적 정보**: 성별·인종·산업·지역·등급 등. 크기에 의미가 없는 정보를 회귀에 넣는 방법이 이 장의 주제.

S5 더미는 독립변수(7.2–7.4)로도, 종속변수(7.5 LPM)로도 쓰임.

S5 **더미변수(dummy)** = 이진변수 = 0/1 변수.

▶ 교수 설명 (수업 전사본 · 판서)

S5 · Qualitative information (1/2)

(10.27 47:10–49:14) 이진/더미변수: **1 또는 0**인 변수. 전에도 조금 다뤘지만 이번에 자세히.

- 예: 성별 → male=1, female=0 또는 그 반대
- 보통은 독립변수로 쓰지만 **이론적으로 종속변수도 가능 → 선형확률모형(LPM)**과 그 문제점을 다룰 예정

TABLE 7.1 A Partial Listing of the Data in WAGE1.RAW

person	*wage*	*educ*	*exper*	*female*	*married*
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
525	11.56	16	5	0	1
526	3.50	14	5	1	0

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In defining a dummy variable, we must decide which event is assigned the value one and which is assigned the value zero. For example, in a study of individual wage determination, we might define *female* to be a binary variable taking on the value one for females and the value zero for males. The name in this case indicates the event with the value one. The same information is captured by defining *male* to be one if the person is male and zero if the person is female. Either of these is better than using *gender* because this name does not make it clear when the dummy variable is one: does *gender* = 1 correspond to male or female? What we call our variables is unimportant for getting regression results, but it always helps to choose names that clarify equations and expositions.

EXPLORING FURTHER 7.1

Suppose that, in a study comparing election outcomes between Democratic and Republican candidates, you wish to indicate the party of each candidate. Is a name such as *party* a wise choice for a binary variable in this case? What would be a better name?

Suppose in the wage example that we have chosen the name *female* to indicate gender. Further, we define a binary variable *married* to equal one if a person is married and zero if otherwise. Table 7.1 gives a partial listing of a wage data set that might result. We see that Person 1 is female and not married, Person 2 is female and married, Person 3 is male and not married, and so on.

Why do we use the values zero and one to describe qualitative information? In a sense, these values are arbitrary: any two different values would do. The real benefit of capturing qualitative information using zero-one variables is that it leads to regression models where the parameters have very natural interpretations, as we will see now.

7.2 A Single Dummy Independent Variable

How do we incorporate binary information into regression models? In the simplest case, with only a single dummy explanatory variable, we just add it as an independent variable in the equation. For example, consider the following simple model of hourly wage determination:

$$wage = \beta_0 + \delta_0 female + \beta_1 educ + u. \qquad [7.1]$$

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S6 어떤 사건이 1인지 먼저 정함. 예: female=1(여성), 0(남성).

REF 변수 이름은 '1인 사건'으로 (gender ✗ → female ○). 해석 실수를 줄임.

S6 wage = β0 + **δ0** · female + β1 · educ + u. δ0 = 같은 교육 수준에서 여성이 남성보다 더(덜) 받는 임금.

S6 S6–S13 = 7.2절.

▶ 교수 설명 (수업 전사본 · 판서)

S6 · Single dummy independent variable

(10.27 49:23–50:15) 더미변수를 넣으면 교육의 효과에 대해 **(집단별로) 서로 다른 직선**을 그릴 수 있음.

We use δ_0 as the parameter on *female* in order to highlight the interpretation of the parameters multiplying dummy variables; later, we will use whatever notation is most convenient.

In model (7.1), only two observed factors affect wage: gender and education. Because *female* = 1 when the person is female, and *female* = 0 when the person is male, the parameter δ_0 has the following interpretation: δ_0 is the difference in hourly wage between females and males, *given* the same amount of education (and the same error term *u*). Thus, the coefficient δ_0 determines whether there is discrimination against women: if $\delta_0 < 0$, then, for the same level of other factors, women earn less than men on average.

In terms of expectations, if we assume the zero conditional mean assumption $E(u|female,educ) = 0$, then

$$\delta_0 = E(wage|female = 1,educ) - E(wage|female = 0,educ).$$

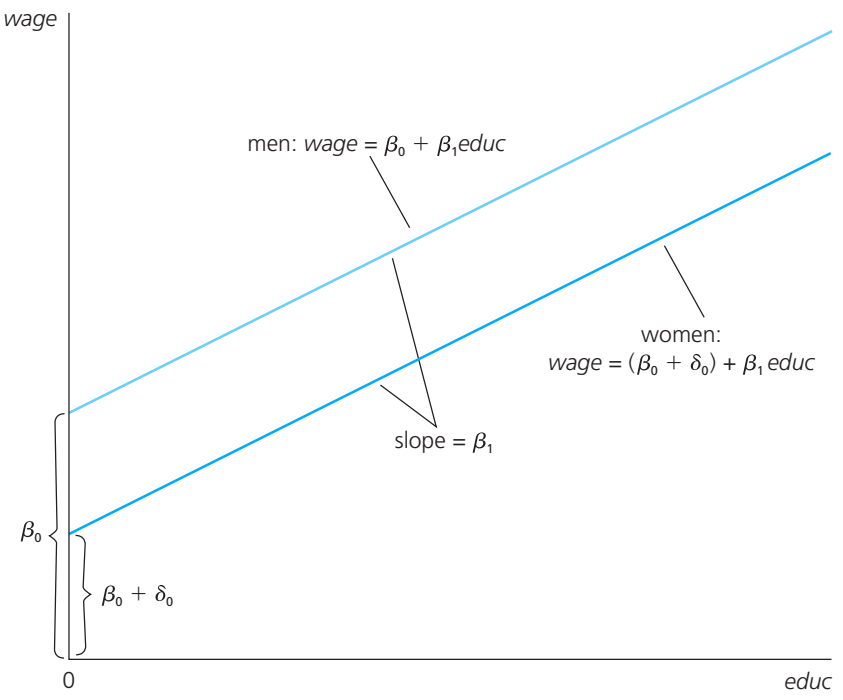
Because *female* = 1 corresponds to females and *female* = 0 corresponds to males, we can write this more simply as

$$\delta_0 = E(wage|female,educ) - E(wage|male,educ). \quad [7.2]$$

The key here is that the level of education is the same in both expectations; the difference, δ_0, is due to gender only.

The situation can be depicted graphically as an **intercept shift** between males and females. In Figure 7.1, the case $\delta_0 < 0$ is shown, so that men earn a fixed amount more per

FIGURE 7.1 Graph of $wage = \beta_0 + \delta_0 female + \beta_1 educ$ for $\delta_0 < 0$.



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S6 S6 δ0 < 0이면 같은 조건에서 여성 임금이 평균적으로 낮음.

S7 **S7 δ0 = E(wage|female=1, educ) − E(wage|female=0, educ)**. educ가 양쪽에서 같다는 점이 핵심.

S7 **S7 절편 이동(intercept shift)**: 남성 절편 β0, 여성 절편 β0+δ0, 기울기 β1은 공통 → 두 직선이 평행 (그림 7.1).

▶ 교수 설명 (수업 전사본 · 판서)

S7 · Graphical illustration: intercept shift

(10.27 52:06–54:27) x축 교육, y축 임금으로 그리면 두 식은 **절편이 다름**. 수업 중 문답:

- 다른 건 x절편? y절편? → **y절편**
- 기울기는 같은가 다른가? → **같음**
- 남녀 모두 교육을 더 받으면 임금 격차는 줄까, 늘까, 그대로일까? → 그대로 (이 모형에서는 격차가 변하지 않음)

hour than women. The difference does not depend on the amount of education, and this explains why the wage-education profiles for women and men are parallel.

At this point, you may wonder why we do not also include in (7.1) a dummy variable, say *male*, which is one for males and zero for females. This would be redundant. In (7.1), the intercept for males is β_0, and the intercept for females is $\beta_0 + \delta_0$. Because there are just two groups, we only need two different intercepts. This means that, in addition to β_0, we need to use only *one* dummy variable; we have chosen to include the dummy variable for females. Using two dummy variables would introduce perfect collinearity because $female + male = 1$, which means that *male* is a perfect linear function of *female*. Including dummy variables for both genders is the simplest example of the so-called **dummy variable trap**, which arises when too many dummy variables describe a given number of groups. We will discuss this problem later.

In (7.1), we have chosen males to be the **base group** or **benchmark group**, that is, the group against which comparisons are made. This is why β_0 is the intercept for males, and δ_0 is the *difference* in intercepts between females and males. We could choose females as the base group by writing the model as

$$wage = \alpha_0 + \gamma_0 male + \beta_1 educ + u,$$

where the intercept for females is α_0 and the intercept for males is $\alpha_0 + \gamma_0$; this implies that $\alpha_0 = \beta_0 + \delta_0$ and $\alpha_0 + \gamma_0 = \beta_0$. In any application, it does not matter how we choose the base group, but it is important to keep track of which group is the base group.

Some researchers prefer to drop the overall intercept in the model and to include dummy variables for each group. The equation would then be $wage = \beta_0 male + \alpha_0 female + \beta_1 educ + u$, where the intercept for men is β_0 and the intercept for women is α_0. There is no dummy variable trap in this case because we do not have an overall intercept. However, this formulation has little to offer, since testing for a difference in the intercepts is more difficult, and there is no generally agreed upon way to compute *R*-squared in regressions without an intercept. Therefore, we will always include an overall intercept for the base group.

Nothing much changes when more explanatory variables are involved. Taking males as the base group, a model that controls for experience and tenure in addition to education is

$$wage = \beta_0 + \delta_0 female + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u. \quad [7.3]$$

If *educ*, *exper*, and *tenure* are all relevant productivity characteristics, the null hypothesis of *no* difference between men and women is H_0: $\delta_0 = 0$. The alternative that there is discrimination against women is H_1: $\delta_0 < 0$.

How can we actually test for wage discrimination? The answer is simple: just estimate the model by OLS, *exactly* as before, and use the usual *t* statistic. Nothing changes about the mechanics of OLS or the statistical theory when some of the independent variables are defined as dummy variables. The only difference with what we have done up until now is in the interpretation of the coefficient on the dummy variable.

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S7 S7 임금 격차가 교육 수준과 무관하게 일정.

S8 S8 **더미변수 함정**: female + male = 1 = 절편항 → 완전공선성 (MLR.3 위배), 추정 불가.

S8 S8 규칙: g개 그룹이면 더미는 **g−1개** (+절편).

S8 S8 **기준집단(base group)** = 더미를 뺀 집단. 계수는 모두 기준집단 대비 차이.

S8 S8 기준을 여성으로 바꾸면 γ0 · male. 관계: γ0 = −δ0. 무엇을 기준으로 잡든 결과는 같지만 해석이 달라짐.

S9 S9 **대안**: 절편을 빼고 더미를 전부 넣기 → 함정은 없음.

S9 S9 단점 ① 집단 간 차이 검정이 번거로움 ② 절편 없는 회귀의 R^2 계산에 합의된 방식 없음 → 항상 절편 포함.

S10 S10 통제변수 추가: educ, exper, tenure. H0: δ0=0 vs H1: δ0<0.

S10 S10 검정은 평소의 t검정. 달라지는 건 계수의 해석뿐.

▶ 교수 설명 (수업 전사본 · 판서)

S8 · Dummy variable trap (1/2)

(10.27 54:27–56:37) male과 female을 둘 다 넣은 회귀를 돌릴 수 있나? → **못 함**. "세션 고정효과 때 이미 했다. 기억하라."

- 세션 더미도 세션 수보다 **하나 적게** 넣었음 (전사본 숫자는 불명확)
- 이유를 한 단어로? → 학생 답: **Perfect collinearity** ("여성 비율은 항상 1 − 남성 비율")
- [시험] 완전공선성을 피하려면 **하나를 빼야** 함

S9 · Dummy variable trap (2/2): no intercept

(10.27 56:45–57:39) 하나를 빼면 **해석이 바뀜**: δ0 = 여성인 것의 효과, γ0 = 남성인 것의 효과.

- 교재가 제시하는 다른 방법: **절편을 제거**
- 교수 의견: **권장하지 않음**. 지금까지 배운 것 대부분이 절편이 있다는 전제. "실제 연구 논문에서 이 방식을 쓴 걸 본 적이 없다." 이론적으로 가능할 뿐

EXAMPLE 7.1 **HOURLY WAGE EQUATION**

Using the data in WAGE1.RAW, we estimate model (7.3). For now, we use *wage*, rather than log(*wage*), as the dependent variable:

$$\widehat{wage} = -1.57 - 1.81\,female + .572\,educ \quad (.72) \quad (.26) \quad (.049)$$
$$+ .025\,exper + .141\,tenure \quad (.012) \quad (.021) \qquad (7.4)$$
$$n = 526,\ R^2 = .364.$$

The negative intercept—the intercept for men, in this case—is not very meaningful because no one has zero values for all of *educ*, *exper*, and *tenure* in the sample. The coefficient on *female* is interesting because it measures the average difference in hourly wage between a man and a woman who have the *same* levels of *educ*, *exper*, and *tenure*. If we take a woman and a man with the same levels of education, experience, and tenure, the woman earns, on average, $1.81 less per hour than the man. (Recall that these are 1976 wages.)

It is important to remember that, because we have performed multiple regression and controlled for *educ*, *exper*, and *tenure*, the $1.81 wage differential cannot be explained by different average levels of education, experience, or tenure between men and women. We can conclude that the differential of $1.81 is due to gender or factors associated with gender that we have not controlled for in the regression. [In 2003 dollars, the wage differential is about 3.23(1.81) ≈ 5.85.]

It is informative to compare the coefficient on *female* in equation (7.4) to the estimate we get when all other explanatory variables are dropped from the equation:

$$\widehat{wage} = 7.10 - 2.51\,female \quad (.21) \quad (.30) \qquad (7.5)$$
$$n = 526,\ R^2 = .116.$$

The coefficients in (7.5) have a simple interpretation. The intercept is the average wage for men in the sample (let *female* = 0), so men earn $7.10 per hour on average. The coefficient on *female* is the difference in the average wage between women and men. Thus, the average wage for women in the sample is 7.10 − 2.51 = 4.59, or $4.59 per hour. (Incidentally, there are 274 men and 252 women in the sample.)

Equation (7.5) provides a simple way to carry out a *comparison-of-means test* between the two groups, which in this case are men and women. The estimated difference, −2.51, has a *t* statistic of −8.37, which is very statistically significant (and, of course, $2.51 is economically large as well). Generally, simple regression on a constant and a dummy variable is a straightforward way to compare the means of two groups. For the usual *t* test to be valid, we must assume that the homoskedasticity assumption holds, which means that the population variance in wages for men is the same as that for women.

The estimated wage differential between men and women is larger in (7.5) than in (7.4) because (7.5) does not control for differences in education, experience, and tenure, and these are lower, on average, for women than for men in this sample. Equation (7.4) gives a more reliable estimate of the ceteris paribus gender wage gap; it still indicates a very large differential.

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p.231 · PPT 필기 S10 S11

S10 — S10 S10–S11 = 예제 7.1 (WAGE1, n=526).

S10 — S10 female −1.81(.26): 교육·경력·근속이 같을 때 여성이 시간당 $1.81 적게 받음. t ≈ −6.84.

S10 — S10 차별의 증거인가? 꼭 그렇진 않음. 통제하지 못한 생산성 특성이 female과 상관일 수 있음.

S11 — S11 통제변수 없이: 7.10 − 2.51 female. 절편 = 남성 평균임금, 계수 = 평균 차이. (SE: 교재 .30, 슬라이드 .26 — 판 차이.)

S11 — S11 상수 + 더미 하나의 단순회귀 = 두 집단 평균 비교 t검정. t = −8.37.

REF — REF 이 t검정은 동분산(남녀 임금 분산 동일) 가정이 필요 → Ch.8과 연결.

S11 — S11 −2.51 → −1.81로 줄어드는 이유: 격차의 일부는 교육·경력·근속 차이 때문 (S11 슬라이드 노트).

▶ 교수 설명 (수업 전사본 · 판서)

S10 · Estimated wage eq. with intercept shift

(10.27 57:39–58:55) female 계수가 유의해 보인다. 그럼 여성 차별이 있다고 말할 수 있나? → [시험] 이것만으로는 말할 수 없음.

- 이유: 여성인 것과 상관되어 있으면서 통제되지 않은 다른 특성이 있을 수 있음

S11 · Comparing means of subpopulations

(10.27 59:30–01:00:23) 다른 독립변수를 모두 빼고 female만 넣으면 단순한 사실만 남음: 여성이 남성보다 2.51 적게 번다. → 표본에서 평균적인 남성과 평균적인 여성의 차이.

In many cases, dummy independent variables reflect choices of individuals or other economic units (as opposed to something predetermined, such as gender). In such situations, the matter of causality is again a central issue. In the following example, we would like to know whether personal computer ownership *causes* a higher college grade point average.

EXAMPLE 7.2 EFFECTS OF COMPUTER OWNERSHIP ON COLLEGE GPA

In order to determine the effects of computer ownership on college grade point average, we estimate the model

$$colGPA = \beta_0 + \delta_0 PC + \beta_1 hsGPA + \beta_2 ACT + u,$$

where the dummy variable *PC* equals one if a student owns a personal computer and zero otherwise. There are various reasons PC ownership might have an effect on *colGPA*. A student's work might be of higher quality if it is done on a computer, and time can be saved by not having to wait at a computer lab. Of course, a student might be more inclined to play computer games or surf the Internet if he or she owns a PC, so it is not obvious that δ_0 is positive. The variables *hsGPA* (high school GPA) and *ACT* (achievement test score) are used as controls: it could be that stronger students, as measured by high school *GPA* and *ACT* scores, are more likely to own computers. We control for these factors because we would like to know the average effect on *colGPA* if a student is picked at random and given a personal computer.

Using the data in GPA1.RAW, we obtain

$$\widehat{colGPA} = 1.26 + .157\,PC + .447\,hsGPA + .0087\,ACT \quad [7.6]$$
$$(.33)\ \ (.057)\qquad (.094)\qquad\ (.0105)$$
$$n = 141,\ R^2 = .219.$$

This equation implies that a student who owns a PC has a predicted GPA about .16 points higher than a comparable student without a PC (remember, both *colGPA* and *hsGPA* are on a four-point scale). The effect is also very statistically significant, with $t_{PC} = .157/.057 \approx 2.75$.

What happens if we drop *hsGPA* and *ACT* from the equation? Clearly, dropping the latter variable should have very little effect, as its coefficient and *t* statistic are very small. But *hsGPA* is very significant, and so dropping it could affect the estimate of β_{PC}. Regressing *colGPA* on *PC* gives an estimate on *PC* equal to about .170, with a standard error of .063; in this case, $\hat{\beta}_{PC}$ and its *t* statistic do not change by much.

In the exercises at the end of the chapter, you will be asked to control for other factors in the equation to see if the computer ownership effect disappears, or if it at least gets notably smaller.

SKIP

Each of the previous examples can be viewed as having relevance for **policy analysis**. In the first example, we were interested in gender discrimination in the workforce. In the second example, we were concerned with the effect of computer ownership on college performance. A special case of policy analysis is **program evaluation**, where we would like to know the effect of economic or social programs on individuals, firms, neighborhoods, cities, and so on.

In the simplest case, there are two groups of subjects. The **control group** does not participate in the program. The **experimental group** or **treatment group** does take part in the program. These names come from literature in the experimental sciences, and they should not be taken literally. Except in rare cases, the choice of the control and treatment

S12

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SKIP 예제 7.2 (PC 보유와 대학 GPA) — PPT 미포함.

S12 **프로그램 평가**: 통제집단(control, 미참여) vs 처치집단(treatment, 참여).

groups is not random. However, in some cases, multiple regression analysis can be used to control for enough other factors in order to estimate the causal effect of the program.

EXAMPLE 7.3 **EFFECTS OF TRAINING GRANTS ON HOURS OF TRAINING**

Using the 1988 data for Michigan manufacturing firms in JTRAIN.RAW, we obtain the following estimated equation:

$$\widehat{hrsemp} = 46.67 + 26.25\ grant - .98\ \log(sales)$$
$$(43.41)\quad (5.59)\quad (3.54)$$
$$- 6.07\ \log(employ) \qquad (7.7)$$
$$(3.88)$$
$$n = 105,\ R^2 = .237.$$

The dependent variable is hours of training per employee, at the firm level. The variable *grant* is a dummy variable equal to one if the firm received a job training grant for 1988 and zero otherwise. The variables *sales* and *employ* represent annual sales and number of employees, respectively. We cannot enter *hrsemp* in logarithmic form because *hrsemp* is zero for 29 of the 105 firms used in the regression.

The variable *grant* is very statistically significant, with $t_{grant} = 4.70$. Controlling for sales and employment, firms that received a grant trained each worker, on average, 26.25 hours more. Because the average number of hours of per worker training in the sample is about 17, with a maximum value of 164, *grant* has a large effect on training, as is expected.

The coefficient on log(*sales*) is small and very insignificant. The coefficient on log(*employ*) means that, if a firm is 10% larger, it trains its workers about .61 hour less. Its *t* statistic is −1.56, which is only marginally statistically significant.

As with any other independent variable, we should ask whether the measured effect of a qualitative variable is causal. In equation (7.7), is the difference in training between firms that receive grants and those that do not due to the grant, or is grant receipt simply an indicator of something else? It might be that the firms receiving grants would have, on average, trained their workers more even in the absence of a grant. Nothing in this analysis tells us whether we have estimated a causal effect; we must know how the firms receiving grants were determined. We can only hope we have controlled for as many factors as possible that might be related to whether a firm received a grant and to its levels of training.

We will return to policy analysis with dummy variables in Section 7.6, as well as in later chapters.

Interpreting Coefficients on Dummy Explanatory Variables When the Dependent Variable Is log(*y*)

A common specification in applied work has the dependent variable appearing in logarithmic form, with one or more dummy variables appearing as independent variables. How do we interpret the dummy variable coefficients in this case? Not surprisingly, the coefficients have a *percentage* interpretation.

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S12 S12 S12 = 예제 7.3 (JTRAIN, n=105).

S12 S12 grant 26.25(5.59), t=4.70: 보조금 받은 기업이 직원 1인당 26.25시간 더 훈련.

S12 S12 인과인가? grant가 다른 것의 지표일 수 있음.

S12 S12 보조금이 어떻게 배정됐는지 알아야 판단 가능. 훈련에 열성적인 기업이 보조금도 받고 훈련도 더 한다면 인과 아님 (슬라이드 노트 'Enthusiasm?').

S13 S13 S13 = 이 소절.

S13 S13 종속변수가 log(y)면 더미 계수 × 100 ≈ % 차이.

S13

▶ 교수 설명 (수업 전사본 · 판서)

S12 · Training grants on hours of training

(10.27 01:01:14–01:02:51) 기업이 훈련 보조금을 받았는지 → 훈련 시간. **인과라고 말할 수 있나? 아마 아님.**

- 문제는 역시 **상관된 다른 변수**
- 예: **열의(enthusiasm)**. 훈련에 더 열성적인 기업이 보조금을 받을 가능성이 높고, 그 열의가 훈련도 늘림 → 이 효과는 인과가 아닐 수 있음

EXAMPLE 7.4 HOUSING PRICE REGRESSION

Using the data in HPRICE1.RAW, we obtain the equation

$$\widehat{\log(price)} = -1.35 + .168\log(lotsize) + .707\log(sqrft)$$
$$(.65)\ \ (.038)\qquad\qquad (.093)$$
$$+ .027\,bdrms + .054\,colonial \qquad [7.8]$$
$$(.029)\qquad\quad (.045)$$
$$n = 88,\ R^2 = .649.$$

All the variables are self-explanatory except *colonial*, which is a binary variable equal to one if the house is of the colonial style. What does the coefficient on *colonial* mean? For given levels of *lotsize*, *sqrft*, and *bdrms*, the difference in $\widehat{\log(price)}$ between a house of colonial style and that of another style is .054. This means that a colonial-style house is predicted to sell for about 5.4% more, holding other factors fixed.

This example shows that, when log(*y*) is the dependent variable in a model, the coefficient on a dummy variable, when multiplied by 100, is interpreted as the percentage difference in *y*, holding all other factors fixed. When the coefficient on a dummy variable suggests a large proportionate change in *y*, the exact percentage difference can be obtained exactly as with the semi-elasticity calculation in Section 6.2.

EXAMPLE 7.5 LOG HOURLY WAGE EQUATION

Let us reestimate the wage equation from Example 7.1, using log(*wage*) as the dependent variable and adding quadratics in *exper* and *tenure*:

$$\widehat{\log(wage)} = .417 - .297\,female + .080\,educ + .029\,exper$$
$$(.099)\ \ (.036)\qquad (.007)\qquad (.005)$$
$$- .00058\,exper^2 + .032\,tenure - .00059\,tenure^2 \qquad [7.9]$$
$$(.00010)\qquad (.007)\qquad (.00023)$$
$$n = 526,\ R^2 = .441.$$

Using the same approximation as in Example 7.4, the coefficient on *female* implies that, for the same levels of *educ*, *exper*, and *tenure*, women earn about 100(.297) = 29.7% less than men. We can do better than this by computing the exact percentage difference in predicted wages. What we want is the proportionate difference in wages between females and males, holding other factors fixed: $(\widehat{wage}_F - \widehat{wage}_M)/\widehat{wage}_M$. What we have from (7.9) is

$$\widehat{\log(wage_F)} - \widehat{\log(wage_M)} = -.297.$$

Exponentiating and subtracting one gives

$$(\widehat{wage}_F - \widehat{wage}_M)/\widehat{wage}_M = \exp(-.297) - 1 \approx -.257.$$

This more accurate estimate implies that a woman's wage is, on average, 25.7% below a comparable man's wage.

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S13

S13 **S13 = 예제 7.4** (HPRICE1, n=88).

S13

S13 colonial .054(.045).

S13

S13 colonial이 0→1이면 가격 약 **5.4%**↑. 슬라이드의 '%Δprice/%Δcolonial' 식은 부적절(더미에 % 변화는 의미 없음), 'percentage points'도 정확히는 '%'.

REF

REF 예제 7.5 (PPT 미포함, S19의 비교 기준): female −.297 → 근사 29.7%.

REF

REF 정확한 % 차이: exp(−.297) − 1 = **−25.7%**. 계수가 클수록 근사와 벌어짐.

▶ 교수 설명 (수업 전사본 · 판서)

S13 · Dummy variables in log(y) equations

(10.27 01:03:01–01:05:44) 독립변수는 log가 아니고 종속변수만 log일 때: colonial이 **0에서 1로** 바뀌면(비-콜로니얼 → 콜로니얼) 집값이 약 **5.4%** 오른다고 말할 수 있음.

- [중요] 교수 지적: **슬라이드의 식(%Δprice/%Δcolonial)은 오류.** 더미변수에 대해 '퍼센트 변화'를 말하는 건 말이 안 됨
- '0→1로 바뀔 때 5.4% 상승'이라는 설명 쪽이 맞다고 함

If we had made the same correction in Example 7.4, we would have obtained $\exp(.054) - 1 \approx .0555$, or about 5.6%. The correction has a smaller effect in Example 7.4 than in the wage example because the magnitude of the coefficient on the dummy variable is much smaller in (7.8) than in (7.9).

Generally, if $\hat{\beta}_1$ is the coefficient on a dummy variable, say x_1, when $\log(y)$ is the dependent variable, the exact percentage difference in the predicted y when $x_1 = 1$ versus when $x_1 = 0$ is

$$100 \cdot [\exp(\hat{\beta}_1) - 1]. \qquad [7.10]$$

The estimate $\hat{\beta}_1$ can be positive or negative, and it is important to preserve its sign in computing (7.10).

The logarithmic approximation has the advantage of providing an estimate between the magnitudes obtained by using each group as the base group. In particular, although equation (7.10) gives us a better estimate than $100 \cdot \hat{\beta}_1$ of the percentage by which y for $x_1 = 1$ is greater than y for $x_1 = 0$, (7.10) is not a good estimate if we switch the base group. In Example 7.5, we can estimate the percentage by which a man's wage exceeds a comparable woman's wage, and this estimate is $100 \cdot [\exp(-\hat{\beta}_1) - 1] = 100 \cdot [\exp(.297) - 1] \approx 34.6$. The approximation, based on $100 \cdot \hat{\beta}_1$, 29.7, is between 25.7 and 34.6 (and close to the middle). Therefore, it makes sense to report that "the difference in predicted wages between men and women is about 29.7%," without having to take a stand on which is the base group.

7.3 Using Dummy Variables for Multiple Categories

We can use several dummy independent variables in the same equation. For example, we could add the dummy variable *married* to equation (7.9). The coefficient on *married* gives the (approximate) proportional differential in wages between those who are and are not married, holding gender, *educ*, *exper*, and *tenure* fixed. When we estimate this model, the coefficient on *married* (with standard error in parentheses) is .053 (.041), and the coefficient on *female* becomes −.290 (.036). Thus, the "marriage premium" is estimated to be about 5.3%, but it is not statistically different from zero ($t = 1.29$). An important limitation of this model is that the marriage premium is assumed to be the same for men and women; this is relaxed in the following example.

EXAMPLE 7.6 LOG HOURLY WAGE EQUATION

Let us estimate a model that allows for wage differences among four groups: married men, married women, single men, and single women. To do this, we must select a base group; we choose single men. Then, we must define dummy variables for each of the remaining groups. Call these *marrmale*, *marrfem*, and *singfem*. Putting these three variables into (7.9) (and, of course, dropping *female*, since it is now redundant) gives

$$\begin{aligned}\widehat{\log(wage)} &= .321 + .213\,marrmale - .198\,marrfem \\ &\qquad (.100)\ \ (.055)\qquad\quad (.058) \\ &\quad - .110\,singfem + .079\,educ + .027\,exper - .00054\,exper^2 \\ &\qquad (.056)\qquad\ (.007)\qquad\ (.005)\qquad\ (.00011) \\ &\quad + .029\,tenure - .00053\,tenure^2 \\ &\qquad (.007)\qquad\ (.00023) \\ n &= 526,\ R^2 = .461.\end{aligned} \qquad [7.11]$$

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p.235 · PPT 필기 S14

REF REF 일반식: 100·[exp(β1) − 1]%. colonial: exp(.054)−1 = 5.6% (근사 5.4%와 거의 같음).

S14 S14 S14–S16 = 7.3절.

REF REF female, married를 따로 넣으면 결혼 프리미엄이 남녀 동일하다고 가정 → 4개 범주로 풀어 완화.

S14 S14 S14 = 예제 7.6.

S14 S14 4개 집단(기혼남·기혼녀·미혼남·미혼녀) → 기준집단 **미혼 남성**을 빼고 더미 3개.

S14 S14 marrmale .213, marrfem −.198, singfem −.110. n=526, R^2=.461. (슬라이드의 n=2,725, R^2=.0422는 오타 — 예제 7.12 수치.)

▶ 교수 설명 (수업 전사본 · 판서)

S14 · Dummy variables for multiple categories

(10.27 01:06:22–01:08:17) 범주가 몇 개인가? **빠진 범주 = 기준 범주**는 무엇인가? (출석 부르는 동안 생각하게 함)

- 답: 미혼 남성(single male). 나머지 조합(기혼 남성, 기혼 여성, 미혼 여성)은 모두 식에 있음

All of the coefficients, with the exception of *singfem*, have *t* statistics well above two in absolute value. The *t* statistic for *singfem* is about −1.96, which is just significant at the 5% level against a two-sided alternative.

To interpret the coefficients on the dummy variables, we must remember that the base group is single males. Thus, the estimates on the three dummy variables measure the proportionate difference in wage *relative* to single males. For example, married men are estimated to earn about 21.3% more than single men, holding levels of education, experience, and tenure fixed. [The more precise estimate from (7.10) is about 23.7%.] A married woman, on the other hand, earns a predicted 19.8% less than a single man with the same levels of the other variables.

Because the base group is represented by the intercept in (7.11), we have included dummy variables for only three of the four groups. If we were to add a dummy variable for single males to (7.11), we would fall into the dummy variable trap by introducing perfect collinearity. Some regression packages will automatically correct this mistake for you, while others will just tell you there is perfect collinearity. It is best to carefully specify the dummy variables because then we are forced to properly interpret the final model.

Even though single men is the base group in (7.11), we can use this equation to obtain the estimated difference between any two groups. Because the overall intercept is common to all groups, we can ignore that in finding differences. Thus, the estimated proportionate difference between single and married women is −.110 − (−.198) = .088, which means that single women earn about 8.8% more than married women. Unfortunately, we cannot use equation (7.11) for testing whether the estimated difference between single and married women is statistically significant. Knowing the standard errors on *marrfem* and *singfem* is not enough to carry out the test (see Section 4.4). The easiest thing to do is to choose one of these groups to be the base group and to reestimate the equation. Nothing substantive changes, but we get the needed estimate and its standard error directly. When we use married women as the base group, we obtain

$$\widehat{\log(wage)} = .123 + .411\ marrmale + .198\ singmale + .088\ singfem + \ldots,$$
$$(.106)\quad (.056)\qquad\qquad (.058)\qquad\qquad (.052)$$

where, of course, none of the unreported coefficients or standard errors have changed. The estimate on *singfem* is, as expected, .088. Now, we have a standard error to go along with this estimate. The *t* statistic for the null that there is no difference in the population between married and single women is $t_{singfem} = .088/.052 \approx 1.69$. This is marginal evidence against the null hypothesis. We also see that the estimated difference between married men and married women is very statistically significant ($t_{marrmale} = 7.34$).

The previous example illustrates a general principle for including dummy variables to indicate different groups: if the regression model is to have different intercepts for, say, *g* groups or categories, we need to include $g - 1$ dummy variables in the model along with an intercept. The intercept for the base group is the overall intercept in the model, and the dummy variable coefficient for a particular group represents the estimated difference in intercepts between that group and the base group. Including *g* dummy variables along with an intercept will result in the dummy variable trap. An alternative is to include *g* dummy variables and to exclude an overall intercept. Including *g* dummies without an overall intercept is sometimes useful, but it has two practical drawbacks. First, it makes it more cumbersome

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S14 모든 더미 계수는 **미혼 남성 대비** 비례 차이.

S14 슬라이드 주석: 기혼 여성은 미혼 남성보다 19.8% 적게 받음.

S8 미혼 남성 더미까지 넣으면 더미변수 함정.

REF 기준집단이 아닌 두 집단 비교: 계수 차이 −.110−(−.198)=.088. 유의성은 기준집단을 바꿔 재추정해야 SE를 얻음.

S14 **일반 원칙**: g개 범주 → 절편 + 더미 g−1개.

to test for differences relative to a base group. Second, regression packages usually change the way *R*-squared is computed when an overall intercept is not included. In particular, in the formula $R^2 = 1 - \text{SSR/SST}$, the total sum of squares, SST, is replaced with a total sum of squares that does not center y_i about its mean, say, $\text{SST}_0 = \sum_{i=1}^{n} y_i^2$. The resulting *R*-squared, say $R_0^2 = 1 - \text{SSR/SST}_0$, is sometimes called the **uncentered *R*-squared**. Unfortunately, R_0^2 is rarely suitable as a goodness of fit measure. It is always true that $\text{SST}_0 \geq \text{SST}$ with equality only if $\bar{y} = 0$. Often, SST_0 is much larger that SST, which means that R_0^2 is much larger than R^2. For example, if in the previous example we regress log(*wage*) on *marrmale*, *singmale*, *marrfem*, *singfem*, and the other explanatory variables—without an intercept—the reported *R*-squared from Stata, which is R_0^2, is .948. This high *R*-squared is an artifact of not centering the total sum of squares in the calculation. The correct *R*-squared is given in equation (7.11) as .461. Some regression packages, including Stata, have an option to force calculation of the centered *R*-squared even though an overall intercept has not been included, and using this option is generally a good idea. In the vast majority of cases, any *R*-squared based on comparing an SSR and SST should have SST computed by centering the y_i about $\bar{y}$. We can think of this SST as the sum of squared residuals obtained if we just use the sample average, $\bar{y}$, to predict each y_i. Surely we are setting the bar pretty low for any model if all we measure is its fit relative to using a constant predictor. For a model without an intercept that fits poorly, it is possible that SSR > SST, which means R^2 would be negative. The uncentered *R*-squared will always be between zero and one, which likely explains why it is usually the default when an intercept is not estimated in regression models.

EXPLORING FURTHER 7.2

In the baseball salary data found in MLB1.RAW, players are given one of six positions: *frstbase*, *scndbase*, *thrdbase*, *shrtstop*, *outfield*, or *catcher*. To allow for salary differentials across position, with outfielders as the base group, which dummy variables would you include as independent variables?

Incorporating Ordinal Information by Using Dummy Variables

Suppose that we would like to estimate the effect of city credit ratings on the municipal bond interest rate (*MBR*). Several financial companies, such as Moody's Investors Service and Standard and Poor's, rate the quality of debt for local governments, where the ratings depend on things like probability of default. (Local governments prefer lower interest rates in order to reduce their costs of borrowing.) For simplicity, suppose that rankings range from zero to four, with zero being the worst credit rating and four being the best. This is an example of an **ordinal variable**. Call this variable *CR* for concreteness. The question we need to address is: How do we incorporate the variable *CR* into a model to explain *MBR*?

One possibility is to just include *CR* as we would include any other explanatory variable:

$$MBR = \beta_0 + \beta_1 CR + other\ factors,$$

where we do not explicitly show what other factors are in the model. Then β_1 is the percentage point change in *MBR* when *CR* increases by one unit, holding other factors fixed. Unfortunately, it is rather hard to interpret a one-unit increase in *CR*. We know the

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p.237 · PPT 필기 S15

REF REF S9 단점 ②의 근거: 절편이 없으면 패키지가 uncentered R^2를 계산 → 부풀려짐(.948 vs 올바른 .461).

S15 S15 S15–S16 = 이 소절 (서열 정보).

S15 S15 예: 도시 신용등급 CR(0=최악 … 4=최고) → 지방채 금리 MBR. **서열변수(ordinal)**: 순서만 의미.

S15 S15 MBR = β0 + β1·CR + … : 한 등급 오를 때 효과가 항상 β1로 같다고 가정하는 셈.

▶ 교수 설명 (수업 전사본 · 판서)

S15 · Ordinal information (1/2)

(10.27 01:08:39–01:10:54) 더미가 적절한 또 다른 경우: 금리와 **신용등급**. 그런데 이 둘이 **선형 관계**인가? 아닐 수 있음.

- 그냥 "1등급, 2등급, 3등급"일 뿐인 관계일 수 있음

quantitative meaning of another year of education, or another dollar spent per student, but things like credit ratings typically have only ordinal meaning. We know that a *CR* of four is better than a *CR* of three, but is the difference between four and three the same as the difference between one and zero? If not, then it might not make sense to assume that a one-unit increase in *CR* has a constant effect on *MBR*.

A better approach, which we can implement because *CR* takes on relatively few values, is to define dummy variables for each value of *CR*. Thus, let $CR_1 = 1$ if $CR = 1$, and $CR_1 = 0$ otherwise; $CR_2 = 1$ if $CR = 2$, and $CR_2 = 0$ otherwise; and so on. Effectively, we take the single credit rating and turn it into five categories. Then, we can estimate the model

$$MBR = \beta_0 + \delta_1 CR_1 + \delta_2 CR_2 + \delta_3 CR_3 + \delta_4 CR_4 + other\ factors. \quad (7.12)$$

Following our rule for including dummy variables in a model, we include four dummy variables because we have five categories. The omitted category here is a credit rating of zero, and so it is the base group. (This is why we do not need to define a dummy variable for this category.) The coefficients are easy to interpret: δ_1 is the difference in *MBR* (other factors fixed) between a municipality with a credit rating of one and a municipality with a credit rating of zero; δ_2 is the difference in *MBR* between a municipality with a credit rating of two and a municipality with a credit rating of zero; and so on. The movement between each credit rating is allowed to have a different effect, so using (7.12) is much more flexible than simply putting *CR* in as a single variable. Once the dummy variables are defined, estimating (7.12) is straightforward.

EXPLORING FURTHER 7.3

In model (7.12), how would you test the null hypothesis that credit rating has no effect on *MBR*?

Equation (7.12) contains the model with a constant partial effect as a special case. One way to write the three restrictions that imply a constant partial effect is $\delta_2 = 2\delta_1$, $\delta_3 = 3\delta_1$, and $\delta_4 = 4\delta_1$. When we plug these into equation (7.12) and rearrange, we get $MBR = \beta_0 + \delta_1(CR_1 + 2CR_2 + 3CR_3 + 4CR_4) + other\ factors$. Now, the term multiplying δ_1 is simply the original credit rating variable, *CR*. To obtain the *F* statistic for testing the constant partial effect restrictions, we obtain the unrestricted *R*-squared from (7.12) and the restricted *R*-squared from the regression of *MBR* on *CR* and the other factors we have controlled for. The *F* statistic is obtained as in equation (4.41) with $q = 3$.

EXAMPLE 7.7 EFFECTS OF PHYSICAL ATTRACTIVENESS ON WAGE

Hamermesh and Biddle (1994) used measures of physical attractiveness in a wage equation. (The file BEAUTY.RAW contains fewer variables but more observations than used by Hamermesh and Biddle. See Computer Exercise C12.) Each person in the sample was ranked by an interviewer for physical attractiveness, using five categories (homely, quite plain, average, good looking, and strikingly beautiful or handsome). Because there are so few people at the two extremes, the authors put people into one of three groups for the regression analysis: average, below average, and above average, where the base group is *average*. Using data from the 1977 Quality of Employment Survey, after controlling for the usual productivity characteristics, Hamermesh and Biddle estimated an equation for men:

$$\widehat{\log(wage)} = \hat{\beta}_0 - .164\ belavg + .016\ abvavg + other\ factors$$
$$(.046) \qquad (.033)$$
$$n = 700, \bar{R}^2 = .403$$

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S15

S15 4와 3의 차이가 1과 0의 차이와 같다는 보장이 없음 → 선형 가정이 부적절할 수 있음.

S16

S16 **해법**: 등급 값마다 더미(CR1…CR4).

S16

S16 MBR = β0 + δ1CR1 + δ2CR2 + δ3CR3 + δ4CR4 + ⋯. 5개 범주 → 더미 4개.

S16

S16 기준 = 등급 0(최악). δj = 등급 j와 등급 0의 MBR 차이.

SKIP

SKIP 예제 7.7 (외모와 임금) — PPT 미포함.

S16

S16 등급 간 이동 효과가 제각각이어도 됨 → CR 하나를 넣는 것보다 유연.

REF

REF 선형 가정 검정: δ2=2δ1, δ3=3δ1, δ4=4δ1 (제약 3개) → F검정. CR 하나짜리 모형이 제약모형.

▶ 교수 설명 (수업 전사본 · 판서)

S16 · Ordinal information (2/2)

(10.27 01:10:58–01:11:47) 선형 관계가 아니라는 증거가 있다면 → **등급마다 더미변수**를 만든다 (1등급, 2등급, ⋯, 4등급).

SKIP

and an equation for women:

$$\widehat{\log(wage)} = \hat{\beta}_0 - .124\ belavg + .035\ abvavg + other\ factors$$
$$(.066) \qquad (.049)$$
$$n = 409, \bar{R}^2 = .330.$$

The other factors controlled for in the regressions include education, experience, tenure, marital status, and race; see Table 3 in Hamermesh and Biddle's paper for a more complete list. In order to save space, the coefficients on the other variables are not reported in the paper and neither is the intercept.

For men, those with below average looks are estimated to earn about 16.4% less than an average-looking man who is the same in other respects (including education, experience, tenure, marital status, and race). The effect is statistically different from zero, with $t = -3.57$. Similarly, men with above average looks earn an estimated 1.6% more, although the effect is not statistically significant ($t < .5$).

A woman with below average looks earns about 12.4% less than an otherwise comparable average-looking woman, with $t = -1.88$. As was the case for men, the estimate on *abvavg* is not statistically different from zero.

In related work, Biddle and Hamermesh (1998) revisit the effects of looks on earnings using a more homogeneous group: graduates of a particular law school. The authors continue to find that physical appearance has an effect on annual earnings, something that is perhaps not too surprising among people practicing law.

REF

In some cases, the ordinal variable takes on too many values so that a dummy variable cannot be included for each value. For example, the file LAWSCH85.RAW contains data on median starting salaries for law school graduates. One of the key explanatory variables is the rank of the law school. Because each law school has a different rank, we clearly cannot include a dummy variable for each rank. If we do not wish to put the rank directly in the equation, we can break it down into categories. The following example shows how this is done.

SKIP

EXAMPLE 7.8 **EFFECTS OF LAW SCHOOL RANKINGS ON STARTING SALARIES**

Define the dummy variables *top10*, *r11_25*, *r26_40*, *r41_60*, *r61_100* to take on the value unity when the variable *rank* falls into the appropriate range. We let schools ranked below 100 be the base group. The estimated equation is

$$\widehat{\log(salary)} = 9.17 + .700\ top10 + .594\ r11_25 + .375\ r26_40$$
$$(.41)\ (.053) \qquad (.039) \qquad (.034)$$
$$+ .263\ r41_60 + .132\ r61_100 + .0057\ LSAT$$
$$(.028) \qquad (.021) \qquad (.0031) \qquad [7.13]$$
$$+ .014\ GPA + .036 \log(libvol) + .0008 \log(cost)$$
$$(.074) \qquad (.026) \qquad (.0251)$$
$$n = 136, R^2 = .911, \bar{R}^2 = .905.$$

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REF

REF 값이 너무 많은 서열변수(예: 로스쿨 순위)는 구간으로 묶어 더미화.

SKIP

SKIP 예제 7.8 (로스쿨 순위와 초봉) — PPT 미포함.

SKIP

We see immediately that all of the dummy variables defining the different ranks are very statistically significant. The estimate on *r61_100* means that, holding *LSAT*, *GPA*, *libvol*, and *cost* fixed, the median salary at a law school ranked between 61 and 100 is about 13.2% higher than that at a law school ranked below 100. The difference between a top 10 school and a below 100 school is quite large. Using the exact calculation given in equation (7.10) gives exp(.700) − 1 ≈ 1.014, and so the predicted median salary is more than 100% higher at a top 10 school than it is at a below 100 school.

As an indication of whether breaking the rank into different groups is an improvement, we can compare the adjusted *R*-squared in (7.13) with the adjusted *R*-squared from including *rank* as a single variable: the former is .905 and the latter is .836, so the additional flexibility of (7.13) is warranted.

Interestingly, once the rank is put into the (admittedly somewhat arbitrary) given categories, all of the other variables become insignificant. In fact, a test for joint significance of *LSAT, GPA,* log(*libvol*), and log(*cost*) gives a *p*-value of .055, which is borderline significant. When *rank* is included in its original form, the *p*-value for joint significance is zero to four decimal places.

One final comment about this example: In deriving the properties of ordinary least squares, we assumed that we had a random sample. The current application violates that assumption because of the way *rank* is defined: a school's rank necessarily depends on the rank of the other schools in the sample, and so the data cannot represent independent draws from the population of all law schools. This does not cause any serious problems provided the error term is uncorrelated with the explanatory variables.

7.4 Interactions Involving Dummy Variables

S17

Interactions among Dummy Variables

OBJ

Just as variables with quantitative meaning can be interacted in regression models, so can dummy variables. We have effectively seen an example of this in Example 7.6, where we defined four categories based on marital status and gender. In fact, we can recast that model by adding an **interaction term** between *female* and *married* to the model where *female* and *married* appear separately. This allows the marriage premium to depend on gender, just as it did in equation (7.11). For purposes of comparison, the estimated model with the *female-married* interaction term is

$$\widehat{\log(wage)} = .321 - .110\,female + .213\,married \\ (.100)\quad(.056)\qquad(.055) \\ -.301\,female{\cdot}married + \ldots, \\ (.072) \qquad [7.14]$$

where the rest of the regression is necessarily identical to (7.11). Equation (7.14) shows explicitly that there is a statistically significant interaction between gender and marital status. This model also allows us to obtain the estimated wage differential among all four groups, but here we must be careful to plug in the correct combination of zeros and ones.

Setting *female* = 0 and *married* = 0 corresponds to the group single men, which is the base group, since this eliminates *female*, *married*, and *female·married*. We can find

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S17 S17–S24 = 7.4절.

OBJ 학습목표 7.2 (더미 간 교호작용) — 슬라이드 본문엔 없음. 단 S14의 4범주 모형(7.11)과 같은 모형의 다른 표기: female·married 계수 −.301 = 결혼 프리미엄의 남녀 차이. 유의 → 결혼 효과가 성별에 따라 다름.

▶ 교수 설명 (수업 전사본 · 판서)

OBJ-7.2 · 학습목표 7.2 Interactions among dummies

수업에서 **언급 없음** (10.27 전사본, 10.29 PPT 모두).

- 다만 S14의 4범주 모형이 같은 내용의 다른 표기 → 개념은 S14로 충분
- 학습목표 7.2에 명시돼 있으므로 'female·married 계수의 해석'은 교재 p.240–241로 확인

the intercept for married men by setting *female* = 0 and *married* = 1 in (7.14); this gives an intercept of .321 + .213 = .534, and so on.

Equation (7.14) is just a different way of finding wage differentials across all gender–marital status combinations. It allows us to easily test the null hypothesis that the gender differential does not depend on marital status (equivalently, that the marriage differential does not depend on gender). Equation (7.11) is more convenient for testing for wage differentials between any group and the base group of single men.

EXAMPLE 7.9 EFFECTS OF COMPUTER USAGE ON WAGES

Krueger (1993) estimates the effects of computer usage on wages. He defines a dummy variable, which we call *compwork*, equal to one if an individual uses a computer at work. Another dummy variable, *comphome*, equals one if the person uses a computer at home. Using 13,379 people from the 1989 Current Population Survey, Krueger (1993, Table 4) obtains

$$\widehat{\log(wage)} = \hat{\beta}_0 + \underset{(.009)}{.177}\, compwork + \underset{(.019)}{.070}\, comphome + \underset{(.023)}{.017}\, compwork{\cdot}comphome + other\ factors. \qquad [7.15]$$

(The other factors are the standard ones for wage regressions, including education, experience, gender, and marital status; see Krueger's paper for the exact list.) Krueger does not report the intercept because it is not of any importance; all we need to know is that the base group consists of people who do not use a computer at home or at work. It is worth noticing that the estimated return to using a computer at work (but not at home) is about 17.7%. (The more precise estimate is 19.4%.) Similarly, people who use computers at home but not at work have about a 7% wage premium over those who do not use a computer at all. The differential between those who use a computer at both places, relative to those who use a computer in neither place, is about 26.4% (obtained by adding all three coefficients and multiplying by 100), or the more precise estimate 30.2% obtained from equation (7.10).

The interaction term in (7.15) is not statistically significant, nor is it very big economically. But it is causing little harm by being in the equation.

Allowing for Different Slopes

We have now seen several examples of how to allow different intercepts for any number of groups in a multiple regression model. There are also occasions for interacting dummy variables with explanatory variables that are not dummy variables to allow for a **difference in slopes**. Continuing with the wage example, suppose that we wish to test whether the return to education is the same for men and women, allowing for a constant wage differential between men and women (a differential for which we have already found evidence). For simplicity, we include only education and gender in the model. What kind of model allows for different returns to education? Consider the model

$$\log(wage) = (\beta_0 + \delta_0 female) + (\beta_1 + \delta_1 female)educ + u. \qquad [7.16]$$

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p.241 · PPT 필기 S17

OBJ

OBJ 절편 계산: 기혼 남성 .321+.213=.534. (7.14)는 '교호항=0' 검정에, (7.11)은 기준집단 대비 비교에 편리.

SKIP

SKIP 예제 7.9 (컴퓨터 사용과 임금) — PPT 미포함.

S17

S17 S17: 더미 × 연속변수 = 기울기 차이.

S17

S17 질문: 교육 수익률이 남녀 같은가?

S17

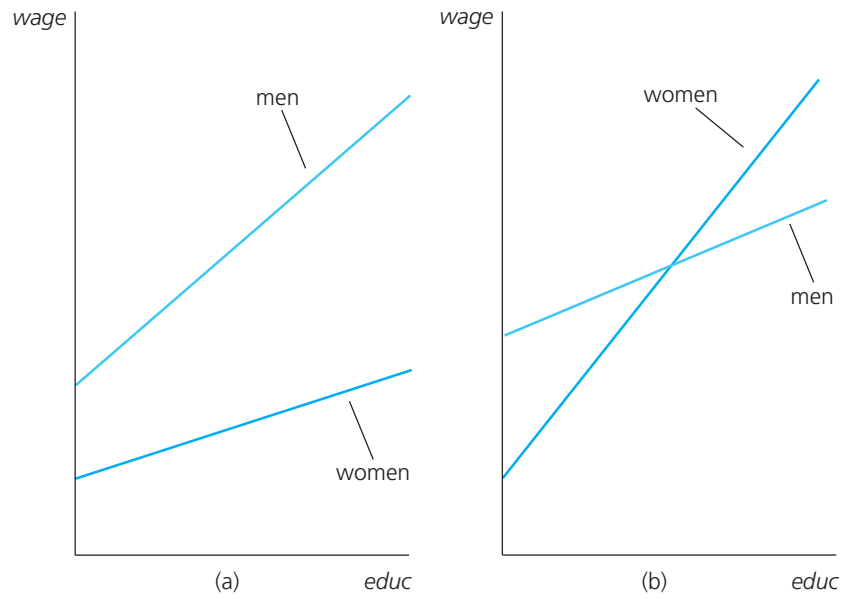
S17 log(wage) = (β0 + δ0·female) + (β1 + δ1·female)·educ + u. 남성: 절편 β0, 기울기 β1 / 여성: 절편 β0+δ0, 기울기 β1+δ1.

▶ 교수 설명 (수업 전사본 · 판서)

S17 · Interactions: allowing different slopes

(10.27 01:11:55–01:12:12) 교호항은 이미 많이 다뤘음. 이번엔 **더미변수 × 교육**의 교호항이 어떤 효과를 내는지.

FIGURE 7.2 Graphs of equation (7.16): (a) $\delta_0 < 0$, $\delta_1 < 0$; (b) $\delta_0 < 0$, $\delta_1 > 0$.



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If we plug *female* = 0 into (7.16), then we find that the intercept for males is β_0, and the slope on education for males is β_1. For females, we plug in *female* = 1; thus, the intercept for females is $\beta_0 + \delta_0$, and the slope is $\beta_1 + \delta_1$. Therefore, δ_0 measures the difference in intercepts between women and men, and δ_1 measures the difference in the return to education between women and men. Two of the four cases for the signs of δ_0 and δ_1 are presented in Figure 7.2.

Graph (a) shows the case where the intercept for women is below that for men, and the slope of the line is smaller for women than for men. This means that women earn less than men at all levels of education, and the gap increases as *educ* gets larger. In graph (b), the intercept for women is below that for men, but the slope on education is larger for women. This means that women earn less than men at low levels of education, but the gap narrows as education increases. At some point, a woman earns more than a man, given the same levels of education (and this point is easily found given the estimated equation).

How can we estimate model (7.16)? To apply OLS, we must write the model with an interaction between *female* and *educ*:

$$\log(wage) = \beta_0 + \delta_0 female + \beta_1 educ + \delta_1 female \cdot educ + u. \qquad [7.17]$$

The parameters can now be estimated from the regression of log(*wage*) on *female*, *educ*, and *female*·*educ*. Obtaining the interaction term is easy in any regression package. Do not be daunted by the odd nature of *female*·*educ*, which is zero for any man in the sample and equal to the level of education for any woman in the sample.

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S17

S17 δ0 = 절편 차이, δ1 = 기울기(교육 수익률) 차이.

S18

S18 그림 7.2(a) δ0<0, δ1<0: 모든 교육 수준에서 여성이 낮고 격차가 확대.

S18

S18 (b) δ0<0, δ1>0: 격차가 축소되다가 교차점 이후 역전. 교차점 educ* = −δ0/δ1.

S17

S17 추정용 표기: female, educ, **female·educ**(교호항)에 회귀. 교호항 = 여성이면 educ, 남성이면 0.

REF

▶ 교수 설명 (수업 전사본 · 판서)

S18 · Graphs: different intercepts and slopes

(10.27 01:15:04–01:15:52) 남녀가 **절편도 다르고 기울기도 다른** 그래프가 가능.

- (a) 교육을 더 받을수록 임금 격차가 **커지는** 경우
- (b) 격차가 **사라지고 여성이 더 잘 벌기 시작**하는 경우
- 여기서 10.27 수업 종료

An important hypothesis is that the return to education is the same for women and men. In terms of model (7.17), this is stated as H_0: $\delta_1 = 0$, which means that the slope of log(*wage*) with respect to *educ* is the same for men and women. Note that this hypothesis puts no restrictions on the difference in intercepts, δ_0. A wage differential between men and women is allowed under this null, but it must be the same at all levels of education. This situation is described by Figure 7.1.

We are also interested in the hypothesis that average wages are identical for men and women who have the same levels of education. This means that δ_0 and δ_1 must *both* be zero under the null hypothesis. In equation (7.17), we must use an *F* test to test H_0: $\delta_0 = 0$, $\delta_1 = 0$. In the model with just an intercept difference, we reject this hypothesis because H_0: $\delta_0 = 0$ is soundly rejected against H_1: $\delta_0 < 0$.

EXAMPLE 7.10 LOG HOURLY WAGE EQUATION

We add quadratics in experience and tenure to (7.17):

$$\widehat{\log(wage)} = .389 - .227\,female + .082\,educ$$
$$(.119)\ \ (.168)\qquad\quad (.008)$$
$$- .0056\,female{\cdot}educ + .029\,exper - .00058\,exper^2$$
$$(.0131)\qquad\quad (.005)\qquad\quad (.00011) \qquad [7.18]$$
$$+ .032\,tenure - .00059\,tenure^2$$
$$(.007)\qquad\quad (.00024)$$
$$n = 526,\ R^2 = .441.$$

The estimated return to education for men in this equation is .082, or 8.2%. For women, it is .082 − .0056 = .0764, or about 7.6%. The difference, −.56%, or just over one-half a percentage point less for women, is not economically large nor statistically significant: the *t* statistic is −.0056/.0131 ≈ −.43. Thus, we conclude that there is no evidence against the hypothesis that the return to education is the same for men and women.

The coefficient on *female*, while remaining economically large, is no longer significant at conventional levels ($t = -1.35$). Its coefficient and *t* statistic in the equation without the interaction were −.297 and −8.25, respectively [see equation (7.9)]. Should we now conclude that there is no statistically significant evidence of lower pay for women at the same levels of *educ*, *exper*, and *tenure*? This would be a serious error. Because we have added the interaction *female*·*educ* to the equation, the coefficient on *female* is now estimated much less precisely than it was in equation (7.9): the standard error has increased by almost fivefold (.168/.036 ≈ 4.67). This occurs because *female* and *female*·*educ* are highly correlated in the sample. In this example, there is a useful way to think about the multicollinearity: in equation (7.17) and the more general equation estimated in (7.18), δ_0 measures the wage differential between women and men when *educ* = 0. Very few people in the sample have very low levels of education, so it is not surprising that we have a difficult time estimating the differential at *educ* = 0 (nor is the differential at zero years of education very informative). More interesting would be to estimate the gender differential at, say, the average education level in the sample (about 12.5). To do this, we would replace *female*·*educ* with *female*·(*educ* − 12.5) and rerun the regression; this only changes the coefficient on *female* and its standard error. (See Computer Exercise C7.)

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p.243 · PPT 필기 S17 S19

S17 S17 가설 ① H0: δ1=0 — 교육 수익률이 남녀 동일(t검정). 절편 차이는 허용.

S17 S17 가설 ② H0: δ0=0, δ1=0 — 임금식 전체가 남녀 동일 → F검정.

S19 S19 = 예제 7.10.

S19 S19 female −.227(.168), female·educ −.0056(.0131). 교호항 t = −.43 → 교육 수익률 차이의 증거 없음 (남 8.2%, 여 7.6%).

S19 female의 t = −1.35(비유의). **그렇다고 '여성 저임금 증거 없음'이라 하면 심각한 오류.**

S19 S19 이유 ① female과 female·educ가 강하게 상관 → SE가 .036 → .168 (약 4.67배).

S19 S19 이유 ② δ0는 educ=0일 때의 격차 — 표본에 거의 없는 지점.

S19 S19 해법: 재중심화. female·(educ−12.5)로 바꾸면 female 계수 = 평균 교육 수준에서의 격차 (슬라이드 노트).

▶ 교수 설명 (수업 전사본 · 판서)

S19 · Estimated wage eq. with interaction term

(10.29 00:00–01:08, 녹음 시작 지점) 슬라이드의 질문 '여성 저임금의 유의한 증거가 없다는 뜻인가?'에 대한 답: "Collinearity is hiding it"

- 교호항들은 서로 상관되어 **표준오차를 키워** 효과를 숨길 수 있음
- female, female×(변수)들은 모두 상관 → 그 때문에 비유의하게 나올 수 있음
- (앞부분 녹음 없음 → 슬라이드 노트: 이 계수는 educ=0에서의 효과일 뿐. educ=12.5(평균) 기준으로 재중심화해야 함)

If we compute the F statistic for H_0: $\delta_0 = 0, \delta_1 = 0$, we obtain $F = 34.33$, which is a huge value for an F random variable with numerator $df = 2$ and denominator $df = 518$: the p-value is zero to four decimal places. In the end, we prefer model (7.9), which allows for a constant wage differential between women and men.

S19

EXPLORING FURTHER 7.4

How would you augment the model estimated in (7.18) to allow the return to *tenure* to differ by gender?

SKIP

As a more complicated example involving interactions, we now look at the effects of race and city racial composition on major league baseball player salaries.

EXAMPLE 7.11 EFFECTS OF RACE ON BASEBALL PLAYER SALARIES

Using MLB1.RAW, the following equation is estimated for the 330 major league baseball players for which city racial composition statistics are available. The variables *black* and *hispan* are binary indicators for the individual players. (The base group is white players.) The variable *percblck* is the percentage of the team's city that is black, and *perchisp* is the percentage of Hispanics. The other variables measure aspects of player productivity and longevity. Here, we are interested in race effects after controlling for these other factors.

In addition to including *black* and *hispan* in the equation, we add the interactions *black*·*percblck* and *hispan*·*perchisp*. The estimated equation is

$$\begin{aligned}\widehat{\log(salary)} &= 10.34 + .0673\,years + .0089\,gamesyr \\ &\quad (2.18)\ (.0129)\qquad (.0034) \\ &\quad + .00095\,bavg + .0146\,hrunsyr + .0045\,rbisyr \\ &\quad (.00151)\qquad (.0164)\qquad (.0076) \\ &\quad + .0072\,runsyr + .0011\,fldperc + .0075\,allstar \\ &\quad (.0046)\qquad (.0021)\qquad (.0029) \\ &\quad - .198\,black - .190\,hispan + .0125\,black{\cdot}percblck \\ &\quad (.125)\qquad (.153)\qquad (.0050) \\ &\quad + .0201\,hispan{\cdot}perchisp \\ &\quad (.0098) \\ n &= 330,\ R^2 = .638.\end{aligned} \tag{7.19}$$

First, we should test whether the four race variables, *black*, *hispan*, *black*·*percblck*, and *hispan*·*perchisp*, are jointly significant. Using the same 330 players, the R-squared when the four race variables are dropped is .626. Since there are four restrictions and $df = 330 - 13$ in the unrestricted model, the F statistic is about 2.63, which yields a p-value of .034. Thus, these variables are jointly significant at the 5% level (though not at the 1% level).

How do we interpret the coefficients on the race variables? In the following discussion, all productivity factors are held fixed. First, consider what happens for black players, holding *perchisp* fixed. The coefficient −.198 on *black* literally means that, if a black player is in a city with no blacks (*percblck* = 0), then the black player earns about 19.8%

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S19 결합검정 H0: δ0=0, δ1=0 → **F=34.33**, p≈0. 개별 t는 비유의여도 결합하면 강하게 기각.

SKIP 예제 7.11 (인종과 야구선수 연봉) — PPT 미포함.

SKIP

less than a comparable white player. As *percblck* increases—which means the white population decreases, since *perchisp* is held fixed—the salary of blacks increases relative to that for whites. In a city with 10% blacks, log(*salary*) for blacks compared to that for whites is $-.198 + .0125(10) = -.073$, so salary is about 7.3% less for blacks than for whites in such a city. When $percblck = 20$, blacks earn about 5.2% more than whites. The largest percentage of blacks in a city is about 74% (Detroit).

Similarly, Hispanics earn less than whites in cities with a low percentage of Hispanics. But we can easily find the value of *perchisp* that makes the differential between whites and Hispanics equal zero: it must make $-.190 + .0201\ perchisp = 0$, which gives $perchisp \approx 9.45$. For cities in which the percentage of Hispanics is less than 9.45%, Hispanics are predicted to earn less than whites (for a given black population), and the opposite is true if the percentage of Hispanics is above 9.45%. Twelve of the 22 cities represented in the sample have Hispanic populations that are less than 9.45% of the total population. The largest percentage of Hispanics is about 31%.

How do we interpret these findings? We cannot simply claim discrimination exists against blacks and Hispanics, because the estimates imply that whites earn less than blacks and Hispanics in cities heavily populated by minorities. The importance of city composition on salaries might be due to player preferences: perhaps the best black players live disproportionately in cities with more blacks and the best Hispanic players tend to be in cities with more Hispanics. The estimates in (7.19) allow us to determine that some relationship is present, but we cannot distinguish between these two hypotheses.

Testing for Differences in Regression Functions across Groups

The previous examples illustrate that interacting dummy variables with other independent variables can be a powerful tool. Sometimes, we wish to test the null hypothesis that two populations or groups follow the same regression function, against the alternative that one or more of the slopes differ across the groups. We will also see examples of this in Chapter 13, when we discuss pooling different cross sections over time.

Suppose we want to test whether the same regression model describes college grade point averages for male and female college athletes. The equation is

$$cumgpa = \beta_0 + \beta_1 sat + \beta_2 hsperc + \beta_3 tothrs + u,$$

where *sat* is SAT score, *hsperc* is high school rank percentile, and *tothrs* is total hours of college courses. We know that, to allow for an intercept difference, we can include a dummy variable for either males or females. If we want any of the slopes to depend on gender, we simply interact the appropriate variable with, say, *female*, and include it in the equation.

If we are interested in testing whether there is *any* difference between men and women, then we must allow a model where the intercept and all slopes can be different across the two groups:

$$cumgpa = \beta_0 + \delta_0 female + \beta_1 sat + \delta_1 female \cdot sat + \beta_2 hsperc + \delta_2 female \cdot hsperc + \beta_3 tothrs + \delta_3 female \cdot tothrs + u. \quad (7.20)$$

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S20

S20 **S20–S24 = 이 소절** (집단 간 회귀함수 차이 검정).

S20

S20 H0: 두 집단이 같은 회귀함수를 따른다.

S20

S20 **비제약모형**: female 더미 + 모든 변수와의 교호항(female·sat, female·hsperc, female·tothrs). cumgpa=누적 GPA, sat, hsperc=고교 석차 백분위, tothrs=총 이수시간.

▶ 교수 설명 (수업 전사본 · 판서)

S20 · Differences across groups (1/3) unrestricted

(10.29 01:10–02:08) 귀무가설이 참이면 female이 들어간 항이 **전부 0** → 지금 보는 비제약모형이 단순해져 **제약모형**이 됨.

The parameter δ_0 is the difference in the intercept between women and men, δ_1 is the slope difference with respect to *sat* between women and men, and so on. The null hypothesis that *cumgpa* follows the same model for males and females is stated as

$$H_0: \delta_0 = 0, \delta_1 = 0, \delta_2 = 0, \delta_3 = 0. \qquad [7.21]$$

If one of the δ_j is different from zero, then the model is different for men and women.

Using the spring semester data from the file GPA3.RAW, the full model is estimated as

$$\widehat{cumgpa} = 1.48 - .353\,female + .0011\,sat + .00075\,female{\cdot}sat$$
$$(0.21)\ (.411)\qquad (.0002)\qquad (.00039)$$
$$-.0085\,hsperc - .00055\,female{\cdot}hsperc + .0023\,tothrs$$
$$(.0014)\qquad (.00316)\qquad (.0009) \qquad [7.22]$$
$$-.00012\,female{\cdot}tothrs$$
$$(.00163)$$
$$n = 366,\ R^2 = .406,\ \bar{R}^2 = .394.$$

None of the four terms involving the female dummy variable is very statistically significant; only the *female·sat* interaction has a *t* statistic close to two. But we know better than to rely on the individual *t* statistics for testing a joint hypothesis such as (7.21). To compute the *F* statistic, we must estimate the restricted model, which results from dropping *female* and all of the interactions; this gives an R^2 (the restricted R^2) of about .352, so the *F* statistic is about 8.14; the *p*-value is zero to five decimal places, which causes us to soundly reject (7.21). Thus, men and women athletes do follow different GPA models, even though each term in (7.22) that allows women and men to be different is individually insignificant at the 5% level.

The large standard errors on *female* and the interaction terms make it difficult to tell exactly how men and women differ. We must be very careful in interpreting equation (7.22) because, in obtaining differences between women and men, the interaction terms must be taken into account. If we look only at the *female* variable, we would wrongly conclude that *cumgpa* is about .353 less for women than for men, holding other factors fixed. This is the estimated difference only when *sat*, *hsperc*, and *tothrs* are all set to zero, which is not close to being a possible scenario. At $sat = 1{,}100$, $hsperc = 10$, and $tothrs = 50$, the predicted *difference* between a woman and a man is $-.353 + .00075(1{,}100) - .00055(10) - .00012(50) \approx .461$. That is, the female athlete is predicted to have a GPA that is almost one-half a point higher than the comparable male athlete.

In a model with three variables, *sat*, *hsperc*, and *tothrs*, it is pretty simple to add all of the interactions to test for group differences. In some cases, many more explanatory variables are involved, and then it is convenient to have a different way to compute the statistic. It turns out that the sum of squared residuals form of the *F* statistic can be computed easily even when many independent variables are involved.

In the general model with *k* explanatory variables and an intercept, suppose we have two groups; call them $g = 1$ and $g = 2$. We would like to test whether the intercept and all slopes are the same across the two groups. Write the model as

$$y = \beta_{g,0} + \beta_{g,1}x_1 + \beta_{g,2}x_2 + \ldots + \beta_{g,k}x_k + u, \qquad [7.23]$$

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S21 **H0: δ0=δ1=δ2=δ3=0** — female이 들어간 항 전부 0 ⇒ 비제약모형이 제약모형(남녀 공통 회귀)으로 축소.

S22 추정 결과 (GPA3, n=366, R^2=.406).

S22 female 항 4개 모두 개별적으로 비유의(슬라이드의 동그라미 친 SE). 결합가설을 개별 t로 판단하면 안 됨.

S22 R^2 형태: 제약 R^2=.352 → F≈8.14, p≈0 → **기각**. 남녀 운동선수의 GPA 모형은 다름.

REF female 계수 −.353은 sat=hsperc=tothrs=0일 때의 차이(무의미). 현실적 값에선 여성이 약 .461 높음.

S23 변수가 많으면 교호항을 다 만드는 대신 SSR 형태로 계산.

S23 집단 g=1, 2 각각의 회귀식. 제약 수 = k+1.

▶ 교수 설명 (수업 전사본 · 판서)

S21 · Differences across groups (2/3) restricted · H0

(10.29 02:08–03:38) 비제약모형을 제약모형에 대해 검정 → 제약모형이 참인지 확인.

- **[시험] 함정 질문**: "귀무가설 하에서 참인 모형은 제약모형인가 비제약모형인가?" → 둘 다 참
- 이유: δ0=δ1=…=0이면 비제약모형이 곧 제약모형이 되므로, H0 하에서는 두 모형 모두 성립
- 우리가 확인하려는 것은 **제약모형도 참인지**

S22 · Differences across groups (3/3) estimates

(10.29 03:44–05:53) 계수 4개와 표준오차 4개만 보면 **다중공선성** 때문에 비유의한 결과가 나올 수 있음.

- 그래서 H0(모든 계수 0)를 **F검정**(전사본 'S test')으로: 제약·비제약 모형의 **잔차제곱합**을 비교
- 함께 검정하면 → 귀무가설 **기각**. 남녀가 같지 않다는 유의한 증거
- (교수는 'equal pay'라고 말했지만 이 예제의 종속변수는 GPA — S19 임금 예와 섞어 말한 것으로 보임)

for $g = 1$ and $g = 2$. The hypothesis that each beta in (7.23) is the same across the two groups involves $k + 1$ restrictions (in the GPA example, $k + 1 = 4$). The unrestricted model, which we can think of as having a group dummy variable and k interaction terms in addition to the intercept and variables themselves, has $n - 2(k + 1)$ degrees of freedom. [In the GPA example, $n - 2(k + 1) = 366 - 2(4) = 358$.] So far, there is nothing new. The key insight is that the sum of squared residuals from the unrestricted model can be obtained from two *separate* regressions, one for each group. Let SSR_1 be the sum of squared residuals obtained estimating (7.23) for the first group; this involves n_1 observations. Let SSR_2 be the sum of squared residuals obtained from estimating the model using the second group (n_2 observations). In the previous example, if group 1 is females, then $n_1 = 90$ and $n_2 = 276$. Now, the sum of squared residuals for the unrestricted model is simply $SSR_{ur} = SSR_1 + SSR_2$. The restricted sum of squared residuals is just the SSR from pooling the groups and estimating a single equation, say SSR_P. Once we have these, we compute the F statistic as usual:

$$F = \frac{[SSR_P - (SSR_1 + SSR_2)]}{SSR_1 + SSR_2} \cdot \frac{[n - 2(k + 1)]}{k + 1}, \qquad [7.24]$$

where n is the *total* number of observations. This particular F statistic is usually called the **Chow statistic** in econometrics. Because the Chow test is just an F test, it is only valid under homoskedasticity. In particular, under the null hypothesis, the error variances for the two groups must be equal. As usual, normality is not needed for asymptotic analysis.

To apply the Chow statistic to the GPA example, we need the SSR from the regression that pooled the groups together: this is $SSR_P = 85.515$. The SSR for the 90 women in the sample is $SSR_1 = 19.603$, and the SSR for the men is $SSR_2 = 58.752$. Thus, $SSR_{ur} = 19.603 + 58.752 = 78.355$. The F statistic is $[(85.515 - 78.355)/78.355](358/4) \approx 8.18$]; of course, subject to rounding error, this is what we get using the R-squared form of the test in the models with and without the interaction terms. (A word of caution: there is no simple R-squared form of the test if separate regressions have been estimated for each group; the R-squared form of the test can be used only if interactions have been included to create the unrestricted model.)

One important limitation of the traditional Chow test, regardless of the method used to implement it, is that the null hypothesis allows for no differences at all between the groups. In many cases, it is more interesting to allow for an intercept difference between the groups and then to test for slope differences; we saw one example of this in the wage equation in Example 7.10. There are two ways to allow the intercepts to differ under the null hypothesis. One is to include the group dummy and all interaction terms, as in equation (7.22), but then test joint significance of the interaction terms only. The second approach, which produces an identical statistic, is to form a sum-of-squared-residuals F statistic, as in equation (7.24), but where the restricted SSR, called "SSR_P" in equation (7.24), is obtained using a regression that contains only an intercept shift. Because we are testing k restrictions, rather than $k + 1$, the F statistic becomes

$$F = \frac{[SSR_P - (SSR_1 + SSR_2)]}{SSR_1 + SSR_2} \cdot \frac{[n - 2(k + 1)]}{k}.$$

Using this approach in the GPA example, SSR_P is obtained from the regression *cumgpa* on *female*, *sat*, *hsperc*, and *tothrs* using the data for both male and female student-athletes.

Because there are relatively few explanatory variables in the GPA example, it is easy to estimate (7.20) and test $H_0: \delta_1 = 0, \delta_2 = 0, \delta_3 = 0$ (with δ_0 unrestricted under the null).

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S23 **S23 핵심**: 비제약 SSR = 집단별로 따로 돌린 두 회귀의 SSR 합. **SSR_ur = SSR1 + SSR2** (판서).

S24 **S24** 제약 SSR = 두 집단을 합쳐(pooled) 한 번 돌린 회귀의 SSR_P.

S23 **S23 Chow 통계량** F = {[SSR_P − (SSR1+SSR2)]/(k+1)} / {(SSR1+SSR2)/[n−2(k+1)]}.

S24 **S24** 이렇게 계산한 F검정 = **Chow 검정**. 전제: 동분산 — 두 집단의 오차분산이 같아야 함.

S23 **S23** 수치: SSR_P=85.515, SSR1=19.603(여 90명), SSR2=58.752(남 276명) → SSR_ur=78.355. F=[(85.515−78.355)/4]/[78.355/358] ≈ **8.18** → 기각.

OBJ **OBJ 학습목표 7.5 후반** ('절편 차이를 허용한' Chow) — 슬라이드 본문엔 없음. 기울기만 같은지 검정: 제약모형에 female 더미만 남김 → 제약 수 k, F의 분모 자유도는 동일.

▶ 교수 설명 (수업 전사본 · 판서)

S23 · Joint test with F-statistic (1/2)

(10.29 05:53–09:06) 같은 검정의 다른 계산법: **집단별로 회귀를 따로** 돌림.

- [판서] Group 1 and 2 (합쳐서): y = β0+β1x1+β2x2+u → **SSR_R**
- Group 1: y = β01+β11x11+β21x21+u → SSR1
- Group 2: y = β02+β12x12+β22x22+u → SSR2
- SSR_UR = SSR1 + SSR2
- SSR_R과 SSR_UR이 있으면 → 검정 이름은? → **F검정**

S24 · Joint test (2/2): Chow test

(10.29 09:06–10:50) 이 방식이 **Chow 검정**(전사본 'child test').

- F검정이 알아내는 것: **제약모형이 참인가**
- 문답: "제약모형에서 남녀는 다른가, 같은가?" → **같음**. 즉 Chow 검정 = 남녀가 같은지(제약모형이 참인지) 검정

The F statistic for the three exclusion restrictions gives a p-value equal to .205, and so we do not reject the null hypothesis at even the 20% significance level.

Failure to reject the hypothesis that the parameters multiplying the interaction terms are all zero suggests that the best model allows for an intercept difference only:

$$\widehat{cumgpa} = 1.39 + .310\,female + .0012\,sat - .0084\,hsperc$$
$$(.18)\ (.059)\qquad (.0002)\qquad (.0012)$$
$$+ .0025\,tothrs \qquad [7.25]$$
$$(.0007)$$
$$n = 366,\ R^2 = .398,\ \bar{R}^2 = .392.$$

The slope coefficients in (7.25) are close to those for the base group (males) in (7.22); dropping the interactions changes very little. However, *female* in (7.25) is highly significant: its t statistic is over 5, and the estimate implies that, at given levels of *sat*, *hsperc*, and *tothrs*, a female athlete has a predicted GPA that is .31 point higher than that of a male athlete. This is a practically important difference.

7.5 A Binary Dependent Variable: The Linear Probability Model

By now, we have learned much about the properties and applicability of the multiple linear regression model. In the last several sections, we studied how, through the use of binary independent variables, we can incorporate qualitative information as explanatory variables in a multiple regression model. In all of the models up until now, the dependent variable y has had *quantitative* meaning (for example, y is a dollar amount, a test score, a percentage, or the logs of these). What happens if we want to use multiple regression to *explain* a qualitative event?

In the simplest case, and one that often arises in practice, the event we would like to explain is a binary outcome. In other words, our dependent variable, y, takes on only two values: zero and one. For example, y can be defined to indicate whether an adult has a high school education; y can indicate whether a college student used illegal drugs during a given school year; or y can indicate whether a firm was taken over by another firm during a given year. In each of these examples, we can let $y = 1$ denote one of the outcomes and $y = 0$ the other outcome.

What does it mean to write down a multiple regression model, such as

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u, \qquad [7.26]$$

when y is a binary variable? Because y can take on only two values, β_j cannot be interpreted as the change in y given a one-unit increase in x_j, holding all other factors fixed: y either changes from zero to one or from one to zero (or does not change). Nevertheless, the β_j still have useful interpretations. If we assume that the zero conditional mean assumption MLR.4 holds, that is, $E(u|x_1, \ldots, x_k) = 0$, then we have, as always,

$$E(y|\mathbf{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k,$$

where $\mathbf{x}$ is shorthand for all of the explanatory variables.

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OBJ

OBJ GPA 예: 교호항 3개 결합검정 p=.205 → 기각 못 함 → 절편 차이만 있는 모형(7.25)이 최선. female .310: 여성 선수가 GPA .31 높음.

S25

S25 S25–S29 = 7.5절 선형확률모형(LPM).

S25

S25 종속변수 y가 0 또는 1.

S25

S25 y = β0 + β1x1 + ⋯ + βkxk + u (y는 이진).

▶ 교수 설명 (수업 전사본 · 판서)

OBJ-7.5 · 학습목표 7.5 절편 차이를 허용한 Chow 검정

학습목표 7.5는 'Chow 검정을 **절편 차이를 허용하는 경우와 아닌 경우** 모두' 계산하는 것.

- 수업에서 다룬 것: **절편까지 전부 같다**는 기본 Chow 검정 (S23–S24)
- '절편 차이는 허용하고 기울기만 검정'하는 버전은 **언급 없음** → 교재 p.247–248로 보충. 시험 범위 여부는 확인 필요

The key point is that when y is a binary variable taking on the values zero and one, it is always true that $P(y = 1|\mathbf{x}) = E(y|\mathbf{x})$: the probability of "success"—that is, the probability that $y = 1$—is the same as the expected value of y. Thus, we have the important equation

$$P(y = 1|\mathbf{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k, \qquad [7.27]$$

which says that the probability of success, say, $p(\mathbf{x}) = P(y = 1|\mathbf{x})$, is a linear function of the x_j. Equation (7.27) is an example of a binary response model, and $P(y = 1|\mathbf{x})$ is also called the **response probability**. (We will cover other binary response models in Chapter 17.) Because probabilities must sum to one, $P(y = 0|\mathbf{x}) = 1 - P(y = 1|\mathbf{x})$ is also a linear function of the x_j.

The multiple linear regression model with a binary dependent variable is called the **linear probability model (LPM)** because the response probability is linear in the parameters β_j. In the LPM, β_j measures the change in the probability of success when x_j changes, holding other factors fixed:

$$\Delta P(y = 1|\mathbf{x}) = \beta_j \Delta x_j. \qquad [7.28]$$

With this in mind, the multiple regression model can allow us to estimate the effect of various explanatory variables on qualitative events. The mechanics of OLS are the same as before.

If we write the estimated equation as

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \ldots + \hat{\beta}_k x_k,$$

we must now remember that $\hat{y}$ is the predicted probability of success. Therefore, $\hat{\beta}_0$ is the predicted probability of success when each x_j is set to zero, which may or may not be interesting. The slope coefficient $\hat{\beta}_1$ measures the predicted change in the probability of success when x_1 increases by one unit.

To correctly interpret a linear probability model, we must know what constitutes a "success." Thus, it is a good idea to give the dependent variable a name that describes the event $y = 1$. As an example, let *inlf* ("in the labor force") be a binary variable indicating labor force participation by a married woman during 1975: *inlf* = 1 if the woman reports working for a wage outside the home at some point during the year, and zero otherwise. We assume that labor force participation depends on other sources of income, including husband's earnings (*nwifeinc*, measured in thousands of dollars), years of education (*educ*), past years of labor market experience (*exper*), *age*, number of children less than six years old (*kidslt6*), and number of kids between 6 and 18 years of age (*kidsge6*). Using the data in MROZ.RAW from Mroz (1987), we estimate the following linear probability model, where 428 of the 753 women in the sample report being in the labor force at some point during 1975:

$$\widehat{inlf} = .586 - .0034\, nwifeinc + .038\, educ + .039\, exper$$
$$(.154)\ (.0014)\qquad (.007)\qquad (.006)$$
$$-.00060\, exper^2 - .016\, age - .262\, kidslt6 + .013\, kidsge6 \qquad [7.29]$$
$$(.00018)\qquad (.002)\qquad (.034)\qquad (.013)$$
$$n = 753,\ R^2 = .264.$$

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S25 **S25 핵심 등식**: E(y|x) = 1·P(y=1|x) + 0·P(y=0|x) = P(y=1|x).

S25 **S25 LPM**: P(y=1|x) = β0 + β1x1 + ⋯ + βkxk. 오차항 u가 사라진 형태(판서 'error disappears').

S25 **S25 ΔP(y=1|x) = βj·Δxj**: 계수 = xj가 1 늘 때 '성공 확률'의 변화(%p 단위로 읽음).

S26 **S26 예**: inlf=1(1975년에 노동시장 참여). MROZ, n=753 중 428명 참여.

S26 S26 nwifeinc −.0034, educ .038, exper .039, kidslt6 −.262 ⋯ R^2=.264. (교재 kidsge6 .013, 슬라이드 .0130.)

S26

▶ 교수 설명 (수업 전사본 · 판서)

S25 · Linear probability model

(10.29 10:50–14:50) LPM: "솔직히 **정말 위험한 모형**". 이론적으로 문제가 많음. 그런데 왜 가르치고 쓰나? → **유용한 근사**일 때가 있어서.

- 종속변수가 이진(더미). 예: 오늘 수업에 **제시간에 도착=1, 지각=0**
- 독립변수: 차가 있는지, 차의 속도, 통학 수단(걷기·버스·운전), 집에서 강의실까지 거리 등
- 이걸 넣으면 **제시간에 도착할 조건부 확률**을 예측하는 선형모형이 됨
- 계수 해석: x1이 차 보유면 "차가 있으면 제시간 도착 확률이 β1만큼 증가"

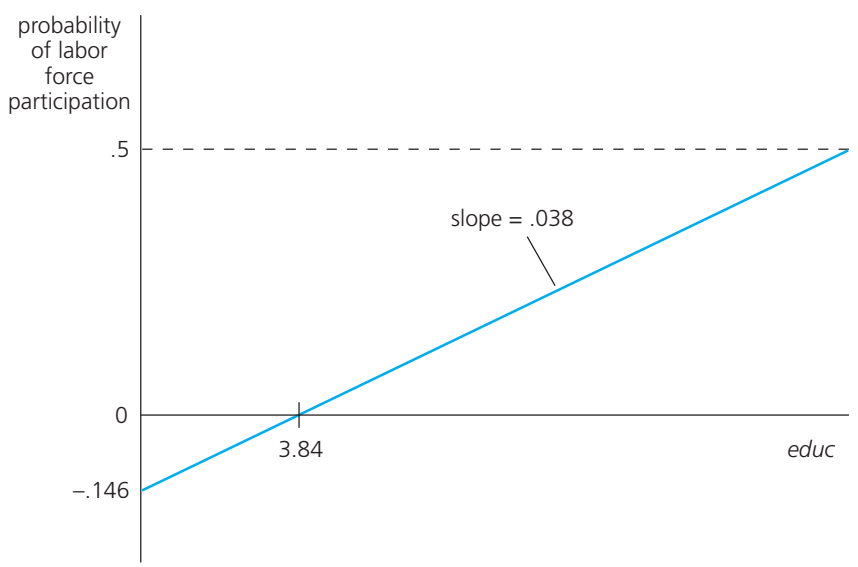
[판서] y = β0+β1x1+u → P(y=1|x) = β0+β1x1 (**error disappears**), y: arrival on time, x1: car ownership

Using the usual t statistics, all variables in (7.29) except *kidsge6* are statistically significant, and all of the significant variables have the effects we would expect based on economic theory (or common sense).

To interpret the estimates, we must remember that a change in the independent variable changes the probability that $inlf = 1$. For example, the coefficient on *educ* means that, everything else in (7.29) held fixed, another year of education increases the probability of labor force participation by .038. If we take this equation literally, 10 more years of education increases the probability of being in the labor force by .038(10) = .38, which is a pretty large increase in a probability. The relationship between the probability of labor force participation and *educ* is plotted in Figure 7.3. The other independent variables are fixed at the values *nwifeinc* = 50, *exper* = 5, *age* = 30, *kidslt6* = 1, and *kidsge6* = 0 for illustration purposes. The predicted probability is negative until education equals 3.84 years. This should not cause too much concern because, in this sample, no woman has less than five years of education. The largest reported education is 17 years, and this leads to a predicted probability of .5. If we set the other independent variables at different values, the range of predicted probabilities would change. But the marginal effect of another year of education on the probability of labor force participation is always .038.

The coefficient on *nwifeinc* implies that, if $\Delta nwifeinc = 10$ (which means an increase of $10,000), the probability that a woman is in the labor force falls by .034. This is not an especially large effect given that an increase in income of $10,000 is substantial in terms of 1975 dollars. Experience has been entered as a quadratic to allow the effect of past experience to have a diminishing effect on the labor force participation probability. Holding other factors fixed, the estimated change in the probability is approximated as $.039 - 2(.0006)exper = .039 - .0012\ exper$. The point at which past experience has no

FIGURE 7.3 Estimated relationship between the probability of being in the labor force and years of education, with other explanatory variables fixed.



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S26

S26 educ 1년↑ → 참여 확률 **.038(3.8%p)**↑.

S27

S27 **그림 7.3** (nwifeinc=50, exper=5, age=30, kidslt6=1, kidsge6=0): educ < 3.84면 예측확률이 음수. 표본 최대 educ=17 → 약 .5.

S27

S27 다른 변수 값을 바꾸면 선이 위아래로 이동할 뿐, 기울기(.038)는 항상 같음.

REF

REF nwifeinc 1만 달러↑ → 확률 .034↓.

▶ 교수 설명 (수업 전사본 · 판서)

S26 · Example: labor force participation (1/2)

(10.29 14:53–15:36) 종속변수 = 노동시장 참여의 조건부 확률. 설명변수에 '자녀가 있는지, 몇 명인지, 6세 미만이 몇 명인지' 등. 회귀하면 슬라이드 같은 결과가 나옴. (수치 해석은 슬라이드 주석 참고)

S27 · Example (2/2): graph · negative probability

(10.29 15:39–17:22) 그래프가 보여주는 문제: **이론상 말이 안 됨** — 확률은 음수가 될 수 없는데, 교육이 아주 낮으면 참여 확률이 음수.

- 슬라이드의 변명 "표본에 그런 사람이 없으니 문제없다"에 대해 교수: 실제 예에선 그럴지 몰라도 **이론적으로는 여전히 문제**
- 다른 데이터셋에선 음의 확률이 나올 수 있고, 표본에 없는 교육 수준에 대한 확률을 추정하라고 요구받을 수도 있음
- "이론적 문제는 존재하고, 때로는 실증적 문제도 존재한다."

effect on the probability of labor force participation is .039/.0012 = 32.5, which is a high level of experience: only 13 of the 753 women in the sample have more than 32 years of experience.

Unlike the number of older children, the number of young children has a huge impact on labor force participation. Having one additional child less than six years old reduces the probability of participation by −.262, at given levels of the other variables. In the sample, just under 20% of the women have at least one young child.

This example illustrates how easy linear probability models are to estimate and interpret, but it also highlights some shortcomings of the LPM. First, it is easy to see that, if we plug certain combinations of values for the independent variables into (7.29), we can get predictions either less than zero or greater than one. Since these are predicted probabilities, and probabilities must be between zero and one, this can be a little embarassing. For example, what would it mean to predict that a woman is in the labor force with a probability of −.10? In fact, of the 753 women in the sample, 16 of the fitted values from (7.29) are less than zero, and 17 of the fitted values are greater than one.

A related problem is that a probability cannot be linearly related to the independent variables for all their possible values. For example, (7.29) predicts that the effect of going from zero children to one young child reduces the probability of working by .262. This is also the predicted drop if the woman goes from having one young child to two. It seems more realistic that the first small child would reduce the probability by a large amount, but subsequent children would have a smaller marginal effect. In fact, when taken to the extreme, (7.29) implies that going from zero to four young children reduces the probability of working by $\Delta\widehat{inlf} = .262(\Delta kidslt6) = .262(4) = 1.048$, which is impossible.

Even with these problems, the linear probability model is useful and often applied in economics. It usually works well for values of the independent variables that are near the averages in the sample. In the labor force participation example, no women in the sample have four young children; in fact, only three women have three young children. Over 96% of the women have either no young children or one small child, and so we should probably restrict attention to this case when interpreting the estimated equation.

Predicted probabilities outside the unit interval are a little troubling when we want to make predictions. Still, there are ways to use the estimated probabilities (even if some are negative or greater than one) to predict a zero-one outcome. As before, let $\hat{y}_i$ denote the fitted values—which may not be bounded between zero and one. Define a predicted value as $\tilde{y}_i = 1$ if $\hat{y}_i \geq .5$ and $\tilde{y}_i = 0$ if $\hat{y}_i < .5$. Now we have a set of predicted values, $\tilde{y}_i$, $i = 1, \ldots, n$, that, like the y_i, are either zero or one. We can use the data on y_I and $\tilde{y}_i$ to obtain the frequencies with which we correctly predict $y_i = 1$ and $y_i = 0$, as well as the proportion of overall correct predictions. The latter measure, when turned into a percentage, is a widely used goodness-of-fit measure for binary dependent variables: the **percent correctly predicted**. An example is given in Computer Exercise C9(v), and further discussion, in the context of more advanced models, can be found in Section 17.1.

Due to the binary nature of y, the linear probability model does violate one of the Gauss-Markov assumptions. When y is a binary variable, its variance, conditional on $\mathbf{x}$, is

$$\text{Var}(y|\mathbf{x}) = p(\mathbf{x})[1 - p(\mathbf{x})], \qquad [7.30]$$

where $p(\mathbf{x})$ is shorthand for the probability of success: $p(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k$. This means that, except in the case where the probability does not depend on any of the independent variables, there *must* be heteroskedasticity in a linear probability model. We know

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S26 S26 슬라이드 주석: 6세 미만 자녀 1명↑ → 참여 확률 **26.2%p**↓.

S28 S28 **단점** ① 예측확률이 0 미만 또는 1 초과 가능.

REF REF 실제로 753명 중 16명 <0, 17명 >1.

S28 S28 **단점** ② 한계효과가 논리적으로 불가능해질 수 있음 (효과가 항상 일정하다는 가정 때문).

S28 S28 예: 자녀 0→4명이면 확률 변화 = −.262×4 = −1.048 (불가능).

S29 S29 **장점**: 추정·해석이 쉽고, 표본 평균 근처 값에서는 잘 맞음.

SKIP SKIP percent correctly predicted (적합도 지표) — PPT 미포함.

S28 S28 **단점** ③ Var(y|x) = p(x)[1−p(x)] (베르누이 분산). x에 따라 달라짐 → **필연적 이분산**. 판서: V(Y)=E(Y²)−E(Y)², Y가 더미면 E(Y²)=E(Y)=P(Y=1).

▶ 교수 설명 (수업 전사본 · 판서)

S28 · LPM disadvantages

(10.29 17:22–33:20) [시험] **LPM의 문제 3가지** (교수가 마지막에 직접 요약)

① **예측확률이 1 초과 또는 0 미만** — 수학적으로 불가능.

② **효과(계수)도 1 초과 또는 −1 미만 가능** [판서] β1 < −1 또는 β1 > 1

- "차를 사면 제시간 도착 확률이 1보다 크게 증가(또는 감소)" → 논리적으로 불가능한 예측

③ **거의 항상 이분산** [판서]

- 통상 식에서 빠진 것은? → **오차항이 사라짐**. 오차는 숨어 있지만 여전히 존재하고 좀 이상하게 작동
- P(y=1|x)=0.5인 사람: u = **0.5**(→ y=1) 또는 **−0.5**(→ y=0)
- P(y=1|x)=0.1인 사람: u = **0.9**(→ y=1) 또는 **−0.1**(→ y=0)
- x에 따라 오차가 가질 수 있는 값(과 분산)이 달라짐 → 이분산

[판서] 분산 유도 ("한국 고교 과정에 있던 공식. 증명도 어렵지 않다")

- $V(X) = E(X^2) - E(X)^2$
- Y가 더미이므로 $E(Y) = P(Y=1)$, $E(Y^2) = P(Y=1)$
- $V(Y) = P(Y=1) - P(Y=1)^2$
- P(Y=1)이 x에 의존하지 않으면 동분산이지만, x에 의존하면(=LPM 식) **확실히 이분산** → OLS 가정 위배

from Chapter 3 that this does not cause bias in the OLS estimators of the β_j. But we also know from Chapters 4 and 5 that homoskedasticity is crucial for justifying the usual t and F statistics, even in large samples. Because the standard errors in (7.29) are not generally valid, we should use them with caution. We will show how to correct the standard errors for heteroskedasticity in Chapter 8. It turns out that, in many applications, the usual OLS statistics are not far off, and it is still acceptable in applied work to present a standard OLS analysis of a linear probability model.

EXAMPLE 7.12 A LINEAR PROBABILITY MODEL OF ARRESTS SKIP

Let *arr86* be a binary variable equal to unity if a man was arrested during 1986, and zero otherwise. The population is a group of young men in California born in 1960 or 1961 who have at least one arrest prior to 1986. A linear probability model for describing *arr86* is

$$arr86 = \beta_0 + \beta_1 pcnv + \beta_2 avgsen + \beta_3 tottime + \beta_4 ptime86 + \beta_5 qemp86 + u,$$

where

pcnv = the proportion of prior arrests that led to a conviction.
avgsen = the average sentence served from prior convictions (in months).
tottime = months spent in prison since age 18 prior to 1986.
ptime86 = months spent in prison in 1986.
qemp86 = the number of quarters (0 to 4) that the man was legally employed in 1986.

The data we use are in CRIME1.RAW, the same data set used for Example 3.5. Here, we use a binary dependent variable because only 7.2% of the men in the sample were arrested more than once. About 27.7% of the men were arrested at least once during 1986. The estimated equation is

$$\widehat{arr86} = .441 - .162\, pcnv + .0061\, avgsen - .0023\, tottime$$
$$(.017)\ \ (.021)\qquad (.0065)\qquad\ (.0050)$$
$$- .022\, ptime86 - .043\, qemp86 \qquad [7.31]$$
$$(.005)\qquad\ (.005)$$
$$n = 2{,}725,\ R^2 = .0474.$$

The intercept, .441, is the predicted probability of arrest for someone who has not been convicted (and so *pcnv* and *avgsen* are both zero), has spent no time in prison since age 18, spent no time in prison in 1986, and was unemployed during the entire year. The variables *avgsen* and *tottime* are insignificant both individually and jointly (the F test gives p-value = .347), and *avgsen* has a counterintuitive sign if longer sentences are supposed to deter crime. Grogger (1991), using a superset of these data and different econometric methods, found that *tottime* has a statistically significant *positive* effect on arrests and concluded that *tottime* is a measure of human capital built up in criminal activity.

Increasing the probability of conviction does lower the probability of arrest, but we must be careful when interpreting the magnitude of the coefficient. The variable *pcnv* is a proportion between zero and one; thus, changing *pcnv* from zero to one essentially means a change from no chance of being convicted to being convicted with certainty. Even this

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S28 이분산이어도 계수 추정량은 불편.

S28 하지만 통상의 t·F 검정은 타당하지 않음 → **이분산-일치(robust) 표준오차** 필요.

S28 → Ch.8 (10.29 수업 후반에서 시작).

S29 실무에선 통상 OLS 통계량도 크게 틀리지 않음 → LPM은 여전히 널리 사용.

SKIP 예제 7.12 (체포의 LPM) — PPT 미포함.

▶ 교수 설명 (수업 전사본 · 판서)

S29 · LPM advantages

(10.29 33:20–34:00) 그럼 왜 이 모형을 쓰나?

- **계산이 쉽다**
- 결과가 **충분히 비슷**할 때가 있다
- 더 나은 방법은 **뒤 장에서** 배움

SKIP

large change reduces the probability of arrest only by .162; increasing *pcnv* by .5 decreases the probability of arrest by .081.

The incarcerative effect is given by the coefficient on *ptime86*. If a man is in prison, he cannot be arrested. Since *ptime86* is measured in months, six more months in prison reduces the probability of arrest by .022(6) = .132. Equation (7.31) gives another example of where the linear probability model cannot be true over all ranges of the independent variables. If a man is in prison all 12 months of 1986, he cannot be arrested in 1986. Setting all other variables equal to zero, the predicted probability of arrest when *ptime86* = 12 is .441 − .022(12) = .177, which is not zero. Nevertheless, if we start from the unconditional probability of arrest, .277, 12 months in prison reduces the probability to essentially zero: .277 − .022(12) = .013.

Finally, employment reduces the probability of arrest in a significant way. All other factors fixed, a man employed in all four quarters is .172 less likely to be arrested than a man who is not employed at all.

We can also include dummy independent variables in models with dummy dependent variables. The coefficient measures the predicted difference in probability relative to the base group. For example, if we add two race dummies, *black* and *hispan*, to the arrest equation, we obtain

$$\widehat{arr86} = .380 - .152\,pcnv + .0046\,avgsen - .0026\,tottime$$
$$(.019)\ \ (.021)\qquad (.0064)\qquad (.0049)$$
$$- .024\,ptime86 - .038\,qemp86 + .170\,black + .096\,hispan \qquad [7.32]$$
$$(.005)\qquad\quad (.005)\qquad\quad (.024)\qquad\ (.021)$$
$$n = 2{,}725,\ R^2 = .0682.$$

EXPLORING FURTHER 7.5

What is the predicted probability of arrest for a black man with no prior convictions—so that *pcnv*, *avgsen*, *tottime*, and *ptime86* are all zero—who was employed all four quarters in 1986? Does this seem reasonable?

The coefficient on *black* means that, all other factors being equal, a black man has a .17 higher chance of being arrested than a white man (the base group). Another way to say this is that the probability of arrest is 17 percentage points higher for blacks than for whites. The difference is statistically significant as well. Similarly, Hispanic men have a .096 higher chance of being arrested than white men.

7.6 More on Policy Analysis and Program Evaluation

S30

S30 S30–S37 = 7.6절.

We have seen some examples of models containing dummy variables that can be useful for evaluating policy. Example 7.3 gave an example of program evaluation, where some firms received job training grants and others did not.

S30

S30 사회과학에서 통제·처치집단은 대개 무작위 배정이 아님.

As we mentioned earlier, we must be careful when evaluating programs because in most examples in the social sciences the control and treatment groups are not randomly assigned. Consider again the Holzer et al. (1993) study, where we are now interested in

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the effect of the job training grants on worker productivity (as opposed to amount of job training). The equation of interest is

$$\log(scrap) = \beta_0 + \beta_1 grant + \beta_2 \log(sales) + \beta_3 \log(employ) + u,$$

where *scrap* is the firm's scrap rate, and the latter two variables are included as controls. The binary variable *grant* indicates whether the firm received a grant in 1988 for job training.

Before we look at the estimates, we might be worried that the unobserved factors affecting worker productivity—such as average levels of education, ability, experience, and tenure—might be correlated with whether the firm receives a grant. Holzer et al. point out that grants were given on a first-come, first-served basis. But this is not the same as giving out grants randomly. It might be that firms with less productive workers saw an opportunity to improve productivity and therefore were more diligent in applying for the grants.

Using the data in JTRAIN.RAW for 1988—when firms actually were eligible to receive the grants—we obtain

$$\widehat{\log(scrap)} = 4.99 - .052\ grant - .455 \log(sales)$$
$$(4.66)\quad (.431)\qquad (.373)$$
$$+ .639 \log(employ) \qquad [7.33]$$
$$(.365)$$
$$n = 50,\ R^2 = .072.$$

(Seventeen out of the 50 firms received a training grant, and the average scrap rate is 3.47 across all firms.) The point estimate of −.052 on *grant* means that, for given *sales* and *employ*, firms receiving a grant have scrap rates about 5.2% lower than firms without grants. This is the direction of the expected effect if the training grants are effective, but the *t* statistic is very small. Thus, from this cross-sectional analysis, we must conclude that the grants had no effect on firm productivity. We will return to this example in Chapter 9 and show how adding information from a prior year leads to a much different conclusion.

Even in cases where the policy analysis does not involve assigning units to a control group and a treatment group, we must be careful to include factors that might be systematically related to the binary independent variable of interest. A good example of this is testing for racial discrimination. Race is something that is not determined by an individual or by government administrators. In fact, race would appear to be the perfect example of an exogenous explanatory variable, given that it is determined at birth. However, for historical reasons, race is often related to other relevant factors: there are systematic differences in backgrounds across race, and these differences can be important in testing for *current* discrimination.

As an example, consider testing for discrimination in loan approvals. If we can collect data on, say, individual mortgage applications, then we can define the dummy dependent variable *approved* as equal to one if a mortgage application was approved, and zero otherwise. A systematic difference in approval rates across races is an indication of discrimination. However, since approval depends on many other factors, including income, wealth, credit ratings, and a general ability to pay back the loan, we must control for them *if* there are systematic differences in these factors across race. A linear probability model to test for discrimination might look like the following:

$$approved = \beta_0 + \beta_1 nonwhite + \beta_2 income + \beta_3 wealth + \beta_4 credrate + other\ factors.$$

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S30 걱정: 생산성에 영향을 주는 비관측 요인이 보조금 수령과 상관.

S30 선착순 배정 ≠ 무작위. 생산성 낮은 기업이 더 적극적으로 신청했을 수 있음 (S30 슬라이드 노트) → grant 계수 편의.

S30 log(scrap) on grant: −.052(.431), n=50, R^2=.072. scrap=불량률(낮을수록 생산성↑).

S30 방향은 기대와 같지만(−5.2%) t≈−.12 → 유의한 효과 없음.

REF 전년도 정보를 추가하면 결론이 달라짐(9장).

REF 인종처럼 '배정'이 아닌 더미도 다른 요인과 체계적으로 상관될 수 있음 → 통제 필요 (대출 승인 차별 예).

▶ 교수 설명 (수업 전사본 · 판서)

S30 · Policy analysis: grants and productivity

(10.29 34:00–43:16) 노동자 생산성과 훈련 보조금 수령 여부. 결과: **유의한 효과 없음**. 이 모형의 문제는?

- grant는 1/0인데 **보조금이 어떻게 주어지는지** 모름
- 보조금이 **정말 도움이 필요한 기업**(생산성이 나쁜 기업)에 주어진다면 → **β1 과소추정**

[판서] productivity = β0 + β1·traininggrant + u (grant는 dummy)
→ 참모형: productivity = β0 + β1·traininggrant + **β2·firmtrouble** + u
"How exactly is the grant given? Is there random assignment?" → **OVB may exist**

- 보조금이 무작위 배정이 아니면 β1을 잘못 추정. 이 현상의 이름은? (출석 부르는 동안 생각) → 누락변수 편의(OVB). [시험] **"중간고사에 나왔던 것"**
- 통제변수를 많이 넣는 이점: 무작위 배정이 없을 때 **OVB 가능성을 줄임**

Discrimination against minorities is indicated by a rejection of H_0: $\beta_1 = 0$ in favor of H_0: $\beta_1 < 0$, because β_1 is the amount by which the probability of a nonwhite getting an approval differs from the probability of a white getting an approval, given the same levels of other variables in the equation. If *income*, *wealth*, and so on are systematically different across races, then it is important to control for these factors in a multiple regression analysis.

Another problem that often arises in policy and program evaluation is that individuals (or firms or cities) choose whether or not to participate in certain behaviors or programs. For example, individuals choose to use illegal drugs or drink alcohol. If we want to examine the effects of such behaviors on unemployment status, earnings, or criminal behavior, we should be concerned that drug usage might be correlated with other factors that can affect employment and criminal outcomes. Children eligible for programs such as Head Start participate based on parental decisions. Since family background plays a role in Head Start decisions and affects student outcomes, we should control for these factors when examining the effects of Head Start [see, for example, Currie and Thomas (1995)]. Individuals selected by employers or government agencies to participate in job training programs can participate or not, and this decision is unlikely to be random [see, for example, Lynch (1992)]. Cities and states choose whether to implement certain gun control laws, and it is likely that this decision is systematically related to other factors that affect violent crime [see, for example, Kleck and Patterson (1993)].

The previous paragraph gives examples of what are generally known as **self-selection** problems in economics. Literally, the term comes from the fact that individuals self-select into certain behaviors or programs: participation is not randomly determined. The term is used generally when a binary indicator of participation might be systematically related to unobserved factors. Thus, if we write the simple model

$$y = \beta_0 + \beta_1 partic + u, \qquad [7.34]$$

where y is an outcome variable and *partic* is a binary variable equal to unity if the individual, firm, or city participates in a behavior or a program or has a certain kind of law, then we are worried that the average value of u depends on participation: $E(u|partic = 1) \neq E(u|partic = 0)$. As we know, this causes the simple regression estimator of β_1 to be biased, and so we will not uncover the true effect of participation. Thus, the self-selection problem is another way that an explanatory variable (*partic* in this case) can be endogenous.

By now, we know that multiple regression analysis can, to some degree, alleviate the self-selection problem. Factors in the error term in (7.34) that are correlated with *partic* can be included in a multiple regression equation, assuming, of course, that we can collect data on these factors. Unfortunately, in many cases, we are worried that unobserved factors are related to participation, in which case multiple regression produces biased estimators.

With standard multiple regression analysis using cross-sectional data, we must be aware of finding spurious effects of programs on outcome variables due to the self-selection problem. A good example of this is contained in Currie and Cole (1993). These authors examine the effect of AFDC (Aid to Families with Dependent Children) participation on the birth weight of a child. Even after controlling for a variety of family and background characteristics, the authors obtain OLS estimates that imply participation in AFDC *lowers* birth weight. As the authors point out, it is hard to believe that AFDC participation itself *causes* lower birth weight. [See Currie (1995) for additional examples.] Using a different econometric method that we will discuss in Chapter 15, Currie and Cole find evidence for either no effect or a *positive* effect of AFDC participation on birth weight.

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S31 **S31 자기선택(self-selection)**: 참여 여부를 스스로 결정 → 참여가 무작위가 아님.

S32 **S32** 예: Head Start 참여는 부모가 결정 → 가정 배경을 통제해야 무작위 배정에 가까워짐.

S32 **S32** 단순회귀 y = β0 + β1·partic + u (슬라이드: y = α + τw + u). 슬라이드 표기 **w**=처치 더미, **τ**=처치효과, y(0)/y(1)=처치 안/받았을 때의 결과.

S32 **S32** E(u|w=1) ≠ E(u|w=0)이면 편의 → 자기선택은 내생성의 한 형태. 단순회귀가 타당하려면 w가 [y(0), y(1)]과 독립(=무작위 배정)이어야 함.

S33 **S33 회귀조정(regression adjustment)**: 통제변수 x1…xk를 넣고 'x 조건부로 w가 [y(0), y(1)]과 독립'이라 가정. y = α + τw + γ1x1 + ⋯ + u.

S33 **S33** 한계: 참여와 상관된 요인이 관측 불가면 다중회귀로도 편의.

REF **REF** 가짜 효과 예: AFDC 참여가 출생체중을 낮춘다는 OLS 결과 — IV(15장)로는 효과 없음/양(+).

▶ 교수 설명 (수업 전사본 · 판서)

S31-33 · Self-selection · regression adjustment

(10.29 43:23–50:47) 결과를 **y(0)**(w=0일 때)과 **y(1)**(w=1일 때)로 나눠 표현. w = 프로그램(처치)에 들어갔는지 여부를 나타내는 **더미**. τ는 w에 붙는 계수.

- Head Start 같은 처치가 **무작위**면 그대로 진행해도 됨
- 아니면 OVB를 피하려고 **통제변수**를 넣어야 함
- 어떤 통제변수? → **처치에 들어갈지 여부에 영향을 주는 것들**. Head Start가 열성적인 가정을 찾는지, 부유한/가난한 가정을 찾는지, 똑똑한/덜 똑똑한 학생을 찾는지에 따라 달라짐. "정확히는 모르지만 **선발 방식에 달렸다**"
- 이 변수들을 통제하고 나면 w는 y(0), y(1)과 독립이라고 말할 수 있음 (조건부 독립)

※ 교재 5판에는 y(0)/y(1), τ 표기와 '회귀조정' 용어가 없음 (8판 내용). 대응 개념은 p.255의 self-selection.

When the self-selection problem causes standard multiple regression analysis to be biased due to a lack of sufficient control variables, the more advanced methods covered in Chapters 13, 14, and 15 can be used instead.

REF

REF 통제변수로 부족하면 13–15장 방법(패널, IV).

7.7 Interpreting Regression Results with Discrete Dependent Variables

OBJ

OBJ **학습목표 7.8** (이산 종속변수의 해석, 7.7절) — 슬라이드 본문 없음. 핵심: 계수는 y의 기댓값(평균)에 대한 효과. 예: educ −.090 → 여성 100명이 교육을 1년씩 더 받으면 자녀 9명 감소.

A binary response is the most extreme form of a discrete random variable: it takes on only two values, zero and one. As we discussed in Section 7.5, the parameters in a linear probability model can be interpreted as measuring the change in the *probability* that $y = 1$ due to a one-unit increase in an explanatory variable. We also discussed that, because y is a zero-one outcome, $P(y = 1) = E(y)$, and this equality continues to hold when we condition on explanatory variables.

Other discrete dependent variables arise in practice, and we have already seen some examples, such as the number of times someone is arrested in a given year (Example 3.5). Studies on factors affecting fertility often use the number of living children as the dependent variable in a regression analysis. As with number of arrests, the number of living children takes on a small set of integer values, and zero is a common value. The data in FERTIL2.RAW, which contains information on a large sample of women in Botswana is one such example. Often demographers are interested in the effects of education on fertility, with special attention to trying to determine whether education has a causal effect on fertility. Such examples raise a question about how one interprets regression coefficients: after all, one cannot have a fraction of a child.

To illustrate the issues, the regression below uses the data in FERTIL2.RAW:

$$\widehat{children} = -1.997 + .175\,age - .090\,educ \qquad [7.35]$$
$$(.094)\quad (.003)\qquad (.006)$$
$$n = 4{,}361,\ R^2 = .560.$$

At this time, we ignore the issue of whether this regression adequately controls for all factors that affect fertility. Instead we focus on interpreting the regression coefficients.

Consider the main coefficient of interest, $\hat{\beta}_{educ} = -.090$. If we take this estimate literally, it says that each additional year of education reduces the estimated number of children by .090—something obviously impossible for any particular woman. A similar problem arises when trying to interpret $\hat{\beta}_{age} = .175$. How can we make sense of these coefficients?

To interpret regression results generally, even in cases where y is discrete and takes on a small number of values, it is useful to remember the interpretation of OLS as estimating the effects of the x_j on the *expected* (or *average*) value of y. Generally, under Assumptions MLR.1 and MLR.4,

$$E(y|x_1, x_2, \ldots, x_k) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k. \qquad [7.36]$$

Therefore, β_j is the effect of a ceteris paribus increase of x_j on the expected value of y. As we discussed in Section 6.4, for a given set of x_j values we interpret the predicted value, $\hat{\beta}_0 + \hat{\beta}_1 x_1 + \ldots + \hat{\beta}_k x_k$, as an estimate of $E(y|x_1, x_2, \ldots, x_k)$. Therefore, $\hat{\beta}_j$ is our estimate of how the *average* of y changes when $\Delta x_j = 1$ (keeping other factors fixed).

Seen in this light, we can now provide meaning to regression results as in equation (7.35). The coefficient $\hat{\beta}_{educ} = -.090$ means that we estimate that *average* fertility falls by .09 children given one more year of education. A nice way to summarize this

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▶ **교수 설명 (수업 전사본 · 판서)**

S34-37 · ATE · URA vs RRA · alternative method

(10.29 50:51–01:02:22) **평균처치효과(ATE)**. 처치효과를 τ 하나로 두는 대신, **처치와 다른 변수의 교호항**을 추가.

학생 질문("이 긴 식이 무슨 뜻인가")에 대한 답 (58:26–01:00:47) — 식은 PPT S34

- yi = α + τwi + γ1xi1 + ⋯ + γkxik + δ1wi(xi1−x̄1) + ⋯ + δkwi(xik−x̄k) + u
- 처치 효과(w가 0→1)가 **통제변수 값에 따라 달라질 수 있다**는 뜻
- w가 0→1일 때의 효과 = τ + 교호항들(δj(xij−x̄j))
- **[중요]** 통제변수가 **표본평균**이면 교호항이 0이 되어 효과 = τ → τ = 통제변수 표본평균에서의 처치효과 = ATE

두 가지 조정

- 처치효과가 개인마다 **다를 수 있는** 경우 (URA, 교호항 포함)
- 처치효과가 개인마다 **같은** 경우 (RRA, 교호항 없음)

ATE를 구하는 다른 방법 (01:00:47–01:02:22): 통제집단과 처치집단을 나눠 **각각 회귀** → 각 회귀로 (모든 사람의) 예측값을 구함 → 두 예측의 차이를 평균 → ATE

- 결과는 같지만 문제: **표준오차는 어떻게 구하나?**

※ 이 부분(S34–S37)은 교재 5판에 없음 → PPT와 이 칸이 유일한 자료.

interpretation is that if each woman in a group of 100 obtains another year of education, we estimate there will be nine fewer children among them.

Adding dummy variables to regressions when y is itself discrete causes no problems when we intepret the estimated effect in terms of average values. Using the data in FERTIL2.RAW we get

$$\widehat{children} = -2.071 + .177\,age - .079\,educ - .362\,electric \qquad [7.37]$$
$$(.095)\quad (.003)\qquad (.006)\qquad\quad (.068)$$
$$n = 4{,}358,\ R^2 = .562,$$

where *electric* is a dummy variable equal to one if the woman lives in a home with electricity. Of course it cannot be true that a particular woman who has electricity has .362 less children than an otherwise comparable woman who does not. But we can say that when comparing 100 women with electricity to 100 women without—at the same age and level of education—we estimate the former group to have about 36 fewer children.

Incidentally, when y is discrete the linear model does not always provide the best estimates of partial effects on $E(y|x_1, x_2, \ldots, x_k)$. Chapter 17 contains more advanced models and estimation methods that tend to fit the data better when the range of y is limited in some substantive way. Nevertheless, a linear model estimated by OLS often provides a good approximation to the true partial effects, at least on average.

Summary

In this chapter, we have learned how to use qualitative information in regression analysis. In the simplest case, a dummy variable is defined to distinguish between two groups, and the coefficient estimate on the dummy variable estimates the ceteris paribus difference between the two groups. Allowing for more than two groups is accomplished by defining a set of dummy variables: if there are g groups, then $g - 1$ dummy variables are included in the model. All estimates on the dummy variables are interpreted relative to the base or benchmark group (the group for which no dummy variable is included in the model).

Dummy variables are also useful for incorporating ordinal information, such as a credit or a beauty rating, in regression models. We simply define a set of dummy variables representing different outcomes of the ordinal variable, allowing one of the categories to be the base group.

Dummy variables can be interacted with quantitative variables to allow slope differences across different groups. In the extreme case, we can allow each group to have its own slope on every variable, as well as its own intercept. The Chow test can be used to detect whether there are any differences across groups. In many cases, it is more interesting to test whether, after allowing for an intercept difference, the slopes for two different groups are the same. A standard F test can be used for this purpose in an unrestricted model that includes interactions between the group dummy and all variables.

The linear probability model, which is simply estimated by OLS, allows us to explain a binary response using regression analysis. The OLS estimates are now interpreted as changes in the probability of "success" ($y = 1$), given a one-unit increase in the corresponding explanatory variable. The LPM does have some drawbacks: it can produce predicted probabilities that are less than zero or greater than one, it implies a constant marginal effect of each explanatory variable that appears in its original form, and it contains heteroskedasticity. The first two problems are often not serious when we are obtaining estimates of the partial effects of the explanatory variables for the middle ranges of the data. Heteroskedasticity does invalidate the

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OBJ

OBJ 더미도 같은 방식: electric −.362 → 전기 있는 여성 100명이 없는 100명보다 자녀 약 36명 적음.

REF

REF S38 정리: 교재 Summary로 복습. 순서 = 더미(절편 차이) → g−1 규칙 → 서열 정보 → 교호항(기울기 차이) → Chow → LPM(단점 3가지) → 프로그램 평가.

▶ 교수 설명 (수업 전사본 · 판서)

OBJ-7.8 · 학습목표 7.8 Discrete dependent variables

수업에서 **언급 없음** (PPT에도 해당 슬라이드 없음).

- 학습목표 7.8에만 있음. 수업 기준 우선순위는 낮지만, 핵심 한 줄("계수 = y의 **평균**에 대한 효과")은 LPM 해석과 같은 논리라 함께 기억해 둘 것

usual OLS standard errors and test statistics, but, as we will see in the next chapter, this is easily fixed in large enough samples.

Section 7.6 provides a discussion of how binary variables are used to evaluate policies and programs. As in all regression analysis, we must remember that program participation, or some other binary regressor with policy implications, might be correlated with unobserved factors that affect the dependent variable, resulting in the usual omitted variables bias.

We ended this chapter with a general discussion of how to interpret regression equations when the dependent variable is discrete. The key is to remember that the coefficients can be interpreted as the effects on the expected value of the dependent variable.

Key Terms

Base Group	Dummy Variables	Policy Analysis
Benchmark Group	Experimental Group	Program Evaluation
Binary Variable	Interaction Term	Response Probability
Chow Statistic	Intercept Shift	Self-Selection
Control Group	Linear Probability Model (LPM)	Treatment Group
Difference in Slopes	Ordinal Variable	Uncentered R-Squared
Dummy Variable Trap	Percent Correctly Predicted	Zero-One Variable

Problems

1 Using the data in SLEEP75.RAW (see also Problem 3 in Chapter 3), we obtain the estimated equation

$$\widehat{sleep} = 3{,}840.83 - .163\ totwrk - 11.71\ educ - 8.70\ age$$
$$(235.11)\ \ (.018)\qquad\quad (5.86)\qquad (11.21)$$
$$+ .128\ age^2 + 87.75\ male$$
$$(.134)\qquad (34.33)$$
$$n = 706,\ R^2 = .123,\ \bar{R}^2 = .117.$$

The variable *sleep* is total minutes per week spent sleeping at night, *totwrk* is total weekly minutes spent working, *educ* and *age* are measured in years, and *male* is a gender dummy.

(i) All other factors being equal, is there evidence that men sleep more than women? How strong is the evidence?

(ii) Is there a statistically significant tradeoff between working and sleeping? What is the estimated tradeoff?

(iii) What other regression do you need to run to test the null hypothesis that, holding other factors fixed, age has no effect on sleeping?

2 The following equations were estimated using the data in BWGHT.RAW:

$$\widehat{\log(bwght)} = 4.66 - .0044\ cigs + .0093\ \log(faminc) + .016\ parity$$
$$(.22)\ \ (.0009)\qquad (.0059)\qquad\quad (.006)$$
$$+ .027\ male + .055\ white$$
$$(.010)\qquad (.013)$$
$$n = 1{,}388,\ R^2 = .0472$$

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p.258 · PPT 필기

KEY

KEY Key Terms: 노랑 = PPT에서 다룬 용어, 파랑 = 교재에만.

SKIP

SKIP Ch.7 연습문제 · Computer Exercises (p.258–267) — 필기 범위 밖.

▶ 교수 설명 (수업 전사본 · 판서)

S38 · Summary

Summary 슬라이드는 따로 다루지 않음. S37 뒤 "다음 장으로 넘어가도 되나?"라고 묻고 곧바로 Ch.8(이분산)으로 넘어감 → 표지 CH8 칸.

and

$$\widehat{\log(bwght)} = 4.65 - .0052\,cigs + .0110\log(faminc) + .017\,parity$$
$$(.38)\ \ (.0010)\qquad (.0085)\qquad\qquad (.006)$$
$$+ .034\,male + .045\,white - .0030\,motheduc + .0032\,fatheduc$$
$$(.011)\qquad (.015)\qquad (.0030)\qquad\qquad (.0026)$$
$$n = 1{,}191,\ R^2 = .0493.$$

The variables are defined as in Example 4.9, but we have added a dummy variable for whether the child is male and a dummy variable indicating whether the child is classified as white.

(i) In the first equation, interpret the coefficient on the variable *cigs*. In particular, what is the effect on birth weight from smoking 10 more cigarettes per day?

(ii) How much more is a white child predicted to weigh than a nonwhite child, holding the other factors in the first equation fixed? Is the difference statistically significant?

(iii) Comment on the estimated effect and statistical significance of *motheduc*.

(iv) From the given information, why are you unable to compute the F statistic for joint significance of *motheduc* and *fatheduc*? What would you have to do to compute the F statistic?

3 Using the data in GPA2.RAW, the following equation was estimated:

$$\widehat{sat} = 1{,}028.10 + 19.30\,hsize - 2.19\,hsize^2 - 45.09\,female$$
$$(6.29)\ \ (3.83)\qquad (.53)\qquad\qquad (4.29)$$
$$- 169.81\,black + 62.31\,female{\cdot}black$$
$$(12.71)\qquad\qquad (18.15)$$
$$n = 4{,}137,\ R^2 = .0858.$$

The variable *sat* is the combined SAT score, *hsize* is size of the student's high school graduating class, in hundreds, *female* is a gender dummy variable, and *black* is a race dummy variable equal to one for blacks and zero otherwise.

(i) Is there strong evidence that $hsize^2$ should be included in the model? From this equation, what is the optimal high school size?

(ii) Holding *hsize* fixed, what is the estimated difference in SAT score between nonblack females and nonblack males? How statistically significant is this estimated difference?

(iii) What is the estimated difference in SAT score between nonblack males and black males? Test the null hypothesis that there is no difference between their scores, against the alternative that there is a difference.

(iv) What is the estimated difference in SAT score between black females and nonblack females? What would you need to do to test whether the difference is statistically significant?

4 An equation explaining chief executive officer salary is

$$\widehat{\log(salary)} = 4.59 + .257\log(sales) + .011\,roe + .158\,finance$$
$$(.30)\ \ (.032)\qquad\qquad (.004)\qquad (.089)$$
$$+ .181\,consprod - .283\,utility$$
$$(.085)\qquad\qquad (.099)$$
$$n = 209,\ R^2 = .357.$$

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The data used are in CEOSAL1.RAW, where *finance*, *consprod*, and *utility* are binary variables indicating the financial, consumer products, and utilities industries. The omitted industry is transportation.

(i) Compute the approximate percentage difference in estimated salary between the utility and transportation industries, holding *sales* and *roe* fixed. Is the difference statistically significant at the 1% level?

(ii) Use equation (7.10) to obtain the exact percentage difference in estimated salary between the utility and transportation industries and compare this with the answer obtained in part (i).

(iii) What is the approximate percentage difference in estimated salary between the consumer products and finance industries? Write an equation that would allow you to test whether the difference is statistically significant.

5 In Example 7.2, let *noPC* be a dummy variable equal to one if the student does not own a PC, and zero otherwise.

(i) If *noPC* is used in place of *PC* in equation (7.6), what happens to the intercept in the estimated equation? What will be the coefficient on *noPC*? (*Hint:* Write $PC = 1 - noPC$ and plug this into the equation $\widehat{colGPA} = \hat{\beta}_0 + \hat{\delta}_0 PC + \hat{\beta}_1 hsGPA + \hat{\beta}_2 ACT$.)

(ii) What will happen to the *R*-squared if *noPC* is used in place of *PC*?

(iii) Should *PC* and *noPC* both be included as independent variables in the model? Explain.

6 To test the effectiveness of a job training program on the subsequent wages of workers, we specify the model

$$\log(wage) = \beta_0 + \beta_1 train + \beta_2 educ + \beta_3 exper + u,$$

where *train* is a binary variable equal to unity if a worker participated in the program. Think of the error term *u* as containing unobserved worker ability. If less able workers have a greater chance of being selected for the program, and you use an OLS analysis, what can you say about the likely bias in the OLS estimator of β_1? (*Hint:* Refer back to Chapter 3.)

7 In the example in equation (7.29), suppose that we define *outlf* to be one if the woman is out of the labor force, and zero otherwise.

(i) If we regress *outlf* on all of the independent variables in equation (7.29), what will happen to the intercept and slope estimates? (*Hint: inlf* = 1 − *outlf*. Plug this into the population equation $inlf = \beta_0 + \beta_1 nwifeinc + \beta_2 educ + \ldots$ and rearrange.)

(ii) What will happen to the standard errors on the intercept and slope estimates?

(iii) What will happen to the *R*-squared?

8 Suppose you collect data from a survey on wages, education, experience, and gender. In addition, you ask for information about marijuana usage. The original question is: "On how many separate occasions last month did you smoke marijuana?"

(i) Write an equation that would allow you to estimate the effects of marijuana usage on wage, while controlling for other factors. You should be able to make statements such as, "Smoking marijuana five more times per month is estimated to change wage by *x*%."

(ii) Write a model that would allow you to test whether drug usage has different effects on wages for men and women. How would you test that there are no differences in the effects of drug usage for men and women?

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(iii) Suppose you think it is better to measure marijuana usage by putting people into one of four categories: nonuser, light user (1 to 5 times per month), moderate user (6 to 10 times per month), and heavy user (more than 10 times per month). Now, write a model that allows you to estimate the effects of marijuana usage on wage.

(iv) Using the model in part (iii), explain in detail how to test the null hypothesis that marijuana usage has no effect on wage. Be very specific and include a careful listing of degrees of freedom.

(v) What are some potential problems with drawing causal inference using the survey data that you collected?

9 Let d be a dummy (binary) variable and let z be a quantitative variable. Consider the model

$$y = \beta_0 + \delta_0 d + \beta_1 z + \delta_1 d \cdot z + u;$$

this is a general version of a model with an interaction between a dummy variable and a quantitative variable. [An example is in equation (7.17).]

(i) Since it changes nothing important, set the error to zero, $u = 0$. Then, when $d = 0$ we can write the relationship between y and z as the function $f_0(z) = \beta_0 + \beta_1 z$. Write the same relationship when $d = 1$, where you should use $f_1(z)$ on the left-hand side to denote the linear function of z.

(ii) Assuming that $\delta_1 \neq 0$ (which means the two lines are not parallel), show that the value of z^* such that $f_0(z^*) = f_1(z^*)$ is $z^* = -\delta_0/\delta_1$. This is the point at which the two lines intersect [as in Figure 7.2(b)]. Argue that z^* is positive if and only if δ_0 and δ_1 have opposite signs.

(iii) Using the data in TWOYEAR.RAW, the following equation can be estimated:

$$\widehat{\log(wage)} = \underset{}{2.289} - \underset{(0.011)}{.357}\, female + \underset{(.015)}{.50}\, totcoll + \underset{(.003)}{.030}\, female \cdot totcoll \underset{(.005)}{}$$
$$n = 6{,}763,\ R^2 = .202,$$

where all coefficients and standard errors have been rounded to three decimal places. Using this equation, find the value of *totcoll* such that the predicted values of log(*wage*) are the same for men and women.

(iv) Based on the equation in part (iii), can women realistically get enough years of college so that their earnings catch up to those of men? Explain.

10 For a child i living in a particular school district, let $voucher_i$ be a dummy variable equal to one if a child is selected to participate in a school voucher program, and let $score_i$ be that child's score on a subsequent standardized exam. Suppose that the participation variable, $voucher_i$, is completely randomized in the sense that it is independent of both observed and unobserved factors that can affect the test score.

(i) If you run a simple regression $score_i$ on $voucher_i$ using a random sample of size n, does the OLS estimator provide an unbiased estimator of the effect of the voucher program?

(ii) Suppose you can collect additional background information, such as family income, family structure (e.g., whether the child lives with both parents), and parents' education levels. Do you need to control for these factors to obtain an unbiased estimator of the effects of the voucher program? Explain.

(iii) Why should you include the family background variables in the regression? Is there a situation in which you would not include the background variables?

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Computer Exercises

C1 Use the data in GPA1.RAW for this exercise.

(i) Add the variables *mothcoll* and *fathcoll* to the equation estimated in (7.6) and report the results in the usual form. What happens to the estimated effect of PC ownership? Is *PC* still statistically significant?

(ii) Test for joint significance of *mothcoll* and *fathcoll* in the equation from part (i) and be sure to report the *p*-value.

(iii) Add $hsGPA^2$ to the model from part (i) and decide whether this generalization is needed.

C2 Use the data in WAGE2.RAW for this exercise.

(i) Estimate the model

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \beta_4 married + \beta_5 black + \beta_6 south + \beta_7 urban + u$$

and report the results in the usual form. Holding other factors fixed, what is the approximate difference in monthly salary between blacks and nonblacks? Is this difference statistically significant?

(ii) Add the variables $exper^2$ and $tenure^2$ to the equation and show that they are jointly insignificant at even the 20% level.

(iii) Extend the original model to allow the return to education to depend on race and test whether the return to education does depend on race.

(iv) Again, start with the original model, but now allow wages to differ across four groups of people: married and black, married and nonblack, single and black, and single and nonblack. What is the estimated wage differential between married blacks and married nonblacks?

C3 A model that allows major league baseball player salary to differ by position is

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 hrunsyr + \beta_5 rbisyr + \beta_6 runsyr + \beta_7 fldperc + \beta_8 allstar + \beta_9 frstbase + \beta_{10} scndbase + \beta_{11} thrdbase + \beta_{12} shrtstop + \beta_{13} catcher + u,$$

where outfield is the base group.

(i) State the null hypothesis that, controlling for other factors, catchers and outfielders earn, on average, the same amount. Test this hypothesis using the data in MLB1.RAW and comment on the size of the estimated salary differential.

(ii) State and test the null hypothesis that there is no difference in average salary across positions, once other factors have been controlled for.

(iii) Are the results from parts (i) and (ii) consistent? If not, explain what is happening.

C4 Use the data in GPA2.RAW for this exercise.

(i) Consider the equation

$$colgpa = \beta_0 + \beta_1 hsize + \beta_2 hsize^2 + \beta_3 hsperc + \beta_4 sat + \beta_5 female + \beta_6 athlete + u,$$

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where *colgpa* is cumulative college grade point average, *hsize* is size of high school graduating class, in hundreds, *hsperc* is academic percentile in graduating class, *sat* is combined SAT score, *female* is a binary gender variable, and *athlete* is a binary variable, which is one for student-athletes. What are your expectations for the coefficients in this equation? Which ones are you unsure about?

(ii) Estimate the equation in part (i) and report the results in the usual form. What is the estimated GPA differential between athletes and nonathletes? Is it statistically significant?

(iii) Drop *sat* from the model and reestimate the equation. Now, what is the estimated effect of being an athlete? Discuss why the estimate is different than that obtained in part (ii).

(iv) In the model from part (i), allow the effect of being an athlete to differ by gender and test the null hypothesis that there is no ceteris paribus difference between women athletes and women nonathletes.

(v) Does the effect of *sat* on *colgpa* differ by gender? Justify your answer.

C5 In Problem 2 in Chapter 4, we added the return on the firm's stock, *ros*, to a model explaining CEO salary; *ros* turned out to be insignificant. Now, define a dummy variable, *rosneg*, which is equal to one if $ros < 0$ and equal to zero if $ros \geq 0$. Use CEOSAL1.RAW to estimate the model

$$\log(salary) = \beta_0 + \beta_1 \log(sales) + \beta_2 roe + \beta_3 rosneg + u.$$

Discuss the interpretation and statistical significance of $\hat{\beta}_3$.

C6 Use the data in SLEEP75.RAW for this exercise. The equation of interest is

$$sleep = \beta_0 + \beta_1 totwrk + \beta_2 educ + \beta_3 age + \beta_4 age^2 + \beta_5 yngkid + u.$$

(i) Estimate this equation separately for men and women and report the results in the usual form. Are there notable differences in the two estimated equations?

(ii) Compute the Chow test for equality of the parameters in the sleep equation for men and women. Use the form of the test that adds *male* and the interaction terms *male*·*totwrk*, …, *male*·*yngkid* and uses the full set of observations. What are the relevant *df* for the test? Should you reject the null at the 5% level?

(iii) Now, allow for a different intercept for males and females and determine whether the interaction terms involving *male* are jointly significant.

(iv) Given the results from parts (ii) and (iii), what would be your final model?

C7 Use the data in WAGE1.RAW for this exercise.

(i) Use equation (7.18) to estimate the gender differential when $educ = 12.5$. Compare this with the estimated differential when $educ = 0$.

(ii) Run the regression used to obtain (7.18), but with *female*·(*educ* − 12.5) replacing *female*·*educ*. How do you interpret the coefficient on *female* now?

(iii) Is the coefficient on *female* in part (ii) statistically significant? Compare this with (7.18) and comment.

C8 Use the data in LOANAPP.RAW for this exercise. The binary variable to be explained is *approve*, which is equal to one if a mortgage loan to an individual was approved. The key explanatory variable is *white*, a dummy variable equal to one if the applicant was white. The other applicants in the data set are black and Hispanic.

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To test for discrimination in the mortgage loan market, a linear probability model can be used:

$$approve = \beta_0 + \beta_1 white + other\ factors.$$

(i) If there is discrimination against minorities, and the appropriate factors have been controlled for, what is the sign of β_1?

(ii) Regress *approve* on *white* and report the results in the usual form. Interpret the coefficient on *white*. Is it statistically significant? Is it practically large?

(iii) As controls, add the variables *hrat*, *obrat*, *loanprc*, *unem*, *male*, *married*, *dep*, *sch*, *cosign*, *chist*, *pubrec*, *mortlat1*, *mortlat2*, and *vr*. What happens to the coefficient on *white*? Is there still evidence of discrimination against nonwhites?

(iv) Now, allow the effect of race to interact with the variable measuring other obligations as a percentage of income (*obrat*). Is the interaction term significant?

(v) Using the model from part (iv), what is the effect of being white on the probability of approval when *obrat* = 32, which is roughly the mean value in the sample? Obtain a 95% confidence interval for this effect.

C9 There has been much interest in whether the presence of 401(k) pension plans, available to many U.S. workers, increases net savings. The data set 401KSUBS.RAW contains information on net financial assets (*nettfa*), family income (*inc*), a binary variable for eligibility in a 401(k) plan (*e401k*), and several other variables.

(i) What fraction of the families in the sample are eligible for participation in a 401(k) plan?

(ii) Estimate a linear probability model explaining 401(k) eligibility in terms of income, age, and gender. Include income and age in quadratic form, and report the results in the usual form.

(iii) Would you say that 401(k) eligibility is independent of income and age? What about gender? Explain.

(iv) Obtain the fitted values from the linear probability model estimated in part (ii). Are any fitted values negative or greater than one?

(v) Using the fitted values $\widehat{e401k}_i$ from part (iv), define $\widetilde{e401k}_i = 1$ if $\widehat{e401k} \geq .5$ and $\widetilde{401k} = 0$ if $\widehat{e401k} < .5$. Out of 9,275 families, how many are predicted to be eligible for a 401(k) plan?

(vi) For the 5,638 families not eligible for a 401(k), what percentage of these are predicted not to have a 401(k), using the predictor $\widetilde{e401k}_i$? For the 3,637 families eligible for a 401(k) plan, what percentage are predicted to have one? (It is helpful if your econometrics package has a "tabulate" command.)

(vii) The overall percent correctly predicted is about 64.9%. Do you think this is a complete description of how well the model does, given your answers in part (vi)?

(viii) Add the variable *pira* as an explanatory variable to the linear probability model. Other things equal, if a family has someone with an individual retirement account, how much higher is the estimated probability that the family is eligible for a 401(k) plan? Is it statistically different from zero at the 10% level?

C10 Use the data in NBASAL.RAW for this exercise.

(i) Estimate a linear regression model relating points per game to experience in the league and position (guard, forward, or center). Include experience in quadratic form and use centers as the base group. Report the results in the usual form.

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(ii) Why do you not include all three position dummy variables in part (i)?

(iii) Holding experience fixed, does a guard score more than a center? How much more? Is the difference statistically significant?

(iv) Now, add marital status to the equation. Holding position and experience fixed, are married players more productive (based on points per game)?

(v) Add interactions of marital status with both experience variables. In this expanded model, is there strong evidence that marital status affects points per game?

(vi) Estimate the model from part (iv) but use assists per game as the dependent variable. Are there any notable differences from part (iv)? Discuss.

C11 Use the data in 401KSUBS.RAW for this exercise.

(i) Compute the average, standard deviation, minimum, and maximum values of *nettfa* in the sample.

(ii) Test the hypothesis that average *nettfa* does not differ by 401(k) eligibility status; use a two-sided alternative. What is the dollar amount of the estimated difference?

(iii) From part (ii) of Computer Exercise C9, it is clear that *e401k* is not exogenous in a simple regression model; at a minimum, it changes by income and age. Estimate a multiple linear regression model for *nettfa* that includes income, age, and *e401k* as explanatory variables. The income and age variables should appear as quadratics. Now, what is the estimated dollar effect of 401(k) eligibility?

(iv) To the model estimated in part (iii), add the interactions $e401k \cdot (age - 41)$ and $e401k \cdot (age - 41)^2$. Note that the average age in the sample is about 41, so that in the new model, the coefficient on *e401k* is the estimated effect of 401(k) eligibility at the average age. Which interaction term is significant?

(v) Comparing the estimates from parts (iii) and (iv), do the estimated effects of 401(k) eligibility at age 41 differ much? Explain.

(vi) Now, drop the interaction terms from the model, but define five family size dummy variables: *fsize1, fsize2, fsize3, fsize4,* and *fsize5*. The variable *fsize5* is unity for families with five or more members. Include the family size dummies in the model estimated from part (iii); be sure to choose a base group. Are the family dummies significant at the 1% level?

(vii) Now, do a Chow test for the model

$$nettfa = \beta_0 + \beta_1 inc + \beta_2 inc^2 + \beta_3 age + \beta_4 age^2 + \beta_5 e401k + u$$

across the five family size categories, allowing for intercept differences. The restricted sum of squared residuals, SSR_r, is obtained from part (vi) because that regression assumes all slopes are the same. The unrestricted sum of squared residuals is $SSR_{ur} = SSR_1 + SSR_2 + ... + SSR_5$, where SSR_f is the sum of squared residuals for the equation estimated using only family size *f*. You should convince yourself that there are 30 parameters in the unrestricted model (5 intercepts plus 25 slopes) and 10 parameters in the restricted model (5 intercepts plus 5 slopes). Therefore, the number of restrictions being tested is $q = 20$, and the *df* for the unrestricted model is $9{,}275 - 30 = 9{,}245$.

C12 Use the data set in BEAUTY.RAW, which contains a subset of the variables (but more usable observations than in the regressions) reported by Hamermesh and Biddle (1994).

(i) Find the separate fractions of men and women that are classified as having above average looks. Are more people rated as having above average or below average looks?

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(ii) Test the null hypothesis that the population fractions of above-average-looking women and men are the same. Report the one-sided p-value that the fraction is higher for women. (*Hint:* Estimating a simple linear probability model is easiest.)

(iii) Now estimate the model

$$\log(wage) = \beta_0 + \beta_1 belavg + \beta_2 abvavg + u$$

separately for men and women, and report the results in the usual form. In both cases, interpret the coefficient on *belavg*. Explain in words what the hypothesis H_0: $\beta_1 = 0$ against H_1: $\beta_1 < 0$ means, and find the p-values for men and women.

(iv) Is there convincing evidence that women with above average looks earn more than women with average looks? Explain.

(v) For both men and women, add the explanatory variables *educ*, *exper*, $exper^2$, *union*, *goodhlth*, *black*, *married*, *south*, *bigcity*, *smllcity*, and *service*. Do the effects of the "looks" variables change in important ways?

(vi) Use the SSR form of the Chow F statistic to test whether the slopes of the regression functions in part (v) differ across men and women. Be sure to allow for an intercept shift under the null.

C13 Use the data in APPLE.RAW to answer this question.

(i) Define a binary variable as $ecobuy = 1$ if $ecolbs > 0$ and $ecobuy = 0$ if $ecolbs = 0$. In other words, *ecobuy* indicates whether, at the prices given, a family would buy any ecologically friendly apples. What fraction of families claim they would buy ecolabeled apples?

(ii) Estimate the linear probability model

$$ecobuy = \beta_0 + \beta_1 ecoprc + \beta_2 regprc + \beta_3 faminc + \beta_4 hhsize + \beta_5 educ + \beta_6 age + u,$$

and report the results in the usual form. Carefully interpret the coefficients on the price variables.

(iii) Are the nonprice variables jointly significant in the LPM? (Use the usual F statistic, even though it is not valid when there is heteroskedasticity.) Which explanatory variable other than the price variables seems to have the most important effect on the decision to buy ecolabeled apples? Does this make sense to you?

(iv) In the model from part (ii), replace *faminc* with log(*faminc*). Which model fits the data better, using *faminc* or log(*faminc*)? Interpret the coefficient on log(*faminc*).

(v) In the estimation in part (iv), how many estimated probabilities are negative? How many are bigger than one? Should you be concerned?

(vi) For the estimation in part (iv), compute the percent correctly predicted for each outcome, $ecobuy = 0$ and $ecobuy = 1$. Which outcome is best predicted by the model?

C14 Use the data in CHARITY.RAW to answer this question. The variable *respond* is a dummy variable equal to one if a person responded with a contribution on the most recent mailing sent by a charitable organization. The variable *resplast* is a dummy variable equal to one if the person responded to the previous mailing, *avggift* is the average of past gifts (in Dutch guilders), and *propresp* is the proportion of times the person has responded to past mailings.

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(i) Estimate a linear probability model relating *respond* to *resplast* and *avggift*. Report the results in the usual form, and interpret the coefficient on *resplast*.
(ii) Does the average value of past gifts seem to affect the probability of responding?
(iii) Add the variable *propresp* to the model, and interpret its coefficient. (Be careful here: an increase of one in *propresp* is the largest possible change.)
(iv) What happened to the coefficient on *resplast* when *propresp* was added to the regression? Does this make sense?
(v) Add *mailsyear*, the number of mailings per year, to the model. How big is its estimated effect? Why might this not be a good estimate of the causal effect of mailings on responding?

C15 Use the data in FERTIL2.RAW to answer this question.
(i) Find the smallest and largest values of *children* in the sample. What is the average of *children*? Does any woman have exactly the average number of children?
(ii) What percentage of women have electricity in the home?
(iii) Compute the average of *children* for those without electricity and do the same for those with electricity. Comment on what you find. Test whether the population means are the same using a simple regression.
(iv) From part (iii), can you infer that having electricity "causes" women to have fewer children? Explain.
(v) Estimate a multiple regression model of the kind reported in equation (7.37), but add age^2, *urban*, and the three religious affiliation dummies. How does the estimated effect of having electricity compare with that in part (iii)? Is it still statistically significant?
(vi) To the equation in part (v), add an interaction between *electric* and *educ*. Is its coefficient statistically significant? What happens to the coefficient on *electric*?
(vii) The median and mode value for *educ* is 7. In the equation from part (vi), use the centered interaction term *electric* • (*educ* – 7) in place of *electric* • *educ*. What happens to the coefficient on *electric* compared with part (vi)? Why? How does the coefficient on *electric* compare with that in part (v)?

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